Supplementary Proof Record for Time-Residual Genesis of Gravity and the Four Fundamental Interactions

Complete Ordered Theorem Registry, Explicit Derivation Ledger, and Executable Certificate Map

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Abstract

This supplementary record provides the exhaustive executable proof index for the accompanying Main Manuscript. It contains the complete causal registry of 525 theorem roots and the complete ledger of 544 explicit derivations generated by the standalone source. The registry is not a substitute for the mathematical exposition in the Main Manuscript. Its purpose is to make every formal claim traceable to one callable proof root, one deterministic output marker, one causal stage, and one declared generated domain. The derivation ledger records, row by row, the generated input, exact operation, generated output, and mathematical consequence used by later stages. The artifact-coherence matrix, source and output hashes, symbol glossary, and data-availability statement complete the reproducibility layer. Internal theorem closure remains distinct from empirical identification, historical priority, unrestricted physical equivalence, and third-party reproduction.

Keywords: executable proof record; theorem registry; derivation ledger; causal dependency; deterministic certificate; covariant residual field equation; four-interaction unification.

Contents

1 How to use this proof record

Reader guide before the proof: Three synchronized evidence layers

The Main Manuscript explains the theory and proves its principal mathematical claims. This document indexes every executable theorem and every explicit derivation. The standalone source and deterministic output provide the complete calculated certificate. A reader should first locate a claim in the Main Manuscript, then use the theorem title or proof-root string in this record, and finally search the deterministic output for the identical root marker.

For each theorem entry, the fields have the following meanings.

Field	Meaning
Ordinal	Unique position in the authoritative interleaved causal order.
Class	Primary executable theorem or narrative theorem tied to the same generated domain.
Stage	One of the nine dependency groups executed by the standalone source.
Executable root	Exact callable/search string identifying the calculation in the Python source and deterministic output.
Reader principle	Plain-language statement of the mathematical or physical role.
Theorem statement	Formal claim evaluated on the declared generated domain.
Certificate	Exact rational, matrix, incidence, path, differential, variational, spectral, or closure calculation printed under the root marker.
Result	Boolean root gate returned by the fresh standalone execution.

2 Authoritative theorem-stage index

Stage	Name	Ordinals	Primary	Narrative	Total
1	SIGNED OCCUPANCY AND TEMPORAL ORIGIN	1–40	36	4	40
2	RESIDUAL DIMENSION AND JS GEOMETRY	41–84	37	7	44
3	FOUR FORCE CARRIERS AND PARENT ACTION	85–134	50	0	50
4	CHIRAL MATTER AND GENERATIONS	135–185	35	16	51
5	ELECTROMAGNETIC AND QUANTUM CONSEQUENCES	186–213	28	0	28
6	NATIVE PROJECTION AND EFFECTIVE DYNAMICS	214–279	66	0	66
7	COVARIANT CONTINUUM FIELD CONSTRUCTION	280–387	98	10	108
8	RECURSIVE QUANTUM AND GRAVITY	388–465	78	0	78
9	GLOBAL CLOSURE AND SINGLE PROOF	466–525	59	1	60

3 Certificate-navigation protocol

To verify one theorem independently, use the following deterministic procedure.

1. Read the theorem's prerequisites and formal proof in the Main Manuscript.
2. Copy the exact executable-root string from this record.
3. Search the standalone Python source to inspect the objects constructed and predicates evaluated by that root.
4. Search the deterministic output for the same root marker and inspect the complete exact certificate printed beneath it.
5. Locate related operations in the explicit derivation ledger by generated input, output, or mathematical consequence.
6. Confirm that the result is TRUE and that no external measured target enters the generated premises unless the theorem is explicitly part of the sealed external-comparison interface.

Logical status of the record

A TRUE executable root proves only the proposition implemented on its declared generated domain. The conjunction of all roots establishes internal closure of that domain. It does not by itself establish empirical identity with nature, unrestricted equivalence to every Standard-Model or Einstein configuration, historical priority, or independent reproduction.

A Complete ordered theorem registry: all 525 roots

The registry follows the same interleaved causal order executed by the standalone source. Every statement is obtained from the current executable output, the current callable docstring, or the current theorem exposition map; no earlier manuscript metadata is used.

Reader guide before the proof: 1. NESTED SIGNED OCCUPANCY ORIGIN THEOREM

Negative, zero, and positive outer states each contain A, B, and C inner signs. Every inner sign occurs in empty and full form, producing exactly eighteen primitive states.

Executable root: `_u905_nested_signed_occupancy_origin_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.1 (1. NESTED SIGNED OCCUPANCY ORIGIN THEOREM)**.** *Execute the exact certificate for NESTED SIGNED OCCUPANCY ORIGIN THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.2. NESTED SIGNED OCCUPANCY ORIGIN THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 2. STRUCTURED ZERO RECURSIVE STATE SEMANTICS THEOREM

Negative, zero, and positive are relational positions, while full and empty are independent occupancy states. Zero can therefore be empty or full without contradiction. A completely saturated layer has no remaining internal contrast and becomes the centered zero reference of the next recursive layer.

Executable root: `theorem_structured_zero_recursive_state_semantics`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.3 (2. STRUCTURED ZERO RECURSIVE STATE SEMANTICS THEOREM)**.** *Prove the complete negative-zero-positive by A-B-C by full-empty state grammar and the distinction between relational zero, empty zero, full zero, and a saturated layer rebased as the next relational zero.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.4. STRUCTURED ZERO RECURSIVE STATE SEMANTICS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 3. COMPLETE EIGHTEEN STATE SEMANTIC BIJECTION THEOREM

Every negative, zero and positive outer relation contains occupied and vacant A-negative, B-zero and C-positive inner relations. The complete semantic table is a bijection with the eighteen primitive tuples.

Executable root: `theorem_complete_eighteen_state_semantic_bijection`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.5 (3. COMPLETE EIGHTEEN STATE SEMANTIC BIJECTION THEOREM)**.** *Prove a one-to-one semantic ledger for every outer sign, inner relation, and occupied-vacant state.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.6. COMPLETE EIGHTEEN STATE SEMANTIC BIJECTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 4. PRIMITIVE RELATIONAL CATEGORY GLOBAL MINIMALITY THEOREM

A reversible sign relation requires one negative direction, one relational zero, and one positive direction. Occupancy requires one empty and one full state. Any larger admissible finite alphabet contains extra independent reversal or complement pairs, while the unique irreducible minimum is three by three by two.

Executable root: `theorem_primitive_relational_category_global_minimality`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.7 (4. PRIMITIVE RELATIONAL CATEGORY GLOBAL MINIMALITY THEOREM)**.** *Classify the finite reversible-sign and occupancy-complement product category and prove that three by three by two is its unique irreducible minimum.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.8. PRIMITIVE RELATIONAL CATEGORY GLOBAL MINIMALITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 5. PRIMITIVE LEDGER MINIMAL UNIQUENESS THEOREM

A reversible primitive sign system requires one fixed zero and one exchanged negative-positive pair, so its minimum size is three. An occupancy complement has no fixed point and one exchanged empty-full pair, so its minimum size is two; the unique minimal nested ledger therefore has eighteen states.

Executable root: `theorem_primitive_ledger_minimal_uniqueness`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.9 (5. PRIMITIVE LEDGER MINIMAL UNIQUENESS THEOREM)**.** *Classify the smallest reversible outer sign, inner relation, and occupancy alphabets and derive the eighteen-state ledger.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.10. PRIMITIVE LEDGER MINIMAL UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 6. PRIMITIVE ARCHITECTURE GLOBAL NECESSITY THEOREM

The reversible outer sign, reversible inner relation, and complementary occupancy requirements are classified before any force role is named. Inside the declared irreducible grammar, only the three-by-three-by-two factorization uses every primitive label and preserves both residual signs and closure complement.

Executable root: `theorem_primitive_architecture_global_necessity`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.11 (6. PRIMITIVE ARCHITECTURE GLOBAL NECESSITY THEOREM)**.** *Execute the exact certificate for PRIMITIVE ARCHITECTURE GLOBAL NECESSITY THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.12. PRIMITIVE ARCHITECTURE GLOBAL NECESSITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 7. RECURSIVE PRIMITIVE GRAMMAR IRREDUCIBILITY THEOREM

Every finite involution decomposes into fixed points and exchanged pairs; requiring one relational zero and one connected reversible pair forces a three-state sign alphabet. Requiring one connected empty-full complement forces two occupancy states, so larger finite competitors are reducible copies rather than new irreducible primitives.

Executable root: `theorem_recursive_primitive_grammar_irreducibility`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.13 (7. RECURSIVE PRIMITIVE GRAMMAR IRREDUCIBILITY THEOREM)**.** *Prove that the recursive outer-sign, inner-sign, and occupancy grammar has no larger irreducible finite competitor.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.14. RECURSIVE PRIMITIVE GRAMMAR IRREDUCIBILITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 8. ZERO STATE STRUCTURE THEOREM

Zero is classified by residual, occupation, boundary support, binding, and internal contrast. Equal opposite content and absolute absence are not the same state. A five-to-five state is an algebraic zero because the exposed residual vanishes, although both negative and positive contents remain present.

Executable root: `_zfr_zero_state_structure_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.15 (8. ZERO STATE STRUCTURE THEOREM)**.** *Separate algebraic zero, occupied balance, bound closure, supported cavity, absolute absence, and saturated relative zero.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.16. ZERO STATE STRUCTURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 9. RESIDUAL BIAS SATURATION THEOREM

A time-ordered one-unit imbalance selects a direction. The conservative transfer law then amplifies the selected occupancy until the opposite side is exhausted. The proof establishes monotone directional ordering and finite saturation; the word entropy is used only as intuition unless a separate statistical measure is supplied.

Executable root: `_zfr_residual_bias_saturation_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.17 (9. RESIDUAL BIAS SATURATION THEOREM)**.** *Prove that one transferred unit from the five-to-five zero selects a preserved direction and reaches one-sided saturation in finitely many exact steps.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.18. RESIDUAL BIAS SATURATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 10. SATURATION RELATIVE ZERO THEOREM

A one-sign saturated state is nonzero from outside but has zero internal contrast because no opposite comparison state remains. The whole saturated layer is therefore reused as the next layer zero baseline while its outer sign record is preserved.

Executable root: `_zfr_saturation_relative_zero_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.19 (10. SATURATION RELATIVE ZERO THEOREM)**.** *Prove that a one-sign saturated state remains signed externally while becoming the zero baseline of its next internal relation layer.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.20. SATURATION RELATIVE ZERO THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 11. RECURSIVE RELATION FRACTAL THEOREM

Fractal self-similarity comes from repetition of the same generation operator: zero, residual, direction, saturation, rebasing, and redifferentiation. The theorem fixes the relation alphabet, not a cone, sphere, onion, or any other unique spatial embedding.

Executable root: `_zfr_recursive_relation_fractal_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.21 (11. RECURSIVE RELATION FRACTAL THEOREM)**.** *Prove exact combinatorial self-similarity when every saturated relative zero carries the same eighteen-seat relation alphabet at the next depth.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.22. RECURSIVE RELATION FRACTAL THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 12. BIDIRECTIONAL FRACTAL LIMIT THEOREM

The inward limit refines every saturated relative zero into eighteen child relations. The outward limit groups each complete family of eighteen children into one higher-layer unit. Both directions obey the same mass-consistent cylinder ledger.

Executable root: `_bidirectional_fractal_limit_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.23 (12. BIDIRECTIONAL FRACTAL LIMIT THEOREM)**.** *Construct compatible inward refinement and outward aggregation limits from the repeated eighteen-seat relation alphabet.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.24. BIDIRECTIONAL FRACTAL LIMIT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 13. ONION SHELL CLOSURE THEOREM

Layering produces exact shell and cumulative-count laws. Complete shells close their signed relation ledger, but the same symbolic shell admits inequivalent spatial embeddings, so no sphere, cone, torus, spiral, or branch is selected without an additional embedding rule.

Executable root: `_onion_shell_closure_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.25 (13. ONION SHELL CLOSURE THEOREM)**.** *Derive shell counts, balanced boundary ledgers, and the exact distinction between symbolic closure and a chosen spatial shape.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.26. ONION SHELL CLOSURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 14. RELATION SCALE TOPOLOGY QUANTIZATION THEOREM

The eighteen-seat alphabet already has a canonical symbolic product coordinate system with three outer signs, three inner signs, and two occupations. This fixes exact combinatorial scale factors and closure periods without fixing a unique physical cone, sphere, torus, or other embedding.

Executable root: `_e895_relation_scale_topology_quantization_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.27 (14. RELATION SCALE TOPOLOGY QUANTIZATION THEOREM)**.** *Derive exact inward scales, outward aggregation, symbolic closure periods, conditional flux laws, and discrete shell levels from the generated three-by-three-by-two relation coordinates.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.28. RELATION SCALE TOPOLOGY QUANTIZATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 15. SYMBOLIC CONTINUUM COORDINATE THEOREM

The repeated three-by-three-by-two alphabet is not only a state count. Each infinite path has two ternary coordinate addresses and one binary coordinate address. Finite prefixes form nested rational boxes that partition one complete three-coordinate carrier; temporal residual order then supplies a separate causal coordinate.

Executable root: `_e896_symbolic_continuum_coordinate_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.29 (15. SYMBOLIC CONTINUUM COORDINATE THEOREM)**.** *Decode infinite three-by-three-by-two relation paths into a complete three-coordinate continuum carrier and append the generated temporal order as a fourth causal coordinate.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.30. SYMBOLIC CONTINUUM COORDINATE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 16. THREE DIMENSIONAL EFFECTIVE WORLD PROJECTION THEOREM

The effective world is a complete three-coordinate quotient layer rather than a literal two-dimensional shadow. Many hidden residual and phase states project to one local world cell, while the quotient remains normalized, surjective, and local on the declared finite carrier.

Executable root: `_e897_three_dimensional_effective_world_projection_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.31 (16. THREE DIMENSIONAL EFFECTIVE WORLD PROJECTION THEOREM)**.** *Construct a surjective many-to-one native-to-three-dimensional-world projection with exact quotient fibers and local descent.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.32. THREE DIMENSIONAL EFFECTIVE WORLD PROJECTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 17. MINIMAL SEMANTIC WORLD RANK THEOREM

Three-dimensionality is not inferred merely from the number eighteen. It is forced inside the declared relation grammar by three independently changeable labeled primitives: outer sign, inner sign, and occupancy. Their local change vectors have rank three, and the unique labeled radix tuple is three, three, two.

Executable root: `_e898_minimal_semantic_world_rank_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.33 (17. MINIMAL SEMANTIC WORLD RANK THEOREM)**.** *Prove the minimal labeled three-coordinate rank of the primitive sign-sign-occupancy relation carrier.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.34. MINIMAL SEMANTIC WORLD RANK THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 18. TIME RESIDUAL NESTED SEAT THEOREM

Generate the three-by-three-by-two relation seats from first and second residuals of one ordered scalar time chain.

Executable root: `_nsr_time_residual_seat_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.35 (18. TIME RESIDUAL NESTED SEAT THEOREM)**.** *Generate the three-by-three-by-two relation seats from first and second residuals of one ordered scalar time chain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.36. TIME RESIDUAL NESTED SEAT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 19. NESTED RESIDUAL READOUT DYNAMICS THEOREM

Compute every outer readout from the current outer residual and the occupied inner residual channels.

Executable root: `_nsr_local_readout_dynamics_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.37 (19. NESTED RESIDUAL READOUT DYNAMICS THEOREM)**.** *Compute every outer readout from the current outer residual and the occupied inner residual channels.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.38. NESTED RESIDUAL READOUT DYNAMICS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 20. MINIMAL TERNARY UPDATE UNIQUENESS THEOREM

The outer readout law is not selected after seeing force data. Every deterministic ternary law is exhausted and only the clipped residual sum survives the native symmetry, cancellation, identity, and monotonicity axioms.

Executable root: `_x893_minimal_ternary_update_uniqueness_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.39 (20. MINIMAL TERNARY UPDATE UNIQUENESS THEOREM)**.** *Exhaust every deterministic ternary update and prove the residual-clipping law is unique under the native axioms.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.40. MINIMAL TERNARY UPDATE UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 21. BINARY TIME RESIDUAL GENESIS THEOREM

Derive the native levels, temporal decay modes, positive weights, and one all-order recurrence from one binary time-depth relation without observable-value input.

Executable root: `_u880_binary_time_residual_genesis_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.41 (21. BINARY TIME RESIDUAL GENESIS THEOREM)**.** *Derive the native levels, temporal decay modes, positive weights, and one all-order recurrence from one binary time-depth relation without observable-value input.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.42. BINARY TIME RESIDUAL GENESIS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 22. DOUBLE ZERO PRIMITIVE ALGEBRA THEOREM

Derive four primitive occupancy states, two inequivalent zero states, and the complete character frame from two polarity projectors.

Executable root: `_double_zero_primitive_algebra_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.43 (22. DOUBLE ZERO PRIMITIVE ALGEBRA THEOREM)**.** *Derive four primitive occupancy states, two inequivalent zero states, and the complete character frame from two polarity projectors.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.44. DOUBLE ZERO PRIMITIVE ALGEBRA THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 23. RESIDUAL CLOSURE CONSERVATION THEOREM

Prove the unique lossless decomposition of every nonnegative polarity ledger into signed residual and closed pairs.

Executable root: `_residual_closure_conservation_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.45 (23. RESIDUAL CLOSURE CONSERVATION THEOREM)**.** *Prove the unique lossless decomposition of every nonnegative polarity ledger into signed residual and closed pairs.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.46. RESIDUAL CLOSURE CONSERVATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 24. RADIX INDEPENDENT RELATION STRUCTURE THEOREM

Prove that numeral bases and state names change representation only, while rank, character orthogonality, relation roots, edges, and facets remain invariant.

Executable root: `_radix_independent_relation_structure_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.47 (24. RADIX INDEPENDENT RELATION STRUCTURE THEOREM)**.** *Prove that numeral bases and state names change representation only, while rank, character orthogonality, relation roots, edges, and facets remain invariant.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.48. RADIX INDEPENDENT RELATION STRUCTURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 25. HANKEL DIMENSION GENESIS THEOREM

Prove that one scalar time stream has exact latent dimension four: Hankel ranks grow one by one through four and the fifth delay adds no new independent direction.

Executable root: `_u880_hankel_dimension_genesis_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.49 (25. HANKEL DIMENSION GENESIS THEOREM)**.** *Prove that one scalar time stream has exact latent dimension four: Hankel ranks grow one by one through four and the fifth delay adds no new independent direction.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.50. HANKEL DIMENSION GENESIS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 26. LAYER TIME DIMENSION ROLE COMPLETION THEOREM

A completed lower-layer process is one higher-layer event. Resolving every layer reveals three new symbolic coordinates per depth, while coarse observation compresses the closed lower hierarchy to one three-coordinate effective carrier. Physical role is selected by closure and projection predicates rather than by an immutable label.

Executable root: `_e895_layer_time_dimension_role_completion_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.51 (26. LAYER TIME DIMENSION ROLE COMPLETION THEOREM)**.** *Relate microscopic saturation time, aggregate layer time, resolved recursive dimension, coarse dimension, and role classification in one exact hierarchy.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.52. LAYER TIME DIMENSION ROLE COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 27. UNIVERSAL CAUSAL UPDATE SPEED THEOREM

The invariant causal ratio is generated by one spatial relation update per one effective temporal update and is preserved when both are coarse-grained by the exact six-step closure period. Object-specific native speeds differ and therefore cannot alone define the universal limit.

Executable root: `_e897_universal_causal_update_speed_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.53 (27. UNIVERSAL CAUSAL UPDATE SPEED THEOREM)**.** *Generate a layer-invariant causal speed from synchronized relation and time updates and prove a no-go for object-specific universality.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.54. UNIVERSAL CAUSAL UPDATE SPEED THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 28. EXACT LORENTZ CAUSAL SUBGROUP THEOREM

The generated causal speed is not only a ratio. A nontrivial exact rational boost and an exact rational spatial rotation preserve the same three-plus-one interval and its null cone. This supplies an internal causal-frame symmetry without inserting a measured speed.

Executable root: `_e898_exact_lorentz_causal_subgroup_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.55 (28. EXACT LORENTZ CAUSAL SUBGROUP THEOREM)**.** *Construct a nontrivial exact rational Lorentz subgroup preserving the generated causal cone and spatial metric.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.56. EXACT LORENTZ CAUSAL SUBGROUP THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 29. COMMON RESIDUAL RENORMALIZATION THEOREM

A complete family of eighteen lower-layer relations has one average common readout and seventeen independent deviations that sum to zero. Exact coarse graining keeps the common mode and hides the residual kernel; exact refinement restores a uniform child family. This supplies a scale-independent one-plus-seventeen decomposition before any physical role name is assigned.

Executable root: `_e896_common_residual_renormalization_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.57 (29. COMMON RESIDUAL RENORMALIZATION THEOREM)**.** *Prove that eighteen child relations split canonically into one coarse common mode and a seventeen-dimensional zero-sum residual kernel preserved by exact refine-coarsen maps.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.58. COMMON RESIDUAL RENORMALIZATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 30. RESIDUAL COORDINATE BASIS THEOREM

Prove that residual orders zero through three form an invertible dimension-generating coordinate basis for every four-sample temporal window.

Executable root: `_u880_residual_coordinate_basis_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.59 (30. RESIDUAL COORDINATE BASIS THEOREM)**.** *Prove that residual orders zero through three form an invertible dimension-generating coordinate basis for every four-sample temporal window.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.60. RESIDUAL COORDINATE BASIS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 31. CAUSAL COMPANION DYNAMICS THEOREM

Derive one unique companion evolution from the temporal recurrence and prove that the bounded forward orientation distinguishes causal time from its expanding inverse.

Executable root: `_u880_causal_companion_dynamics_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.61 (31. CAUSAL COMPANION DYNAMICS THEOREM)**.** *Derive one unique companion evolution from the temporal recurrence and prove that the bounded forward orientation distinguishes causal time from its expanding inverse.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.62. CAUSAL COMPANION DYNAMICS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 32. TEMPORAL RESIDUAL GENESIS THEOREM

One ordered temporal chain generates the outer sign as a first residual and the inner sign as a second residual. The occupancy transition generates the common, signed, and curvature modes with exact rational eigenvalues.

Executable root: `_u905_temporal_residual_genesis_theorem`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.63 (32. TEMPORAL RESIDUAL GENESIS THEOREM)**.** *Derive the complete three-residual jet and its unique contractive spectral ratio from one ordered scalar chain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.64. TEMPORAL RESIDUAL GENESIS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 33. TIME ORIGIN CAUSAL PRIORITY THEOREM

The eighteen nested signed-occupancy states first form one ordered scalar history. Three finite residuals of that history generate the spatial carrier before any force sector is read.

Executable root: `theorem_time_origin_causal_priority`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.65 (33. TIME ORIGIN CAUSAL PRIORITY THEOREM)**.** *Prove that the eighteen nested signed-occupancy states first generate one ordered temporal line, and only its three nonconstant residuals generate the spatial carrier and four force sectors.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.66. TIME ORIGIN CAUSAL PRIORITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 34. TIME RESIDUAL DIMENSION STRICT DEPENDENCY THEOREM

Time is not appended to a pre-existing space. Directed update generates one temporal order, its three independent residual records generate spatial rank three, and those directions close into the JS shell.

Executable root: `theorem_time_residual_dimension_strict_dependency`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.67 (34. TIME RESIDUAL DIMENSION STRICT DEPENDENCY THEOREM)**.** *Prove the strict dependency time first, three residuals second, spatial rank three third, and shell closure fourth.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.68. TIME RESIDUAL DIMENSION STRICT DEPENDENCY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 35. TEMPORAL RESIDUAL ADDITIVE DEPTH ORIGIN THEOREM

The primitive one-third contraction composes multiplicatively, while its uniquely normalized depth composes additively. This is the origin of the logarithmic coordinate class without a logarithm call.

Executable root: `theorem_temporal_residual_additive_depth_origin`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.69 (35. TEMPORAL RESIDUAL ADDITIVE DEPTH ORIGIN THEOREM)**.** *Derive the logarithmic-class additive depth as the unique normalized homomorphism of the primitive temporal contraction orbit without evaluating a logarithm.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.70. TEMPORAL RESIDUAL ADDITIVE DEPTH ORIGIN THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 36. REGULAR SOURCE MULTIAXIS ORBIT THEOREM

The primitive eighteen-state signed occupancy ledger supplies a regular temporal order and three independent residual directions. One native step is linear in depth while its bounded inward, bounded outward, and multiplicative readouts change simultaneously.

Executable root: `theorem_regular_source_multiaxis_orbit`

Class and stage: Primary; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.71 (36. REGULAR SOURCE MULTIAXIS ORBIT THEOREM)**.** *Derive one regular temporal source orbit and its simultaneous linear, bounded, mirrored, and multiplicative readouts.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.72. REGULAR SOURCE MULTIAXIS ORBIT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 37. ZERO BIPOLAR ORIGIN ALGEBRA

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_zero_bipolar_origin_algebra`

Class and stage: Narrative; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.73 (37. ZERO BIPOLAR ORIGIN ALGEBRA)**.** *Execute the exact certificate for ZERO BIPOLAR ORIGIN ALGEBRA on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.74. ZERO BIPOLAR ORIGIN ALGEBRA holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 38. FINITE MIRROR CLOSURE

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_finite_mirror_closure`

Class and stage: Narrative; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.75 (38. FINITE MIRROR CLOSURE)**.** *Execute the exact certificate for FINITE MIRROR CLOSURE on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.76. FINITE MIRROR CLOSURE holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 39. CAUSAL DRIVE LAG ORDER

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_causal_drive_lag_order`

Class and stage: Narrative; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.77 (39. CAUSAL DRIVE LAG ORDER)**.** *Execute the exact certificate for CAUSAL DRIVE LAG ORDER on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.78. CAUSAL DRIVE LAG ORDER holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 40. NATIVE CONSTANT ORIGIN AND CIRCLE CLOSURE

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_native_constant_origin_circle_closure`

Class and stage: Narrative; 1. SIGNED OCCUPANCY AND TEMPORAL ORIGIN

Theorem A.79 (40. NATIVE CONSTANT ORIGIN AND CIRCLE CLOSURE)**.** *Execute the exact certificate for NATIVE CONSTANT ORIGIN AND CIRCLE CLOSURE on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.80. NATIVE CONSTANT ORIGIN AND CIRCLE CLOSURE holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 41. TWELVE SIGNATURE JS SHELL THEOREM

Compress all five hundred twelve simultaneous occupancy arrays into twelve exact transition signatures and derive the twelve, one hundred forty-four, and sixteen multiplicities.

Executable root: `_nsr_global_signature_js_shell_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.81 (41. TWELVE SIGNATURE JS SHELL THEOREM)**.** *Compress all five hundred twelve simultaneous occupancy arrays into twelve exact transition signatures and derive the twelve, one hundred forty-four, and sixteen multiplicities.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.82. TWELVE SIGNATURE JS SHELL THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 42. RELATION ROOT JS CELL GENESIS THEOREM

Generate the complete twelve-seat JS shell as the A3 relation-root polytope of the four primitive double-zero states.

Executable root: `_relation_root_js_cell_genesis_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.83 (42. RELATION ROOT JS CELL GENESIS THEOREM)**.** *Generate the complete twelve-seat JS shell as the A3 relation-root polytope of the four primitive double-zero states.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.84. RELATION ROOT JS CELL GENESIS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 43. JS CHAIN HODGE LOCAL ACTION THEOREM

Prove the closed JS surface chain, its exact homology, the complete edge Hodge dimension count, and positive local channel actions.

Executable root: `_js_chain_hodge_local_action_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.85 (43. JS CHAIN HODGE LOCAL ACTION THEOREM)**.** *Prove the closed JS surface chain, its exact homology, the complete edge Hodge dimension count, and positive local channel actions.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.86. JS CHAIN HODGE LOCAL ACTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 44. TEMPORAL DOUBLE ZERO JS DYNAMICS THEOREM

Transport the primitive temporal companion through residual generation, double-zero occupancy, the twelve relation roots, and complete four-force observation.

Executable root: `_temporal_double_zero_js_dynamics_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.87 (44. TEMPORAL DOUBLE ZERO JS DYNAMICS THEOREM)**.** *Transport the primitive temporal companion through residual generation, double-zero occupancy, the twelve relation roots, and complete four-force observation.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.88. TEMPORAL DOUBLE ZERO JS DYNAMICS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 45. TIME DOUBLE ZERO RELATION CELL FORCE COMPLETION THEOREM

Integrate primitive time, polarity occupancy, double-zero conservation, radix independence, dimension genesis, relation-root JS geometry, local action, and four-force dynamics.

Executable root: `_time_double_zero_relation_cell_force_completion_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.89 (45. TIME DOUBLE ZERO RELATION CELL FORCE COMPLETION THEOREM)**.** *Integrate primitive time, polarity occupancy, double-zero conservation, radix independence, dimension genesis, relation-root JS geometry, local action, and four-force dynamics.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.90. TIME DOUBLE ZERO RELATION CELL FORCE COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 46. THREE SLOT COMPLETE STATE THEOREM

Prove that the complete three-slot state cube has six single-polarity states and two additional balanced-zero states.

Executable root: `_u882_three_slot_complete_state_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.91 (46. THREE SLOT COMPLETE STATE THEOREM)**.** *Prove that the complete three-slot state cube has six single-polarity states and two additional balanced-zero states.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.92. THREE SLOT COMPLETE STATE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 47. RESIDUAL CLOSURE PHASE LEDGER THEOREM

Separate conserved polarity content, closed-pair depth, and the binary open-closed phase without deleting balanced content.

Executable root: `_u882_residual_closure_phase_ledger_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.93 (47. RESIDUAL CLOSURE PHASE LEDGER THEOREM)**.** *Separate conserved polarity content, closed-pair depth, and the binary open-closed phase without deleting balanced content.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.94. RESIDUAL CLOSURE PHASE LEDGER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 48. OPEN CLOSED ZERO PARTICLE WAVE THEOREM

Distinguish the four zero preimages and derive exact open two-channel and closed one-channel operators.

Executable root: `_u882_open_closed_zero_particle_wave_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.95 (48. OPEN CLOSED ZERO PARTICLE WAVE THEOREM)**.** *Distinguish the four zero preimages and derive exact open two-channel and closed one-channel operators.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.96. OPEN CLOSED ZERO PARTICLE WAVE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 49. INFORMATION PRESERVING DETECTION THEOREM

Prove that closure observation can create a localized branch while the complementary branch preserves the complete open state.

Executable root: `_u882_information_preserving_detection_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.97 (49. INFORMATION PRESERVING DETECTION THEOREM). *Prove that closure observation can create a localized branch while the complementary branch preserves the complete open state.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.98. INFORMATION PRESERVING DETECTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 50. FOUR FORCE PARTICLE WAVE READOUT THEOREM

Generate four force readouts and particle-wave readouts from the complete three-slot state without loss.

Executable root: `_u882_four_force_particle_wave_readout_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.99 (50. FOUR FORCE PARTICLE WAVE READOUT THEOREM). *Generate four force readouts and particle-wave readouts from the complete three-slot state without loss.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.100. FOUR FORCE PARTICLE WAVE READOUT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 51. DOUBLE JS LAYER CLOSURE LIFT THEOREM

Lift the twelve-relation JS cell through the open and closed closure layers without changing its latent incidence.

Executable root: `_u882_double_js_layer_closure_lift_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.101 (51. DOUBLE JS LAYER CLOSURE LIFT THEOREM). *Lift the twelve-relation JS cell through the open and closed closure layers without changing its latent incidence.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.102. DOUBLE JS LAYER CLOSURE LIFT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 52. TEMPORAL WAVE PARTICLE CYCLE THEOREM

Generate the minimal open-closed manifestation cycle from one ordered temporal step while preserving carrier and signed readouts.

Executable root: `_u882_temporal_wave_particle_cycle_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.103 (52. TEMPORAL WAVE PARTICLE CYCLE THEOREM)**.** *Generate the minimal open-closed manifestation cycle from one ordered temporal step while preserving carrier and signed readouts.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.104. TEMPORAL WAVE PARTICLE CYCLE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 53. THREE SLOT PARTICLE WAVE FORCE COMPLETION THEOREM

Integrate time, three-slot occupancy, four zero preimages, particle-wave closure, JS geometry, and four-force readout.

Executable root: `_u882_three_slot_particle_wave_force_completion_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.105 (53. THREE SLOT PARTICLE WAVE FORCE COMPLETION THEOREM)**.** *Integrate time, three-slot occupancy, four zero preimages, particle-wave closure, JS geometry, and four-force readout.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.106. THREE SLOT PARTICLE WAVE FORCE COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 54. RESIDUAL DIMENSION GENESIS THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_residual_dimension_genesis_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.107 (54. RESIDUAL DIMENSION GENESIS THEOREM)**.** *Execute the exact certificate for RESIDUAL DIMENSION GENESIS THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.108. RESIDUAL DIMENSION GENESIS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 55. FUNDAMENTAL RESIDUAL DYNAMICS THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_fundamental_residual_dynamics_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.109 (55. FUNDAMENTAL RESIDUAL DYNAMICS THEOREM)**.** *Execute the exact certificate for FUNDAMENTAL RESIDUAL DYNAMICS THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.110. FUNDAMENTAL RESIDUAL DYNAMICS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 56. RESIDUAL FIELD DYNAMICS THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_residual_field_dynamics_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.111 (56. RESIDUAL FIELD DYNAMICS THEOREM)**.** *Execute the exact certificate for RESIDUAL FIELD DYNAMICS THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.112. RESIDUAL FIELD DYNAMICS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 57. RESIDUAL CONTINUUM ACTION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_residual_continuum_action_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.113 (57. RESIDUAL CONTINUUM ACTION THEOREM)**.** *Execute the exact certificate for RESIDUAL CONTINUUM ACTION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.114. RESIDUAL CONTINUUM ACTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 58. PRIMITIVE EVENT AXIS REDISTRIBUTION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_primitive_event_axis_redistribution_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.115 (58. PRIMITIVE EVENT AXIS REDISTRIBUTION THEOREM)**.** *Execute the exact certificate for PRIMITIVE EVENT AXIS REDISTRIBUTION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.116. PRIMITIVE EVENT AXIS REDISTRIBUTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 59. EVENT AXIS SPATIAL RECONSTRUCTION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_event_axis_spatial_reconstruction_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.117 (59. EVENT AXIS SPATIAL RECONSTRUCTION THEOREM)**.** *Execute the exact certificate for EVENT AXIS SPATIAL RECONSTRUCTION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.118. EVENT AXIS SPATIAL RECONSTRUCTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 60. MASTER TIME RESIDUAL UNIFIED ACTION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_master_time_residual_unified_action_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.119 (60. MASTER TIME RESIDUAL UNIFIED ACTION THEOREM)**.** *Execute the exact certificate for MASTER TIME RESIDUAL UNIFIED ACTION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.120. MASTER TIME RESIDUAL UNIFIED ACTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 61. GENERAL REDISTRIBUTION SOURCE THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_general_redistribution_source_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.121 (61. GENERAL REDISTRIBUTION SOURCE THEOREM)**.** *Execute the exact certificate for GENERAL REDISTRIBUTION SOURCE THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.122. GENERAL REDISTRIBUTION SOURCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 62. GENERAL EVENT MEASURE CONTINUUM DESCENT THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_general_event_measure_continuum_descent_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.123 (62. GENERAL EVENT MEASURE CONTINUUM DESCENT THEOREM)**.** *Execute the exact certificate for GENERAL EVENT MEASURE CONTINUUM DESCENT THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.124. GENERAL EVENT MEASURE CONTINUUM DESCENT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 63. FOUR FORCE TEMPORAL TOMOGRAPHY THEOREM

Generate all four force readouts from temporal residual coordinates and prove exact two-way tomography from one scalar time history.

Executable root: `_u880_four_force_temporal_tomography_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.125 (63. FOUR FORCE TEMPORAL TOMOGRAPHY THEOREM)**.** *Generate all four force readouts from temporal residual coordinates and prove exact two-way tomography from one scalar time history.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.126. FOUR FORCE TEMPORAL TOMOGRAPHY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 64. TEMPORAL LORENTZIAN MASS THEOREM

Generate a positive mass form from temporal correlations and a one-time-plus-three-residual Lorentzian form, then prove exact preservation under complete force projection.

Executable root: `_u880_temporal_lorentzian_mass_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.127 (64. TEMPORAL LORENTZIAN MASS THEOREM)**.** *Generate a positive mass form from temporal correlations and a one-time-plus-three-residual Lorentzian form, then prove exact preservation under complete force projection.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.128. TEMPORAL LORENTZIAN MASS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 65. TEMPORAL VARIATIONAL ACTION THEOREM

Derive the companion evolution as the unique zero of a positive temporal action and prove exact variational equivalence after residual and force projection.

Executable root: `_u880_temporal_variational_action_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.129 (65. TEMPORAL VARIATIONAL ACTION THEOREM)**.** *Derive the companion evolution as the unique zero of a positive temporal action and prove exact variational equivalence after residual and force projection.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.130. TEMPORAL VARIATIONAL ACTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 66. TIME RESIDUAL DIMENSION FORCE COMPLETION THEOREM

Integrate one primitive time depth, cumulative residual generation, exact dimension genesis, causal evolution, four-force tomography, Lorentzian mass transport, and variational dynamics.

Executable root: `_u880_time_residual_dimension_force_completion_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.131 (66. TIME RESIDUAL DIMENSION FORCE COMPLETION THEOREM)**.** *Integrate one primitive time depth, cumulative residual generation, exact dimension genesis, causal evolution, four-force tomography, Lorentzian mass transport, and variational dynamics.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.132. TIME RESIDUAL DIMENSION FORCE COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 67. RECURSIVE DIMENSION GENESIS THEOREM

Two ternary coordinates and one binary coordinate generate three independent spatial directions. Temporal residual depth supplies causal order and every depth contains eighteen to the depth cells.

Executable root: `_u905_recursive_dimension_genesis_theorem`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.133 (67. RECURSIVE DIMENSION GENESIS THEOREM)**.** *Construct the three-dimensional residual fiber as the exact quotient of four ordered samples by the constant temporal line.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.134. RECURSIVE DIMENSION GENESIS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 68. PARENT SHADOW CANONICAL DECOMPOSITION THEOREM

The source does not choose one force path. One complete product grammar is decomposed simultaneously by its unique uniform conditional expectations. Main effects, pair interactions and the scalar plus traceless parts of the full triple interaction are the canonical anonymous shadows.

Executable root: `theorem_parent_shadow_canonical_decomposition`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.135 (68. PARENT SHADOW CANONICAL DECOMPOSITION THEOREM)**.** *Derive the anonymous ranks five, eight, one and three from the unique interaction-order decomposition of the three-by-three-by-two grammar.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.136. PARENT SHADOW CANONICAL DECOMPOSITION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 69. PARENT SHADOW ALGEBRA REALIZATION THEOREM

The anonymous blocks are not selected by matching four known force dimensions. The three primitive factor complements and the generated spatial rank canonically supply one symmetric tracefree spatial response, one central scalar, one traceless two-state endomorphism algebra and one traceless three-state endomorphism algebra.

Executable root: `theorem_parent_shadow_algebra_realization`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.137 (69. PARENT SHADOW ALGEBRA REALIZATION THEOREM)**.** *Realize the four canonical anonymous shadow blocks as the unique generated matrix-response types before physical names are attached.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.138. PARENT SHADOW ALGEBRA REALIZATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 70. PARENT SHADOW NO CHOICE EQUIVALENCE THEOREM

A single parent object is not selected from several force worlds. Every admissible simultaneous relabeling is a coordinate change of the same parent, and the four canonical nonconstant blocks occur together. Merging or splitting the canonical ranges changes the shadow resolution or basis bookkeeping; it does not create another parent object or another dynamical source.

Executable root: `theorem_parent_shadow_no_choice_equivalence`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.139 (70. PARENT SHADOW NO CHOICE EQUIVALENCE THEOREM)**.** *Prove that admissible relabelings change only coordinates, while every complete interaction-order shadow retains the same canonical block ranks.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.140. PARENT SHADOW NO CHOICE EQUIVALENCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 71. REFINED SHADOW DISPERSION ISOTROPY THEOREM

Refinement is the normalized convolution of independent primitive relation steps. The quadratic kinetic tensor is exactly isotropic at every level. The direction-dependent fourth cumulant is recomputed from the convolution law and falls as one over the actual path count, rather than being assigned an external contraction factor.

Executable root: `theorem_refined_shadow_dispersion_isotropy`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.141 (71. REFINED SHADOW DISPERSION ISOTROPY THEOREM)**.** *Derive the refinement decay of the actual convolution kinetic symbol from the primitive twelve-direction step distribution.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.142. REFINED SHADOW DISPERSION ISOTROPY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 72. CAUSAL SIGNATURE UNIQUENESS THEOREM

The source supplies one distinguished ordered coordinate and three permutation-equivalent residual coordinates. After removing one positive overall scale, nondegeneracy plus a nontrivial null boundary uniquely selects one opposite temporal sign and three equal spatial signs.

Executable root: `theorem_causal_signature_uniqueness`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.143 (72. CAUSAL SIGNATURE UNIQUENESS THEOREM)**.** *Classify the Lorentz signature as one global-sign equivalence class on the temporal-line plus residual-fiber decomposition.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.144. CAUSAL SIGNATURE UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 73. FOUR DIMENSIONAL SHADOW AXIS RANK THEOREM

One ordered temporal axis and three independent residual readouts form a nondegenerate four-coordinate chart. Every single-axis deletion has the expected nondegenerate lower-rank minor, and interior bounded compression preserves rank.

Executable root: `theorem_four_dimensional_shadow_axis_rank`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.145 (73. FOUR DIMENSIONAL SHADOW AXIS RANK THEOREM)**.** *Prove a nondegenerate four-coordinate jet carrier and preservation of its rank under interior bounded projection.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.146. FOUR DIMENSIONAL SHADOW AXIS RANK THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 74. MINIMAL SOURCE GEOMETRY NATURAL SELECTION THEOREM

The three generated residual axes and the negative-zero-positive alphabet leave one minimal reversal- and relabeling-closed orbit with two active unit coordinates. That orbit has twelve directions, twenty-four edges, degree four, zero first moment, and isotropic second moment; no particular cone or illumination model is assumed.

Executable root: `theorem_minimal_source_geometry_natural_selection`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.147 (74. MINIMAL SOURCE GEOMETRY NATURAL SELECTION THEOREM)**.** *Select the twelve-direction shell as the unique minimal signed two-axis orbit generated after spatial rank three.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.148. MINIMAL SOURCE GEOMETRY NATURAL SELECTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 75. LORENTZ REFINEMENT ISOTROPY THEOREM

The twelve generated directions have zero first moment and an identity normalized second moment, so every quadratic spatial kinetic form is direction independent. Rational boosts preserve the Minkowski metric exactly, compose inside the same parameterization, and quadratic local kinetic operators are exact at every refinement.

Executable root: `theorem_lorentz_refinement_isotropy`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.149 (75. LORENTZ REFINEMENT ISOTROPY THEOREM)**.** *Prove exact second-moment spatial isotropy, rational Lorentz covariance, and refinement-exact quadratic kinetic operators.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.150. LORENTZ REFINEMENT ISOTROPY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 76. LORENTZ ANISOTROPY SCALE CLOSURE THEOREM

The twelve source directions have exact quadratic isotropy. Their first higher-order directional defect is displayed rather than hidden and obeys the exact native blocking contraction at every scale.

Executable root: `theorem_lorentz_anisotropy_scale_closure`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.151 (76. LORENTZ ANISOTROPY SCALE CLOSURE THEOREM)**.** *Execute the exact certificate for LORENTZ ANISOTROPY SCALE CLOSURE THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.152. LORENTZ ANISOTROPY SCALE CLOSURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 77. ALL RATIONAL LORENTZ GROUP IDENTITY THEOREM

The rational boost formula is cleared of denominators and verified as a polynomial identity, so the metric and determinant statements hold for every admissible rational parameter rather than for a finite sample. The same cross-multiplied calculation proves the exact velocity-composition law for every admissible rational pair.

Executable root: `theorem_all_rational_lorentz_group_identity`

Class and stage: Primary; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.153 (77. ALL RATIONAL LORENTZ GROUP IDENTITY THEOREM)**.** *Prove the Lorentz metric, determinant, and composition identities as polynomial identities for every admissible rational boost parameter.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.154. ALL RATIONAL LORENTZ GROUP IDENTITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 78. DIMENSIONAL BRIDGE QUOTIENT

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_dimensional_bridge_quotient`

Class and stage: Narrative; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.155 (78. DIMENSIONAL BRIDGE QUOTIENT)**.** *Execute the exact certificate for DIMENSIONAL BRIDGE QUOTIENT on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.156. DIMENSIONAL BRIDGE QUOTIENT holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 79. GRAVITY CONFIGURATION ORBIT AND C12 SPECTRAL GEOMETRY

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_gravity_orbit_shell_spectral_geometry`

Class and stage: Narrative; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.157 (79. GRAVITY CONFIGURATION ORBIT AND C12 SPECTRAL GEOMETRY)**.** *Execute the exact certificate for GRAVITY CONFIGURATION ORBIT AND C12 SPECTRAL GEOMETRY on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.158. GRAVITY CONFIGURATION ORBIT AND C12 SPECTRAL GEOMETRY holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 80. FOUR DIMENSIONAL PROJECTIVE CONTINUUM AND QUANTUM GEOMETRY

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_four_dimensional_projective_quantum_geometry`

Class and stage: Narrative; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.159 (80. FOUR DIMENSIONAL PROJECTIVE CONTINUUM AND QUANTUM GEOMETRY)**.** *Execute the exact certificate for FOUR DIMENSIONAL PROJECTIVE CONTINUUM AND QUANTUM GEOMETRY on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.160. FOUR DIMENSIONAL PROJECTIVE CONTINUUM AND QUANTUM GEOMETRY holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 81. CYLINDER FUNCTIONAL FLUX NORMALIZATION AND NONLINEAR COVARIANCE

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_cylinder_flux_nonlinear_covariance`

Class and stage: Narrative; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.161 (81. CYLINDER FUNCTIONAL FLUX NORMALIZATION AND NONLINEAR COVARIANCE)**.** *Execute the exact certificate for CYLINDER FUNCTIONAL FLUX NORMALIZATION AND NONLINEAR COVARIANCE on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.162. CYLINDER FUNCTIONAL FLUX NORMALIZATION AND NONLINEAR COVARIANCE holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 82. CYLINDER COMPLETION TREE AUTOMORPHISM AND CURVED SEMICLASSICAL LIMIT

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_completion_tree_curved_semiclassical_limit`

Class and stage: Narrative; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.163 (82. CYLINDER COMPLETION TREE AUTOMORPHISM AND CURVED SEMICLASSICAL LIMIT)**.** *Execute the exact certificate for CYLINDER COMPLETION TREE AUTOMORPHISM AND CURVED SEMICLASSICAL LIMIT on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.164. CYLINDER COMPLETION TREE AUTOMORPHISM AND CURVED SEMICLASSICAL LIMIT holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 83. PHASE ATLAS COBORDISM SURGERY AND EINSTEIN BRIDGE

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_phase_atlas_cobordism_einstein_bridge`

Class and stage: Narrative; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.165 (83. PHASE ATLAS COBORDISM SURGERY AND EINSTEIN BRIDGE)**.** *Execute the exact certificate for PHASE ATLAS COBORDISM SURGERY AND EINSTEIN BRIDGE on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.166. PHASE ATLAS COBORDISM SURGERY AND EINSTEIN BRIDGE holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 84. PHASE PRESENTATION AND INFINITE ATLAS COMPLETION

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_phase_presentation_infinite_atlas_completion`

Class and stage: Narrative; 2. RESIDUAL DIMENSION AND JS GEOMETRY

Theorem A.167 (84. PHASE PRESENTATION AND INFINITE ATLAS COMPLETION)**.** *Execute the exact certificate for PHASE PRESENTATION AND INFINITE ATLAS COMPLETION on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.168. PHASE PRESENTATION AND INFINITE ATLAS COMPLETION holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 85. ORDERED ZERO BIPOLAR ACTION SELECTOR THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_ordered_zero_bipolar_action_selector_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.169 (85. ORDERED ZERO BIPOLAR ACTION SELECTOR THEOREM)**.** *Execute the exact certificate for ORDERED ZERO BIPOLAR ACTION SELECTOR THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.170. ORDERED ZERO BIPOLAR ACTION SELECTOR THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 86. CONNECTED QUOTIENT DIRICHLET ACTION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_connected_quotient_dirichlet_action_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.171 (86. CONNECTED QUOTIENT DIRICHLET ACTION THEOREM)**.** *Execute the exact certificate for CONNECTED QUOTIENT DIRICHLET ACTION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.172. CONNECTED QUOTIENT DIRICHLET ACTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 87. EXACT SCHUR PROJECTIVE ACTION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_exact_schur_projective_action_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.173 (87. EXACT SCHUR PROJECTIVE ACTION THEOREM)**.** *Execute the exact certificate for EXACT SCHUR PROJECTIVE ACTION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.174. EXACT SCHUR PROJECTIVE ACTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 88. TWENTY TWO CHANNEL COUPLED ACTION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_twenty_two_channel_coupled_action_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.175 (88. TWENTY TWO CHANNEL COUPLED ACTION THEOREM)**.** *Execute the exact certificate for TWENTY TWO CHANNEL COUPLED ACTION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.176. TWENTY TWO CHANNEL COUPLED ACTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 89. FOUR DIMENSIONAL FIELD REALIZATION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_four_dimensional_field_realization_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.177 (89. FOUR DIMENSIONAL FIELD REALIZATION THEOREM)**.** *Execute the exact certificate for FOUR DIMENSIONAL FIELD REALIZATION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.178. FOUR DIMENSIONAL FIELD REALIZATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 90. DOUBLE ZERO FOUR FORCE READOUT THEOREM

Derive gravity, electromagnetic, weak zero-type, and strong closure readouts from the four primitive states and prove complete recovery.

Executable root: `_double_zero_four_force_readout_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.179 (90. DOUBLE ZERO FOUR FORCE READOUT THEOREM)**.** *Derive gravity, electromagnetic, weak zero-type, and strong closure readouts from the four primitive states and prove complete recovery.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.180. DOUBLE ZERO FOUR FORCE READOUT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 91. NESTED CHARACTER FORCE DECOMPOSITION THEOREM

Decompose the eighteen relation seats by orthogonal ternary degree and empty-full parity into one common mode and force dimensions five, one, three, and eight.

Executable root: `_nsr_character_force_decomposition_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.181 (91. NESTED CHARACTER FORCE DECOMPOSITION THEOREM)**.** *Decompose the eighteen relation seats by orthogonal ternary degree and empty-full parity into one common mode and force dimensions five, one, three, and eight.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.182. NESTED CHARACTER FORCE DECOMPOSITION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 92. FORCE SECTOR SELECTOR UNIQUENESS THEOREM

Once the eighteen orthogonal characters have been generated, occupation parity, ternary degree, common-mode exclusion, and completeness leave exactly one sector label for every character.

Executable root: `_x893_force_sector_selector_uniqueness_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.183 (92. FORCE SECTOR SELECTOR UNIQUENESS THEOREM)**.** *Prove the one-plus-five-one-three-eight split is the unique invariant assignment on the generated character basis.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.184. FORCE SECTOR SELECTOR UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 93. ANONYMOUS FORCE ROLE PERMUTATION UNIQUENESS THEOREM

The four carriers are generated and anonymized before physical names are allowed. All twenty-four assignments are tested using only dimension, centrality, chirality, color tracelessness, and universal spin-two metric action. Exactly one assignment survives.

Executable root: `theorem_anonymous_force_role_permutation_uniqueness`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.185 (93. ANONYMOUS FORCE ROLE PERMUTATION UNIQUENESS THEOREM)**.** *Remove all force names from the generated carriers, test every permutation, and recover exactly one electromagnetic, weak, strong, and gravitational assignment from algebraic signatures alone.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.186. ANONYMOUS FORCE ROLE PERMUTATION UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 94. ORIGIN FORCE CARRIER DIMENSION THEOREM

Derive the dimensions five, one, three, and eight from primitive relation spaces rather than from decimal labels.

Executable root: `_ofc_origin_force_carrier_dimension_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.187 (94. ORIGIN FORCE CARRIER DIMENSION THEOREM)**.** *Derive the dimensions five, one, three, and eight from primitive relation spaces rather than from decimal labels.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.188. ORIGIN FORCE CARRIER DIMENSION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 95. EXACT COMPACT ALGEBRA THEOREM

Construct rational anti-Hermitian bases for the weak and color algebras and prove exact closure, Jacobi identity, and compact conjugation covariance.

Executable root: `_ofc_exact_compact_algebra_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.189 (95. EXACT COMPACT ALGEBRA THEOREM)**.** *Construct rational anti-Hermitian bases for the weak and color algebras and prove exact closure, Jacobi identity, and compact conjugation covariance.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.190. EXACT COMPACT ALGEBRA THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 96. SINGLE PARENT LOCAL ACTION THEOREM

Construct one seventeen-component local cochain action and derive all four sector actions, equations, gauge identities, and coefficients by orthogonal projection.

Executable root: `_ofc_parent_local_action_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.191 (96. SINGLE PARENT LOCAL ACTION THEOREM)**.** *Construct one seventeen-component local cochain action and derive all four sector actions, equations, gauge identities, and coefficients by orthogonal projection.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.192. SINGLE PARENT LOCAL ACTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 97. NONABELIAN PARENT INTERACTION THEOREM

Generate exact local weak and color commutator curvature, compact covariance, positive action, and Jacobi-Bianchi closure inside the same parent carrier.

Executable root: `_ofc_nonabelian_parent_interaction_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.193 (97. NONABELIAN PARENT INTERACTION THEOREM)**.** *Generate exact local weak and color commutator curvature, compact covariance, positive action, and Jacobi-Bianchi closure inside the same parent carrier.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.194. NONABELIAN PARENT INTERACTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 98. PARENT CONSTRAINT ASSEMBLY THEOREM

Assemble gauge-source conservation, nonabelian Bianchi closure, and Lorentzian constraint propagation into one direct-sum parent identity.

Executable root: `_ofc_parent_constraint_assembly_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.195 (98. PARENT CONSTRAINT ASSEMBLY THEOREM)**.** *Assemble gauge-source conservation, nonabelian Bianchi closure, and Lorentzian constraint propagation into one direct-sum parent identity.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.196. PARENT CONSTRAINT ASSEMBLY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 99. POLYNOMIAL PARENT REFINEMENT THEOREM

Extend the single parent action to an exact polynomial continuum partition in all seventeen components and prove sector projections commute with every generated refinement.

Executable root: `_ofc_polynomial_parent_refinement_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.197 (99. POLYNOMIAL PARENT REFINEMENT THEOREM)**.** *Extend the single parent action to an exact polynomial continuum partition in all seventeen components and prove sector projections commute with every generated refinement.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.198. POLYNOMIAL PARENT REFINEMENT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 100. UNIFIED RATIO SEAL THEOREM

Seal dimensionless parent ratios before external comparison and require independent signed records for empirical success.

Executable root: `_ofc_unified_ratio_seal_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.199 (100. UNIFIED RATIO SEAL THEOREM)**.** *Seal dimensionless parent ratios before external comparison and require independent signed records for empirical success.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.200. UNIFIED RATIO SEAL THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 101. ORIGIN PARENT FIELD COMPLETION THEOREM

Integrate origin-derived carrier dimensions, exact compact algebras, one local parent action, nonabelian interactions, constraints, continuum refinement, and sealed ratios.

Executable root: `_origin_parent_field_completion_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.201 (101. ORIGIN PARENT FIELD COMPLETION THEOREM)**.** *Integrate origin-derived carrier dimensions, exact compact algebras, one local parent action, nonabelian interactions, constraints, continuum refinement, and sealed ratios.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.202. ORIGIN PARENT FIELD COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 102. RESIDUAL NONABELIAN MATTER THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_residual_nonabelian_matter_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.203 (102. RESIDUAL NONABELIAN MATTER THEOREM)**.** *Execute the exact certificate for RESIDUAL NONABELIAN MATTER THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.204. RESIDUAL NONABELIAN MATTER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 103. RESIDUAL MULTI PROJECTION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_residual_multi_projection_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.205 (103. RESIDUAL MULTI PROJECTION THEOREM)**.** *Execute the exact certificate for RESIDUAL MULTI PROJECTION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.206. RESIDUAL MULTI PROJECTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 104. FOUR FORCE CARRIER PROJECTION THEOREM

The eighteen-character carrier splits orthogonally into one common mode and ranks five, one, three, and eight. The force projectors are complete, idempotent, and pairwise orthogonal.

Executable root: `_u905_force_carrier_projection_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.207 (104. FOUR FORCE CARRIER PROJECTION THEOREM)**.** *Execute the exact certificate for FOUR FORCE CARRIER PROJECTION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.208. FOUR FORCE CARRIER PROJECTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 105. NATIVE FORCE READOUT GENERATION THEOREM

Carrier rank and temporal residual signatures generate the four dimensionless readouts. Exhaustive signature assignment leaves one rank-compatible force dictionary.

Executable root: `_u905_native_force_readout_generation_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.209 (105. NATIVE FORCE READOUT GENERATION THEOREM)**.** *Execute the exact certificate for NATIVE FORCE READOUT GENERATION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.210. NATIVE FORCE READOUT GENERATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 106. INVARIANT READOUT GRAMMAR UNIQUENESS THEOREM

The readout grammar is restricted by role: signed endpoint, oriented reclosure, traceless closure, and global burden each select one lowest-degree invariant monomial. All four role permutations are enumerated and the generated carrier ranks leave one assignment.

Executable root: `theorem_invariant_readout_grammar_uniqueness`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.211 (106. INVARIANT READOUT GRAMMAR UNIQUENESS THEOREM)**.** *Exhaust the lowest-degree invariant monomial grammar and select one readout law for each force role.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.212. INVARIANT READOUT GRAMMAR UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 107. NATIVE PARENT ACTION THEOREM

The generated readouts enter one positive parent action. Stationary elimination gives the exact positive Schur action.

Executable root: `_u905_parent_action_from_native_readouts_theorem`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.213 (107. NATIVE PARENT ACTION THEOREM)**.** *Execute the exact certificate for NATIVE PARENT ACTION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.214. NATIVE PARENT ACTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 108. PARENT ACTION GLOBAL UNIQUENESS THEOREM

Every positive rational permutation-symmetric star action with one common scalar is classified after unit contrast normalization. The required common eigenvalue leaves one coefficient pair, giving the unique positive parent action and its Schur Hessian.

Executable root: `theorem_parent_action_global_uniqueness`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.215 (108. PARENT ACTION GLOBAL UNIQUENESS THEOREM)**.** *Classify the permutation-symmetric one-common-carrier quadratic star actions and prove a unique normalized positive action.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.216. PARENT ACTION GLOBAL UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 109. PARENT ACTION POSITIVE FIELD ANALYTIC UNIQUENESS THEOREM

The parent coefficients are not selected by a bounded search. Unit contrast fixes the star coefficient to one. The normalized common eigenvalue then gives a linear identity whose only characteristic-zero solution is the common coefficient one.

Executable root: `theorem_parent_action_positive_field_analytic_uniqueness`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.217 (109. PARENT ACTION POSITIVE FIELD ANALYTIC UNIQUENESS THEOREM)**.** *Replace bounded coefficient enumeration by an analytic proof that the normalized positive parent star action has the unique coefficient pair one and one over every positive characteristic-zero ordered field.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.218. PARENT ACTION POSITIVE FIELD ANALYTIC UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 110. SINGLE COVARIANT PARENT ACTION SYNTHESIS THEOREM

The local seventeen-component action, common-carrier selector, Schur descent, matter action, metric variation, and projective continuum field are one dependency chain rather than separate sector assumptions.

Executable root: `theorem_covariant_parent_action_synthesis`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.219 (110. SINGLE COVARIANT PARENT ACTION SYNTHESIS THEOREM)**.** *Conjoin the finite local cochain action, global selector, Schur descent, chiral matter, metric variation, and projective continuum field into one parent-action dependency theorem.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.220. SINGLE COVARIANT PARENT ACTION SYNTHESIS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 111. FOUR SECTOR RESPONSE RATIO THEOREM

A propagating sector response must vanish when either its native action readout or its Green residue vanishes. The unique minimum-degree response is therefore the bilinear product, whose normalized four-sector ratios are common-scale invariant.

Executable root: `theorem_four_sector_response_ratio`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.221 (111. FOUR SECTOR RESPONSE RATIO THEOREM)**.** *Execute the exact certificate for FOUR SECTOR RESPONSE RATIO THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.222. FOUR SECTOR RESPONSE RATIO THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 112. FORCE ROLE DYNAMIC SIGNATURE THEOREM

The carrier roles are not selected by dimensions alone. Their center, compact commutator, traceless non-Abelian algebra, spin-two module and parent-action behavior make the four assignments mutually distinguishable.

Executable root: `theorem_force_role_dynamic_signature`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.223 (112. FORCE ROLE DYNAMIC SIGNATURE THEOREM)**.** *Identify four internal carrier roles by computed algebraic invariants, not by dimension labels alone.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.224. FORCE ROLE DYNAMIC SIGNATURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 113. OPERATIONAL FOUR FORCE IDENTITY THEOREM

The one-, three-, eight-, and five-dimensional sectors are identified by abelian signed charge, chiral compact action, traceless color action with a center-flux gap, and universal spin-two metric response. Their names are attached only after these operational signatures are distinct.

Executable root: `theorem_operational_four_force_identity`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.225 (113. OPERATIONAL FOUR FORCE IDENTITY THEOREM)**.** *Identify the four generated sectors by their complete operational action on matter, curvature, propagation, and metric response rather than by dimensions or conventional labels alone.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.226. OPERATIONAL FOUR FORCE IDENTITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 114. FOUR FORCE AXIS READOUT MASTER THEOREM

The four sectors occupy four independent readout axes of the same uniquely selected parent action. Their exact ratio packet remains normalized and nondegenerate while gravity receives the shell-flux readout proved earlier.

Executable root: `theorem_four_force_axis_readout_master`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.227 (114. FOUR FORCE AXIS READOUT MASTER THEOREM)**.** *Place four dynamically distinguished carrier roles on orthogonal readout axes of one parent action.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.228. FOUR FORCE AXIS READOUT MASTER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 115. FOUR FORCE COMMON COMPRESSION ORBIT THEOREM

The four internally distinguished sectors are not assigned four unrelated scale laws. One primitive bounded-shadow semigroup transports every sector, while the generated four-force ratios appear only as distinct initial odds on that common orbit.

Executable root: `theorem_four_force_common_compression_orbit`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.229 (115. FOUR FORCE COMMON COMPRESSION ORBIT THEOREM)**.** *Transport all four dynamically identified force sectors through one primitive bounded-shadow semigroup with sector-specific initial odds only.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.230. FOUR FORCE COMMON COMPRESSION ORBIT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 116. OPERATOR SOURCE BOUNDARY COMPRESSION THEOREM

The four generated interaction sectors are not assigned four unrelated scale laws. Their internally generated response ratios are the four eigenvalues of one exact operator on the complete seventeen-component parent carrier. The generated sector projectors are mutually orthogonal, sum to the parent identity, and give an exact power calculus for every native depth.

Executable root: `theorem_operator_source_boundary_compression`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.231 (116. OPERATOR SOURCE BOUNDARY COMPRESSION THEOREM)**.** *Lift the common source-shadow compression law to one exact spectral operator on the complete seventeen-component parent carrier.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.232. OPERATOR SOURCE BOUNDARY COMPRESSION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 117. LOG FREE SHADOW RESIDUAL ISOMORPHISM THEOREM

The native state is the exact residual ratio (one minus shadow) divided by shadow. Projective shadow compression is therefore multiplication in residual space. Additive depth is an optional observer coordinate and is not needed by the source dynamics.

Executable root: `theorem_log_free_shadow_residual_isomorphism`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.233 (117. LOG FREE SHADOW RESIDUAL ISOMORPHISM THEOREM)**.** *Prove the exact rational bijection between bounded shadow states and multiplicative residual states, including the source-shadow compression law without evaluating a logarithm.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.234. LOG FREE SHADOW RESIDUAL ISOMORPHISM THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 118. PARENT COVARIANT RESIDUAL FIELD EQUATION THEOREM

One equation evolves the exact residual carried by the bounded source-shadow state. The compression operator changes residual magnitude, the parent transport changes local frame, and the inverse residual map returns the next bounded shadow. The scalar master map and the four common force orbits are exact reductions.

Executable root: `theorem_parent_covariant_residual_field_equation`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.235 (118. PARENT COVARIANT RESIDUAL FIELD EQUATION THEOREM)**.** *Lift the scalar source-shadow compression identity to one exact rational covariant residual field equation on the complete seventeen-component parent carrier.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.236. PARENT COVARIANT RESIDUAL FIELD EQUATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 119. GENERATED SECTOR SPECTRAL EDGE COCYCLE THEOREM

The parent edge operator is not an arbitrary matrix. Every sector factor is generated from its exact response eigenvalue, while the gravitational factor also carries the unique target-over-source temporal scale ratio. Orthogonal projectors make identity, reversal and path composition exact.

Executable root: `theorem_generated_sector_spectral_edge_cocycle`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.237 (119. GENERATED SECTOR SPECTRAL EDGE COCYCLE THEOREM)**.** *Derive the complete four-sector positive edge-compression operator from generated response eigenvalues and the finite temporal scale ratio, and prove identity, reversal, projector commutation, and path composition.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.238. GENERATED SECTOR SPECTRAL EDGE COCYCLE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 120. COVARIANT RESIDUAL UPDATE EXISTENCE UNIQUENESS THEOREM

Existence is constructive: multiply the transported source residual by the generated edge operator and invert I plus that residual. Uniqueness follows because both the transport and edge operator are invertible and the residual-shadow map is a bijection.

Executable root: `theorem_covariant_residual_update_existence_uniqueness`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.239 (120. COVARIANT RESIDUAL UPDATE EXISTENCE UNIQUENESS THEOREM)**.** *Prove that the generated positive edge operator and invertible parent transport determine one and only one residual target and one and only one bounded shadow target.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.240. COVARIANT RESIDUAL UPDATE EXISTENCE UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 121. COVARIANT RESIDUAL BOUNDED DOMAIN PRESERVATION THEOREM

The source residual is positive on each canonical block. Orthogonal transport preserves its quadratic form, and the generated edge operator is a positive block scalar that commutes with the transported block. Therefore the target residual remains positive and X=(I+R)^-1 remains a strict bounded shadow.

Executable root: `theorem_covariant_residual_bounded_domain_preservation`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.241 (121. COVARIANT RESIDUAL BOUNDED DOMAIN PRESERVATION THEOREM)**.** *Prove positivity of the residual and strict boundedness of the shadow under every generated sector-spectral edge update on the declared block carrier.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.242. COVARIANT RESIDUAL BOUNDED DOMAIN PRESERVATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 122. SECTOR SPECTRAL GAUGE COVARIANCE THEOREM

Every local gauge frame preserves the canonical sector projectors. The generated edge operator is scalar on each projector block and therefore commutes with both endpoint frames. The complete non-scalar four-sector equation consequently transforms by target conjugation.

Executable root: `theorem_sector_spectral_gauge_covariance`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.243 (122. SECTOR SPECTRAL GAUGE COVARIANCE THEOREM)**.** *Prove full two-end gauge covariance with the generated non-scalar sector-spectral edge operator rather than only a common scalar compression.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.244. SECTOR SPECTRAL GAUGE COVARIANCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 123. COVARIANT RESIDUAL EXACT PATH GROUPOID THEOREM

Generated edge compressions form a multiplicative cocycle and generated frame transports form an ordered groupoid. Their combined residual update has exact identity, inverse and composition laws. A closed path carries no net compression but may retain a nontrivial frame holonomy, which is the finite curvature signal.

Executable root: `theorem_covariant_residual_exact_path_groupoid`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.245 (123. COVARIANT RESIDUAL EXACT PATH GROUPOID THEOREM)**.** *Prove identity edges, inverse edges, arbitrary three-edge composition and closed-loop residual holonomy for the generated sector-spectral covariant update.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.246. COVARIANT RESIDUAL EXACT PATH GROUPOID THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 124. COVARIANT RESIDUAL EDGE DOMAIN COMPLETION THEOREM

The central residual equation defines a complete edge dynamics: its generated spectrum composes, every admissible source has one target, positivity remains inside the bounded-shadow domain, endpoint frames transform covariantly, and reversed or concatenated paths are exact.

Executable root: `theorem_covariant_residual_edge_domain_completion`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.247 (124. COVARIANT RESIDUAL EDGE DOMAIN COMPLETION THEOREM)**.** *Conjoin generated edge spectrum, constructive update, bounded positivity, covariance, inversion, composition, and holonomy.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.248. COVARIANT RESIDUAL EDGE DOMAIN COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 125. PLAQUETTE HOLONOMY CURVATURE COEFFICIENT THEOREM

Write each link as I plus epsilon times its exact rational generator and invert it coefficient by coefficient. The constant plaquette coefficient is I, the linear coefficient cancels, and the quadratic coefficient is the generator commutator. Adding edge variation gives the exterior-difference part, so the continuum parent curvature has the form dOmega plus Omega wedge Omega.

Executable root: `theorem_plaquette_holonomy_curvature_coefficient`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.249 (125. PLAQUETTE HOLONOMY CURVATURE COEFFICIENT THEOREM)**.** *Extract the exact second-order commutator coefficient of a plaquette holonomy in a truncated rational polynomial ring without exponential or logarithmic functions.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.250. PLAQUETTE HOLONOMY CURVATURE COEFFICIENT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 126. PARENT COVARIANT RESIDUAL GAUGE COVARIANCE THEOREM

Changing the local parent frame independently at the source and target conjugates the residual states and changes the edge transport by the usual two-end rule. The compression equation transforms by target conjugation and therefore represents one frame-independent parent law.

Executable root: `theorem_parent_covariant_residual_gauge_covariance`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.251 (126. PARENT COVARIANT RESIDUAL GAUGE COVARIANCE THEOREM)**.** *Prove exact vertex-gauge covariance of the parent covariant residual field equation under independently generated rational frame changes at both ends of an edge.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.252. PARENT COVARIANT RESIDUAL GAUGE COVARIANCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 127. PARENT COVARIANT RESIDUAL PATH COMPOSITION THEOREM

Successive residual transports compose by multiplying their compression operators and ordered frame transports. A closed path returns exactly only when its parent holonomy is the identity; otherwise the finite loop defect is the curvature precursor.

Executable root: `theorem_parent_covariant_residual_path_composition`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.253 (127. PARENT COVARIANT RESIDUAL PATH COMPOSITION THEOREM)**.** *Prove the exact path-composition law, sector-preserving cocycle, and finite holonomy defect of the covariant residual field equation.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.254. PARENT COVARIANT RESIDUAL PATH COMPOSITION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 128. TEMPORAL EXTERNAL INTERNAL CURVATURE SPLIT THEOREM

One temporal compression operator splits into three internal gauge projectors and one universal metric projector. A common unit shift is flat, whereas local relative compression creates curvature.

Executable root: `theorem_temporal_external_internal_curvature_split`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.255 (128. TEMPORAL EXTERNAL INTERNAL CURVATURE SPLIT THEOREM)**.** *Split one temporal compression operator into a universal external metric projector and three orthogonal internal gauge projectors, with no independently inserted force generator.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.256. TEMPORAL EXTERNAL INTERNAL CURVATURE SPLIT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 129. SINGLE TEMPORAL PARENT VARIATION FOUR FORCE THEOREM

One parent action and one time-generated compression operator yield one universal metric variation and three internal gauge variations; four unrelated force actions are not inserted.

Executable root: `theorem_single_temporal_parent_variation_four_force`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.257 (129. SINGLE TEMPORAL PARENT VARIATION FOUR FORCE THEOREM)**.** *Prove that one time-generated parent operator and one covariant parent action split under a single variation into the universal gravity equation and the three internal gauge equations.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.258. SINGLE TEMPORAL PARENT VARIATION FOUR FORCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 130. RESPONSE FUNCTION CONTINUOUS BILINEAR UNIQUENESS THEOREM

A propagating response must disappear if either the action burden or the Green residue disappears. Separate unit homogeneity then transports the normalized value at one and one to every positive pair, forcing the product law rather than selecting it from a short list of monomials.

Executable root: `theorem_response_function_continuous_bilinear_uniqueness`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.259 (130. RESPONSE FUNCTION CONTINUOUS BILINEAR UNIQUENESS THEOREM)**.** *Prove that separate degree-one homogeneity, zero-factor vanishing, positivity, and unit normalization force the action-residue response to be the bilinear product on positive rationals and its continuous positive-real extension.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.260. RESPONSE FUNCTION CONTINUOUS BILINEAR UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 131. PRESHIELDED FORMULA GRAMMAR THEOREM

Only typed maps that preserve the parent meaning are permitted. Unrelated primitives cannot be combined, exponents must equal an actual interaction multiplicity, response must vanish when either required factor vanishes, and the only symmetric scale-free unit-sum decoder divides by the total response.

Executable root: `theorem_preshielded_formula_grammar`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.261 (131. PRESHIELDED FORMULA GRAMMAR THEOREM)**.** *Freeze the typed formula grammar, derive the response normalization uniquely, and seal new target-free parent-shadow invariants before any external comparison.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.262. PRESHIELDED FORMULA GRAMMAR THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 132. GAUGE QUOTIENT FULL FAITHFUL TRANSPORT THEOREM

Gauge transformations are transported by conjugation through the complete projection rather than guessed independently in the shadow. The transport preserves products, inverses, orbits, stabilizers, constraints, invariant actions, and physical observable algebras.

Executable root: `theorem_gauge_quotient_full_faithful_transport`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.263 (132. GAUGE QUOTIENT FULL FAITHFUL TRANSPORT THEOREM)**.** *Prove that every generated gauge morphism, orbit, invariant action, and physical observable transports bijectively through a complete source-shadow projection.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.264. GAUGE QUOTIENT FULL FAITHFUL TRANSPORT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 133. SOURCE ACTION VARIATION PUSHFORWARD THEOREM

The shadow Hessian is the exact inverse-congruence image of the source Hessian. Actions agree and pulling the shadow gradient back gives the source variation for every source vector.

Executable root: `theorem_source_action_variation_pushforward`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.265 (133. SOURCE ACTION VARIATION PUSHFORWARD THEOREM)**.** *Prove exact commutation of source variation, shadow pushforward, and measure reconstruction for an invertible complete projection family.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.266. SOURCE ACTION VARIATION PUSHFORWARD THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 134. CANONICAL FORCE ALGEBRA REPRESENTATION COMPLETION THEOREM

The canonical anonymous decomposition is not identified by dimension alone. Exact commutator closure, Jacobi identities, invariant trace form, spin-two action and anomaly cancellation supply the algebraic operational signatures.

Executable root: `theorem_canonical_force_algebra_representation_completion`

Class and stage: Primary; 3. FOUR FORCE CARRIERS AND PARENT ACTION

Theorem A.267 (134. CANONICAL FORCE ALGEBRA REPRESENTATION COMPLETION THEOREM)**.** *Prove that the anonymous ranks five, one, three and eight admit the exact gravity, U(1), SU(2), and SU(3) algebraic roles before measured couplings are used.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.268. CANONICAL FORCE ALGEBRA REPRESENTATION COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 135. MINIMAL CHIRAL MATTER CARRIER THEOREM

Derive one minimal sixteen-state chiral matter carrier from the eight double-zero microstates, two orientations, and the generated weak and color module dimensions.

Executable root: `_mfc_minimal_chiral_matter_carrier_theorem`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.269 (135. MINIMAL CHIRAL MATTER CARRIER THEOREM)**.** *Derive one minimal sixteen-state chiral matter carrier from the eight double-zero microstates, two orientations, and the generated weak and color module dimensions.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.270. MINIMAL CHIRAL MATTER CARRIER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 136. HYPERCHARGE AND ANOMALY CANCELLATION THEOREM

Derive the primitive charge-ratio solution and prove exact gauge, gravitational, cubic, and global weak anomaly cancellation in the declared minimal matter carrier.

Executable root: `_mfc_hypercharge_anomaly_cancellation_theorem`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.271 (136. HYPERCHARGE AND ANOMALY CANCELLATION THEOREM)**.** *Derive the primitive charge-ratio solution and prove exact gauge, gravitational, cubic, and global weak anomaly cancellation in the declared minimal matter carrier.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.272. HYPERCHARGE AND ANOMALY CANCELLATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 137. THREE GENERATION PROJECTOR THEOREM

Derive exactly three nonvacuum closure-depth sectors from the primitive time spectrum and construct their exact spectral projectors.

Executable root: `_mfc_three_generation_projector_theorem`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.273 (137. THREE GENERATION PROJECTOR THEOREM)**.** *Derive exactly three nonvacuum closure-depth sectors from the primitive time spectrum and construct their exact spectral projectors.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.274*.* THREE GENERATION PROJECTOR THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 138. TIME SPECTRUM THREE GENERATION GLOBAL UNIQUENESS THEOREM

The unique unit vacuum pole is removed from the generated four-pole temporal spectrum, leaving exactly three nonvacuum poles, equal to the generated spatial rank. The nonzero Vandermonde determinant makes every degree-below-three spectral projector polynomial unique and reconstructs the transfer operator exactly.

Executable root: `theorem_time_spectrum_three_generation_global_uniqueness`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.275 (138. TIME SPECTRUM THREE GENERATION GLOBAL UNIQUENESS THEOREM)**.** *Prove that the three nonvacuum temporal projectors are the unique spectral idempotents of the generated simple three-pole spectrum.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.276*.* TIME SPECTRUM THREE GENERATION GLOBAL UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 139. NATIVE GENERATION COUNT PROVENANCE THEOREM

The three generations are not entered as a number. They are the common rank of the three outer source blocks, the three nonvacuum temporal projectors, and the generated spatial residual basis.

Executable root: `theorem_native_generation_count_provenance`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.277 (139. NATIVE GENERATION COUNT PROVENANCE THEOREM)**.** *Prove that the generation count is the common rank of the three outer source blocks, the nonvacuum temporal spectrum, and generated space.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.278. NATIVE GENERATION COUNT PROVENANCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 140. TARGET FREE STANDARD FIELD DECODER THEOREM

The sixteen-state chiral carrier is first partitioned into anonymous algebraic roles. Conventional particle names are attached only after multiplicity and the primitive anomaly-free charge ray have already been derived.

Executable root: `theorem_target_free_standard_field_decoder`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.279 (140. TARGET FREE STANDARD FIELD DECODER THEOREM)**.** *Generate the complete chiral role multiplicities and charge ratios before assigning conventional field names.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.280. TARGET FREE STANDARD FIELD DECODER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 141. TEMPORAL GENERATION SPECTRAL RATIO THEOREM

The three nonvacuum transfer poles are converted to generator gaps by the unique minimum-degree semigroup gap law. Their exact normalized native signature is generated without observed masses or a unit calibration.

Executable root: `theorem_temporal_generation_spectral_ratio`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.281 (141. TEMPORAL GENERATION SPECTRAL RATIO THEOREM)**.** *Execute the exact certificate for TEMPORAL GENERATION SPECTRAL RATIO THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.282. TEMPORAL GENERATION SPECTRAL RATIO THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 142. NATIVE MIXING INVARIANT THEOREM

Ordered generation gaps determine an antisymmetric overlap generator with no fitted angle. The Cayley transform gives an exact rational determinant-one mixing matrix and a doubly normalized probability matrix.

Executable root: `theorem_native_mixing_invariant`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.283 (142. NATIVE MIXING INVARIANT THEOREM)**.** *Execute the exact certificate for NATIVE MIXING INVARIANT THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.284. NATIVE MIXING INVARIANT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 143. MASS MIXING AND CP RATIO THEOREM

Generate positive three-sector mass ledgers, exact rational unitary mixing matrices, and nonzero CP-oriented invariants without importing observed masses or angles.

Executable root: `_mfc_mass_mixing_cp_ratio_theorem`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.285 (143. MASS MIXING AND CP RATIO THEOREM)**.** *Generate positive three-sector mass ledgers, exact rational unitary mixing matrices, and nonzero CP-oriented invariants without importing observed masses or angles.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.286. MASS MIXING AND CP RATIO THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 144. REPRESENTATION INDEX SCALE FLOW THEOREM

Derive one-loop representation-index coefficients from the generated matter carrier and prove an exact rational discrete inverse-coupling flow without logarithms.

Executable root: `_mfc_representation_index_scale_flow_theorem`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.287 (144. REPRESENTATION INDEX SCALE FLOW THEOREM)**.** *Derive one-loop representation-index coefficients from the generated matter carrier and prove an exact rational discrete inverse-coupling flow without logarithms.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.288. REPRESENTATION INDEX SCALE FLOW THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 145. PARENT MATTER COUPLING THEOREM

Couple the generated chiral matter ledger to the parent force carrier and prove exact charge balance, Yukawa closure, generation replication, and source assembly.

Executable root: `_mfc_parent_matter_coupling_theorem`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.289 (145. PARENT MATTER COUPLING THEOREM)**.** *Couple the generated chiral matter ledger to the parent force carrier and prove exact charge balance, Yukawa closure, generation replication, and source assembly.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.290. PARENT MATTER COUPLING THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 146. MATTER RATIO SEAL THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_mfc_matter_ratio_seal_theorem`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.291 (146. MATTER RATIO SEAL THEOREM)**.** *Execute the exact certificate for MATTER RATIO SEAL THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.292. MATTER RATIO SEAL THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 147. FORWARD CHIRAL PREDICTION PACKET THEOREM

The packet is generated entirely from the native carrier, anomaly equations, temporal projectors, and projection decoder. It is frozen before any external ratio table is supplied.

Executable root: `_x893_forward_chiral_prediction_packet_theorem`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.293 (147. FORWARD CHIRAL PREDICTION PACKET THEOREM)**.** *Generate a sealed forward matter-and-observable ratio packet without importing measured values.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.294. FORWARD CHIRAL PREDICTION PACKET THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 148. ORIGIN MATTER FIELD COMPLETION THEOREM

Integrate minimal chiral matter, primitive charge ratios, exact anomaly cancellation, three nonvacuum generations, mixing, scale flow, parent coupling, and sealed ratio predictions.

Executable root: `_origin_matter_field_completion_theorem`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.295 (148. ORIGIN MATTER FIELD COMPLETION THEOREM)**.** *Integrate minimal chiral matter, primitive charge ratios, exact anomaly cancellation, three nonvacuum generations, mixing, scale flow, parent coupling, and sealed ratio predictions.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.296. ORIGIN MATTER FIELD COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 149. GENERATION READOUT SEPARATION THEOREM

Keep outer sign as a generated readout while deriving three generations only from the three nonvacuum primitive-time projectors.

Executable root: `_nsr_generation_readout_separation_theorem`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.297 (149. GENERATION READOUT SEPARATION THEOREM)**.** *Keep outer sign as a generated readout while deriving three generations only from the three nonvacuum primitive-time projectors.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.298. GENERATION READOUT SEPARATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 150. NESTED PARENT MATTER ACTION THEOREM

Place the forty-eight-state matter burden in the common nested mode and retain the complete seventeen-dimensional force residual action.

Executable root: `_nsr_parent_matter_action_theorem`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.299 (150. NESTED PARENT MATTER ACTION THEOREM)**.** *Place the forty-eight-state matter burden in the common nested mode and retain the complete seventeen-dimensional force residual action.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.300. NESTED PARENT MATTER ACTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 151. FERMION CAR SPIN BRIDGE THEOREM

Three Jordan-Wigner modes satisfy every canonical anticommutation relation and carry exact fermion parity. The nontrivial center of the spin double cover reverses spinors while leaving vector observables unchanged, joining half-integer spin and odd fermion parity.

Executable root: `theorem_fermion_car_spin_bridge`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.301 (151. FERMION CAR SPIN BRIDGE THEOREM)**.** *Construct three exact fermion modes, canonical anticommutation, parity, chirality, and the spinorial double-cover sign.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.302. FERMION CAR SPIN BRIDGE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 152. NATIVE HIGGS POTENTIAL UNIQUENESS THEOREM

The lowest radial invariant polynomial with one normalized closed vacuum and stationary closure is uniquely one quarter of the squared closure defect. Its Hessian has one positive radial mode and one gauge-orbit zero mode; generated Abelian and weak readouts then give one neutral and one massive direction.

Executable root: `theorem_native_higgs_potential_uniqueness`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.303 (152. NATIVE HIGGS POTENTIAL UNIQUENESS THEOREM)**.** *Classify the lowest invariant radial potential and derive one vacuum orbit, one radial mode, one gauge mode, and the neutral split.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.304. NATIVE HIGGS POTENTIAL UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 153. NATIVE MATTER FINGERPRINT GENERATION THEOREM

The generated chiral carrier is anonymized before any observed particle name is admitted. Color rank, weak rank, central character, closure origin, orientation support, multiplicity, Higgs incidence and neutrality form six exact distinct native role fingerprints.

Executable root: `theorem_native_matter_fingerprint_generation`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.305 (153. NATIVE MATTER FINGERPRINT GENERATION THEOREM)**.** *Generate anonymous matter roles from native projectors, charges, closure origins, and temporal generations before observed names are introduced.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.306. NATIVE MATTER FINGERPRINT GENERATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 154. NATIVE ROLE EXISTENCE THEOREM

Prove that every generated anonymous role is nonempty and that the six roles exhaust the complete sixteen-state chiral carrier in each of three temporal generations.

Executable root: `theorem_native_role_existence`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.307 (154. NATIVE ROLE EXISTENCE THEOREM)**.** *Prove that every generated anonymous role is nonempty and that the six roles exhaust the complete sixteen-state chiral carrier in each of three temporal generations.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.308. NATIVE ROLE EXISTENCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 155. NATIVE ROLE ORBIT INJECTIVITY THEOREM

Every permutation of the six role orbits is exhausted after internal seat permutations are quotiented. Only the identity preserves all fingerprints, and the inverse fingerprint map reconstructs every role exactly.

Executable root: `theorem_native_role_orbit_injectivity`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.309 (155. NATIVE ROLE ORBIT INJECTIVITY THEOREM)**.** *Quotient internal seat permutations and prove that no nontrivial permutation of the six anonymous role orbits preserves the complete native fingerprint map.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.310. NATIVE ROLE ORBIT INJECTIVITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 156. NATIVE ROLE SEPARATION STABILITY THEOREM

The exact minimum distance between distinct role fingerprints and between temporal generation gaps is computed. Triangle inequality gives a positive rational neighborhood in which the inverse role and generation assignment remains single valued.

Executable root: `theorem_native_role_separation_stability`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.311 (156. NATIVE ROLE SEPARATION STABILITY THEOREM)**.** *Compute an exact positive fingerprint separation margin and prove nearest-orbit stability under admissible rank-preserving rational perturbations.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.312. NATIVE ROLE SEPARATION STABILITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 157. ADMISSIBLE OBSERVABLE GRAMMAR COMPLETENESS THEOREM

Polynomial semigroup multiplicativity forces one monomial, endpoint normalization fixes its coefficient, and unit tangent fixes its degree. Separately multi-affine response laws that vanish when a required factor vanishes are uniquely the normalized product of all factors.

Executable root: `theorem_admissible_observable_grammar_completeness`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.313 (157. ADMISSIBLE OBSERVABLE GRAMMAR COMPLETENESS THEOREM)**.** *Prove representation theorems for every polynomial semigroup gap and every normalized separately multi-affine response functional admitted by the native observable axioms.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.314. ADMISSIBLE OBSERVABLE GRAMMAR COMPLETENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 158. NATIVE MASS DEFINITION EQUIVALENCE THEOREM

Transfer defect, correlation decrement, generator-resolvent pole, parent curvature and projected action burden are derived independently. All five paths yield the same exact scale-free generation mass ratio rather than selecting one convenient formula after comparison.

Executable root: `theorem_native_mass_definition_equivalence`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.315 (158. NATIVE MASS DEFINITION EQUIVALENCE THEOREM)**.** *Derive one native mass-ratio spectrum independently as a transfer defect, a one-step correlation decrement, a generator-resolvent pole, a parent quadratic curvature, and a projected action burden.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.316. NATIVE MASS DEFINITION EQUIVALENCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 159. SHADOW MASS COMPRESSION RATE EQUIVALENCE THEOREM

The already independent transfer, correlation, resolvent, curvature, and action definitions give the same 4:6:7 mass signature. The same signature is now proved to be both an additive compression depth ratio and a common continuum-generator rate ratio.

Executable root: `theorem_shadow_mass_compression_rate_equivalence`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.317 (159. SHADOW MASS COMPRESSION RATE EQUIVALENCE THEOREM)**.** *Transport the independently derived native mass ratio into equivalent discrete compression-depth and continuous generator-rate representations.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.318. SHADOW MASS COMPRESSION RATE EQUIVALENCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 160. ANONYMOUS MATTER SPECTRUM MIXING COMPLETION THEOREM

Matter roles are first distinguished by anonymous gauge and occupancy fingerprints. Temporal pole order then fixes the three generation order uniquely. Positive mass ledgers, unitary mixing matrices, and nonzero orientation invariants are generated before any experimental particle name is attached.

Executable root: `theorem_anonymous_matter_spectrum_mixing_completion`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.319 (160. ANONYMOUS MATTER SPECTRUM MIXING COMPLETION THEOREM)**.** *Generate anonymous chiral roles, three causal generations, sector mass ledgers, unitary mixing, and nonzero orientation invariants before any observed particle name or measured value is supplied.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.320. ANONYMOUS MATTER SPECTRUM MIXING COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 161. PARENT ACTION CHIRAL VERTEX COMPLETION THEOREM

The sixteen-state chiral carrier is placed in one-to-one correspondence with six anonymous role orbits. Their color, weak, central-charge and universal metric indices determine every parent gauge incidence, while the exact charge ray determines four closed Yukawa vertex types in each of the three temporal generations.

Executable root: `theorem_parent_action_chiral_vertex_completion`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.321 (161. PARENT ACTION CHIRAL VERTEX COMPLETION THEOREM)**.** *Derive every chiral gauge and Yukawa vertex from the generated parent action, anonymous role fingerprints, exact charges and three temporal generation projectors.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.322. PARENT ACTION CHIRAL VERTEX COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 162. PARENT VERTEX DECAY AMPLITUDE COMPLETION THEOREM

The parent gauge and Yukawa vertices select the chiral role orbit. The exact unitary generation rotations supply one-step transition probabilities. Strict generation descent makes the transition operator nilpotent, so the complete direct-plus-intermediate path sum is the finite inverse of one minus the weighted transition operator.

Executable root: `theorem_parent_vertex_decay_amplitude_completion`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.323 (162. PARENT VERTEX DECAY AMPLITUDE COMPLETION THEOREM)**.** *Derive direct and intermediate generation-changing channel weights from parent vertices and the exact rational unitary quark-like and lepton-like mixing operators.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.324. PARENT VERTEX DECAY AMPLITUDE COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 163. PARENT DECAY BRANCHING RATIO COMPLETION THEOREM

Every branch numerator is the unique separately homogeneous product of a complete parent-vertex path weight, the native generation-gap phase-space ratio and the generated role multiplicity. Dividing by the complete width at fixed initial state gives a unique normalized branching packet.

Executable root: `theorem_parent_decay_branching_ratio_completion`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.325 (163. PARENT DECAY BRANCHING RATIO COMPLETION THEOREM)**.** *Derive normalized two-body generation-branch ratios and total-width ratios directly from the parent-vertex path weights, native mass gaps and role multiplicities.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.326. PARENT DECAY BRANCHING RATIO COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 164. PARENT DECAY RG SCATTERING COMPLETION THEOREM

The four parent-sector coordinates run with the generated thresholds. At each scale they rescale the complete width of every role while leaving the fixed-initial normalized branching packet unchanged. The same scale set carries the exact unitary differential scattering ratios and the projective common-compression invariant.

Executable root: `theorem_parent_decay_rg_scattering_completion`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.327 (164. PARENT DECAY RG SCATTERING COMPLETION THEOREM)**.** *Transport parent-derived branch and scattering ratios through every generated threshold scale while preserving normalization, channel support and the projective common-compression invariant.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.328. PARENT DECAY RG SCATTERING COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 165. MATTER CHANNEL DYNAMICS COMPLETION THEOREM

The primitive relational grammar generates time, three temporal generations and a minimal chiral carrier. The single parent action then supplies the role vertices, unitary mixing supplies descending amplitudes, native gaps supply phase-space ratios, and the threshold flow transports the resulting branching and scattering packets.

Executable root: `theorem_matter_channel_dynamics_completion`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.329 (165. MATTER CHANNEL DYNAMICS COMPLETION THEOREM)**.** *Close one dependency chain from the eighteen-state origin through chiral matter, parent vertices, decay amplitudes, branching ratios, differential scattering and scale transport.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.330. MATTER CHANNEL DYNAMICS COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 166. MATTER ANOMALY GENERATION RATIO COMPLETION THEOREM

The matter branch is generated after the parent carrier: chiral roles, exact charge ratios, anomaly cancellation, three temporal generations, mixing, complete finite paths, normalized branching and common ratio flow form one dependency chain.

Executable root: `theorem_matter_anomaly_generation_ratio_completion`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.331 (166. MATTER ANOMALY GENERATION RATIO COMPLETION THEOREM)**.** *Close chiral carrier, charge ratios, anomaly cancellation, three temporal generations, mixing, decay, branching, scattering and ratio flow.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.332. MATTER ANOMALY GENERATION RATIO COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 167. NATIVE DECAY SELECTION AND CHANNEL COMPLETENESS THEOREM

Every ordered role-generation pair is classified by fingerprint conservation and strict temporal descent. Nilpotence closes the direct and intermediate path expansion exactly, so no admitted native path is omitted.

Executable root: `theorem_native_decay_selection_channel_completeness`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.333 (167. NATIVE DECAY SELECTION AND CHANNEL COMPLETENESS THEOREM)**.** *Classify every ordered role-generation transition, impose native conservation rules, and sum all direct and intermediate descending paths by an exact nilpotent path expansion.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.334. NATIVE DECAY SELECTION AND CHANNEL COMPLETENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 168. NATIVE BRANCHING FUNCTIONAL UNIQUENESS THEOREM

Each branch numerator is the unique normalized multi-affine product of path weight, role multiplicity, native phase space and symmetry factor. Every unstable state has a complete branching packet summing to one, while the lightest state in each role has zero declared width.

Executable root: `theorem_native_branching_functional_uniqueness`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.335 (168. NATIVE BRANCHING FUNCTIONAL UNIQUENESS THEOREM)**.** *Use the unique normalized multi-affine product rule to derive every native generation-branch numerator, normalized branching packet, and scale-free total-width ratio.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.336. NATIVE BRANCHING FUNCTIONAL UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 169. NESTED RESIDUAL RATIO SEAL THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_nsr_ratio_seal_theorem`

Class and stage: Primary; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.337 (169. NESTED RESIDUAL RATIO SEAL THEOREM)**.** *Execute the exact certificate for NESTED RESIDUAL RATIO SEAL THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.338. NESTED RESIDUAL RATIO SEAL THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 170. MASS IONIZATION SEPARATION

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_mass_ionization_separation`

Class and stage: Narrative; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.339 (170. MASS IONIZATION SEPARATION)**.** *Execute the exact certificate for MASS IONIZATION SEPARATION on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.340. MASS IONIZATION SEPARATION holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 171. GRAVITY INERTIA PROJECTION

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_gravity_inertia_projection`

Class and stage: Narrative; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.341 (171. GRAVITY INERTIA PROJECTION)**.** *Execute the exact certificate for GRAVITY INERTIA PROJECTION on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.342. GRAVITY INERTIA PROJECTION holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 172. STRONG WEAK PROJECTION

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_strong_weak_projection`

Class and stage: Narrative; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.343 (172. STRONG WEAK PROJECTION)**.** *Execute the exact certificate for STRONG WEAK PROJECTION on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.344. STRONG WEAK PROJECTION holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 173. FOUR FORCE MASTER RATIO MAP

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_four_force_master_ratio_map`

Class and stage: Narrative; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.345 (173. FOUR FORCE MASTER RATIO MAP)**.** *Execute the exact certificate for FOUR FORCE MASTER RATIO MAP on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.346. FOUR FORCE MASTER RATIO MAP holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 174. ALL FORCE DEPENDENCY CLOSURE

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_all_force_dependency_closure`

Class and stage: Narrative; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.347 (174. ALL FORCE DEPENDENCY CLOSURE)**.** *Execute the exact certificate for ALL FORCE DEPENDENCY CLOSURE on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.348. ALL FORCE DEPENDENCY CLOSURE holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 175. FORCE DICTIONARY GAUGE NORMALIZATION AND FAMILY SELECTOR

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_force_dictionary_family_selector`

Class and stage: Narrative; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.349 (175. FORCE DICTIONARY GAUGE NORMALIZATION AND FAMILY SELECTOR)**.** *Execute the exact certificate for FORCE DICTIONARY GAUGE NORMALIZATION AND FAMILY SELECTOR on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.350. FORCE DICTIONARY GAUGE NORMALIZATION AND FAMILY SELECTOR holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 176. MULTI ELECTRON MATTER AND NON-ABELIAN GRAVITY MEASURE

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_multi_electron_nonabelian_gravity_measure`

Class and stage: Narrative; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.351 (176. MULTI ELECTRON MATTER AND NONABELIAN GRAVITY MEASURE)**.** *Execute the exact certificate for MULTI ELECTRON MATTER AND NONABELIAN GRAVITY MEASURE on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.352. MULTI ELECTRON MATTER AND NONABELIAN GRAVITY MEASURE holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 177. UNIFIED SHADOW AND NATIVE LATENT OBJECT PROJECTION

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_unified_shadow_latent_projection`

Class and stage: Narrative; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.353 (177. UNIFIED SHADOW AND NATIVE LATENT OBJECT PROJECTION)**.** *Execute the exact certificate for UNIFIED SHADOW AND NATIVE LATENT OBJECT PROJECTION on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.354. UNIFIED SHADOW AND NATIVE LATENT OBJECT PROJECTION holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 178. ATOMIC SPECTRAL MATTER MIXING AND JOINT CONTINUUM

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_atomic_spectral_mixing_joint_continuum`

Class and stage: Narrative; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.355 (178. ATOMIC SPECTRAL MATTER MIXING AND JOINT CONTINUUM)**.** *Execute the exact certificate for ATOMIC SPECTRAL MATTER MIXING AND JOINT CONTINUUM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.356. ATOMIC SPECTRAL MATTER MIXING AND JOINT CONTINUUM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 179. PROJECTION RECONSTRUCTION LIE GAUGE HIGGS COFRAME AND CONTINUUM

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_projection_lie_higgs_coframe_continuum`

Class and stage: Narrative; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.357 (179. PROJECTION RECONSTRUCTION LIE GAUGE HIGGS COFRAME AND CONTINUUM)**.** *Execute the exact certificate for PROJECTION RECONSTRUCTION LIE GAUGE HIGGS COFRAME AND CONTINUUM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.358. PROJECTION RECONSTRUCTION LIE GAUGE HIGGS COFRAME AND CONTINUUM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 180. LOCAL GAUGE BUNDLE CONFINEMENT AFFINE COVARIANCE AND LORENTZ ACTION

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_local_bundle_confinement_affine_lorentz_action`

Class and stage: Narrative; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.359 (180. LOCAL GAUGE BUNDLE CONFINEMENT AFFINE COVARIANCE AND LORENTZ ACTION)**.** *Execute the exact certificate for LOCAL GAUGE BUNDLE CONFINEMENT AFFINE COVARIANCE AND LORENTZ ACTION on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.360. LOCAL GAUGE BUNDLE CONFINEMENT AFFINE COVARIANCE AND LORENTZ ACTION holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 181. CHARGE NULLSPACE OPERATOR ALPHABET AND PALATINI SELECTION

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_charge_nullspace_palation_selection`

Class and stage: Narrative; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.361 (181. CHARGE NULLSPACE OPERATOR ALPHABET AND PALATINI SELECTION)**.** *Execute the exact certificate for CHARGE NULLSPACE OPERATOR ALPHABET AND PALATINI SELECTION on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.362. CHARGE NULLSPACE OPERATOR ALPHABET AND PALATINI SELECTION holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 182. DE RHAM MAXWELL PULLBACK NATURALITY AND GALERKIN REFINEMENT

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_de_rham_maxwell_galerkin_refinement`

Class and stage: Narrative; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.363 (182. DE RHAM MAXWELL PULLBACK NATURALITY AND GALERKIN REFINEMENT)**.** *Execute the exact certificate for DE RHAM MAXWELL PULLBACK NATURALITY AND GALERKIN REFINEMENT on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.364. DE RHAM MAXWELL PULLBACK NATURALITY AND GALERKIN REFINEMENT holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 183. CHIRAL GAUGE GENERATOR WILSON SELECTOR AND PROJECTIVE DYNAMICS

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_chiral_generator_wilson_projective_dynamics`

Class and stage: Narrative; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.365 (183. CHIRAL GAUGE GENERATOR WILSON SELECTOR AND PROJECTIVE DYNAMICS)**.** *Execute the exact certificate for CHIRAL GAUGE GENERATOR WILSON SELECTOR AND PROJECTIVE DYNAMICS on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.366. CHIRAL GAUGE GENERATOR WILSON SELECTOR AND PROJECTIVE DYNAMICS holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 184. GLOBAL STRONG FIELD CURVATURE AND VARIATIONAL REFINEMENT

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_global_strong_field_variational_refinement`

Class and stage: Narrative; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.367 (184. GLOBAL STRONG FIELD CURVATURE AND VARIATIONAL REFINEMENT)**.** *Execute the exact certificate for GLOBAL STRONG FIELD CURVATURE AND VARIATIONAL REFINEMENT on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.368. GLOBAL STRONG FIELD CURVATURE AND VARIATIONAL REFINEMENT holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 185. CONTINUUM CHIRAL PACKET NATIVE ACTION AND EINSTEIN HIERARCHY

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_continuum_chiral_action_einstein_hierarchy`

Class and stage: Narrative; 4. CHIRAL MATTER AND GENERATIONS

Theorem A.369 (185. CONTINUUM CHIRAL PACKET NATIVE ACTION AND EINSTEIN HIERARCHY)**.** *Execute the exact certificate for CONTINUUM CHIRAL PACKET NATIVE ACTION AND EINSTEIN HIERARCHY on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.370. CONTINUUM CHIRAL PACKET NATIVE ACTION AND EINSTEIN HIERARCHY holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 186. OPEN BALANCED ZERO MAXWELL MODE THEOREM

Derive exact vacuum Maxwell identities from the open balanced two-polarization carrier.

Executable root: `_u883_maxwell_from_open_balanced_zero_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.371 (186. OPEN BALANCED ZERO MAXWELL MODE THEOREM)**.** *Derive exact vacuum Maxwell identities from the open balanced two-polarization carrier.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.372. OPEN BALANCED ZERO MAXWELL MODE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 187. TRANSVERSE POLARIZATION HELICITY THEOREM

Derive two exact transverse helicity sectors without trigonometric input.

Executable root: `_u883_polarization_helicity_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.373 (187. TRANSVERSE POLARIZATION HELICITY THEOREM)**.** *Derive two exact transverse helicity sectors without trigonometric input.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.374. TRANSVERSE POLARIZATION HELICITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 188. RATIONAL ORTHOGONAL BORN WEIGHT THEOREM

Derive normalized branch probabilities from additive conserved quadratic burden on rational orthogonal refinements.

Executable root: `_u883_born_quadratic_probability_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.375 (188. RATIONAL ORTHOGONAL BORN WEIGHT THEOREM)**.** *Derive normalized branch probabilities from additive conserved quadratic burden on rational orthogonal refinements.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.376. RATIONAL ORTHOGONAL BORN WEIGHT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 189. WAVE PARTICLE PROJECTION DUALITY THEOREM

One norm-preserving native carrier has a distributed two-channel amplitude readout and two rank-one detector readouts. Wave-like distribution and particle-like localization are therefore compatible projection resolutions rather than a forced duplication of native objects.

Executable root: `_e897_wave_particle_projection_duality_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.377 (189. WAVE PARTICLE PROJECTION DUALITY THEOREM)**.** *Construct one exact rational norm-preserving carrier with distributed wave and localized detector projections.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.378. WAVE PARTICLE PROJECTION DUALITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 190. BELL NO SIGNALLING PROJECTION THEOREM

A many-to-one native projection can support nonclassical Bell correlations while every local marginal remains exactly one half and independent of the remote setting. Hidden native fibers therefore do not create controllable superluminal signalling in the declared dual-channel quantum readout.

Executable root: `_e898_bell_no_signalling_projection_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.379 (190. BELL NO SIGNALLING PROJECTION THEOREM)**.** *Construct exact Q(sqrt(2)) Bell correlations with normalized positive probabilities and no-signalling marginals.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.380. BELL NO SIGNALLING PROJECTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 191. DOUBLE PATH INTERFERENCE RATIO THEOREM

Generate exact two-path interference ratios and complementarity from internally generated rational phases.

Executable root: `_u883_double_slit_ratio_prediction_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.381 (191. DOUBLE PATH INTERFERENCE RATIO THEOREM)**.** *Generate exact two-path interference ratios and complementarity from internally generated rational phases.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.382. DOUBLE PATH INTERFERENCE RATIO THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 192. FINITE ABELIAN WARD DYSON THEOREM

Prove all-order Ward invariance and exact Dyson resummation in the declared finite transverse abelian class.

Executable root: `_u883_finite_abelian_ward_dyson_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.383 (192. FINITE ABELIAN WARD DYSON THEOREM)**.** *Prove all-order Ward invariance and exact Dyson resummation in the declared finite transverse abelian class.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.384. FINITE ABELIAN WARD DYSON THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 193. INTERNAL PHOTON CANDIDATE CORRESPONDENCE THEOREM

Select the unique internal open and closed balanced carrier states satisfying neutral two-mode propagation and localized detection conditions.

Executable root: `_u883_photon_candidate_correspondence_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.385 (193. INTERNAL PHOTON CANDIDATE CORRESPONDENCE THEOREM)**.** *Select the unique internal open and closed balanced carrier states satisfying neutral two-mode propagation and localized detection conditions.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.386. INTERNAL PHOTON CANDIDATE CORRESPONDENCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 194. PHOTON CANDIDATE EXTERNAL VALIDATION PACKET THEOREM

Seal all internally generated photon-candidate ratios and require independent signed evidence for external completion.

Executable root: `_u883_external_photon_validation_packet_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.387 (194. PHOTON CANDIDATE EXTERNAL VALIDATION PACKET THEOREM)**.** *Seal all internally generated photon-candidate ratios and require independent signed evidence for external completion.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.388. PHOTON CANDIDATE EXTERNAL VALIDATION PACKET THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 195. PHOTON MAXWELL BORN QED COMPLETION THEOREM

Integrate the complete internal photon-candidate chain from time and double zero to Maxwell, Born, interference, helicity, and finite abelian scattering.

Executable root: `_u883_photon_maxwell_born_qed_completion_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.389 (195. PHOTON MAXWELL BORN QED COMPLETION THEOREM)**.** *Integrate the complete internal photon-candidate chain from time and double zero to Maxwell, Born, interference, helicity, and finite abelian scattering.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.390. PHOTON MAXWELL BORN QED COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 196. RATIONAL LOCAL PHASE CONNECTION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u884_rational_local_phase_connection_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.391 (196. RATIONAL LOCAL PHASE CONNECTION THEOREM)**.** *Execute the exact certificate for RATIONAL LOCAL PHASE CONNECTION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.392. RATIONAL LOCAL PHASE CONNECTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 197. CUBICAL MAXWELL VARIATION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u884_cubical_maxwell_variation_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.393 (197. CUBICAL MAXWELL VARIATION THEOREM)**.** *Execute the exact certificate for CUBICAL MAXWELL VARIATION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.394. CUBICAL MAXWELL VARIATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 198. MASSLESS SPIN ONE FORMAL IDENTITY THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u884_massless_spin_one_formal_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.395 (198. MASSLESS SPIN ONE FORMAL IDENTITY THEOREM)**.** *Execute the exact certificate for MASSLESS SPIN ONE FORMAL IDENTITY THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.396*.* MASSLESS SPIN ONE FORMAL IDENTITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 199. RATIONAL BORN UNIQUENESS THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u884_rational_born_uniqueness_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.397 (199. RATIONAL BORN UNIQUENESS THEOREM)**.** *Execute the exact certificate for RATIONAL BORN UNIQUENESS THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.398*.* RATIONAL BORN UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 200. GENERAL RATIONAL TWO PATH PHASE THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u884_general_two_path_phase_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.399 (200. GENERAL RATIONAL TWO PATH PHASE THEOREM)**.** *Execute the exact certificate for GENERAL RATIONAL TWO PATH PHASE THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.400. GENERAL RATIONAL TWO PATH PHASE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 201. TWO HELICITY BOSONIC FOCK THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u884_two_helicity_bosonic_fock_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.401 (201. TWO HELICITY BOSONIC FOCK THEOREM)**.** *Execute the exact certificate for TWO HELICITY BOSONIC FOCK THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.402. TWO HELICITY BOSONIC FOCK THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 202. ALL POLYNOMIAL ORDER WARD THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u884_all_polynomial_order_ward_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.403 (202. ALL POLYNOMIAL ORDER WARD THEOREM)**.** *Execute the exact certificate for ALL POLYNOMIAL ORDER WARD THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.404. ALL POLYNOMIAL ORDER WARD THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 203. AFFINE MAXWELL CONTINUUM REFINEMENT THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u884_affine_maxwell_continuum_refinement_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.405 (203. AFFINE MAXWELL CONTINUUM REFINEMENT THEOREM)**.** *Execute the exact certificate for AFFINE MAXWELL CONTINUUM REFINEMENT THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.406. AFFINE MAXWELL CONTINUUM REFINEMENT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 204. PHOTON QED CONTINUUM COMPLETION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u884_photon_qed_continuum_completion_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.407 (204. PHOTON QED CONTINUUM COMPLETION THEOREM)**.** *Execute the exact certificate for PHOTON QED CONTINUUM COMPLETION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.408. PHOTON QED CONTINUUM COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 205. RATIONAL POLYNOMIAL MAXWELL COMPLEX THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u885_polynomial_maxwell_complex_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.409 (205. RATIONAL POLYNOMIAL MAXWELL COMPLEX THEOREM)**.** *Execute the exact certificate for RATIONAL POLYNOMIAL MAXWELL COMPLEX THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.410. RATIONAL POLYNOMIAL MAXWELL COMPLEX THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 206. POLYNOMIAL CONTINUUM PARTITION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u885_polynomial_continuum_partition_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.411 (206. POLYNOMIAL CONTINUUM PARTITION THEOREM)**.** *Execute the exact certificate for POLYNOMIAL CONTINUUM PARTITION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.412. POLYNOMIAL CONTINUUM PARTITION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 207. ALGEBRAIC TWO HELICITY FOCK DIRECT LIMIT THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u885_algebraic_fock_direct_limit_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.413 (207. ALGEBRAIC TWO HELICITY FOCK DIRECT LIMIT THEOREM)**.** *Execute the exact certificate for ALGEBRAIC TWO HELICITY FOCK DIRECT LIMIT THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.414. ALGEBRAIC TWO HELICITY FOCK DIRECT LIMIT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 208. RATIONAL FOREST RENORMALIZATION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u885_rational_forest_renormalization_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.415 (208. RATIONAL FOREST RENORMALIZATION THEOREM)**.** *Execute the exact certificate for RATIONAL FOREST RENORMALIZATION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.416. RATIONAL FOREST RENORMALIZATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 209. FORMAL ALL ORDER WARD RENORMALIZATION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u885_formal_all_order_ward_renormalized_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.417 (209. FORMAL ALL ORDER WARD RENORMALIZATION THEOREM)**.** *Execute the exact certificate for FORMAL ALL ORDER WARD RENORMALIZATION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.418. FORMAL ALL ORDER WARD RENORMALIZATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 210. RATIONAL UNITARY SCATTERING THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u885_rational_unitary_scattering_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.419 (210. RATIONAL UNITARY SCATTERING THEOREM)**.** *Execute the exact certificate for RATIONAL UNITARY SCATTERING THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.420. RATIONAL UNITARY SCATTERING THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 211. EXTERNAL RATIO REPLICATION BOUNDARY THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u885_external_ratio_replication_boundary_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.421 (211. EXTERNAL RATIO REPLICATION BOUNDARY THEOREM)**.** *Execute the exact certificate for EXTERNAL RATIO REPLICATION BOUNDARY THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.422. EXTERNAL RATIO REPLICATION BOUNDARY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 212. CONTINUUM QED BOUNDARY COMPLETION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u885_continuum_qed_boundary_completion_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.423 (212. CONTINUUM QED BOUNDARY COMPLETION THEOREM)**.** *Execute the exact certificate for CONTINUUM QED BOUNDARY COMPLETION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.424. CONTINUUM QED BOUNDARY COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 213. NATIVE BELL PROJECTOR THEOREM

The Bell joint law is calculated from the singlet density matrix and local projectors. The exact correlations give CHSH fourteen-fifths and local marginals one-half.

Executable root: `_u905_native_bell_projector_theorem`

Class and stage: Primary; 5. ELECTROMAGNETIC AND QUANTUM CONSEQUENCES

Theorem A.425 (213. NATIVE BELL PROJECTOR THEOREM)**.** *Execute the exact certificate for NATIVE BELL PROJECTOR THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.426. NATIVE BELL PROJECTOR THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 214. NATIVE PROJECTION READOUT GENERATION THEOREM

The cone is one example only. The theorem concerns arbitrary native states and independently specified projection states.

Executable root: `_nsp_native_projection_readout_generation_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.427 (214. NATIVE PROJECTION READOUT GENERATION THEOREM)**.** *Prove the general native-state, projection-state, and readout hierarchy without fixing any native shape.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.428. NATIVE PROJECTION READOUT GENERATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 215. NATIVE PROJECTION DYNAMICS PRODUCT RULE THEOREM

A change of readout does not by itself identify whether the carrier moved, the projection changed, or both changed together.

Executable root: `_nsp_native_projection_dynamics_product_rule_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.429 (215. NATIVE PROJECTION DYNAMICS PRODUCT RULE THEOREM)**.** *Decompose one readout change into native motion, projection motion, and their discrete interaction term.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.430. NATIVE PROJECTION DYNAMICS PRODUCT RULE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 216. PROJECTION SYNCHRONIZATION AND LIGHT ROLE THEOREM

A physical light hypothesis must distinguish native-object motion, projection-channel motion, their product-rule interaction, and the universal time-to-space conversion ratio. The declared electromagnetic mode can saturate the generated causal boundary, but empirical light is not identified before an external decoder.

Executable root: `_e897_projection_synchronization_light_role_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.431 (216. PROJECTION SYNCHRONIZATION AND LIGHT ROLE THEOREM)**.** *Classify native-motion, projection-motion, coupled, and time-space-decoder light hypotheses under exact product-rule synchronization.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.432. PROJECTION SYNCHRONIZATION AND LIGHT ROLE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 217. FOUR FORCE PROJECTION FRAME THEOREM

Prove that four native force readouts form a complete orthogonal projection frame while every proper subset loses latent information.

Executable root: `_u876_four_force_projection_frame_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.433 (217. FOUR FORCE PROJECTION FRAME THEOREM)**.** *Prove that four native force readouts form a complete orthogonal projection frame while every proper subset loses latent information.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.434. FOUR FORCE PROJECTION FRAME THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 218. NATIVE PROJECTION FOUR FORCE BRIDGE THEOREM

The four force equations are treated as complementary effective projections of one parent dynamics, not as four independent native objects and not as direct native-value identities.

Executable root: `_nsp_native_projection_four_force_bridge_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.435 (218. NATIVE PROJECTION FOUR FORCE BRIDGE THEOREM)**.** *Connect one native parent dynamics to four readout channels without identifying any readout with the complete native carrier.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.436. NATIVE PROJECTION FOUR FORCE BRIDGE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 219. LAYER ROLE TRANSMUTATION THEOREM

The theorem does not assert that force must become matter. It proves a conditional layer effect: a resolved observer sees internal motion, whereas a coarse observer can identify the same closed periodic carrier as one persistent object, source, or distributed field readout.

Executable root: `_layer_role_transmutation_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.437 (219. LAYER ROLE TRANSMUTATION THEOREM)**.** *Prove that one closed lower-layer dynamics can be read as flow internally and as a persistent object or source after coarse graining.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.438. LAYER ROLE TRANSMUTATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 220. MULTISCALE FORCE MODE THEOREM

The same ternary transition eigenmodes that generate local readouts define a tensor-product scale operator on all eighteen characters. Each generated force sector receives an exact spectrum, while layer-role interpretation remains a separate projection question.

Executable root: `_multiscale_force_mode_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.439 (220. MULTISCALE FORCE MODE THEOREM)**.** *Diagonalize one layer-renormalization operator on the generated eighteen-character force decomposition.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.440. MULTISCALE FORCE MODE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 221. EFFECTIVE SCALE LIMIT THEOREM

The large-scale world is not obtained by discarding the native theory arbitrarily. Repeated coarse graining preserves the common carrier and the two-step electromagnetic causal mode while every gravity, strong, and weak microscopic scale eigenmode contracts in magnitude inside the declared scale operator.

Executable root: `_e898_effective_scale_limit_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.441 (221. EFFECTIVE SCALE LIMIT THEOREM)**.** *Prove exact finite-depth contraction of nonpersistent residual modes and stability of the effective quotient laws.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.442. EFFECTIVE SCALE LIMIT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 222. NATIVE EFFECTIVE PUSHFORWARD THEOREM

The effective equation is not inserted from the observed layer. One native update acts on the complete eighteen-character carrier, and each effective sector law is obtained by projecting that update. Exact intertwining proves that the projected law and the native-first calculation give the same result.

Executable root: `_e895_native_effective_pushforward_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.443 (222. NATIVE EFFECTIVE PUSHFORWARD THEOREM)**.** *Prove that every generated sector evolution is the exact pushforward of one native eighteen-character scale dynamics through its sector projection.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.444. NATIVE EFFECTIVE PUSHFORWARD THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 223. EFFECTIVE QUOTIENT DYNAMICS THEOREM

An effective law can be exact because native evolution descends to equivalence classes modulo the projection kernel. Distinct native states in the same class produce the same complete effective trajectory whenever the native update preserves that kernel. Accuracy of the effective equation therefore does not by itself identify a unique native origin.

Executable root: `_e896_effective_quotient_dynamics_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.445 (223. EFFECTIVE QUOTIENT DYNAMICS THEOREM)**.** *Prove that every generated effective sector is a quotient dynamics on native equivalence classes and that exact effective prediction does not imply unique native reconstruction.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.446. EFFECTIVE QUOTIENT DYNAMICS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 224. UNIVERSAL ROLE PREDICATE THEOREM

Force, energy, matter, particle, field, and gravity-like source are separated by explicit closure and observation predicates. They are not hard-coded as an irreversible sequence. One carrier may satisfy several roles at one resolution and a different subset after coarse graining.

Executable root: `_e896_universal_role_predicate_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.447 (224. UNIVERSAL ROLE PREDICATE THEOREM)**.** *Exhaust the declared closure-resolution predicates and classify when one carrier is read as force, energy, matter, particle, field, or gravity-like source without imposing a mandatory role ladder.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.448. UNIVERSAL ROLE PREDICATE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 225. DYNAMICAL FIBER CLOSURE THEOREM

A projected equation is an exact autonomous law if and only if native evolution preserves every fiber of its projection. Equivalently the projection kernel is invariant and an effective operator intertwines the native update. When this fails, no equation using only the current projected state can reproduce every native trajectory.

Executable root: `_dynamical_fiber_closure_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.449 (225. DYNAMICAL FIBER CLOSURE THEOREM)**.** *Prove the exact necessary-and-sufficient finite linear fiber-closure rule and apply it to all generated force sectors.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.450. DYNAMICAL FIBER CLOSURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 226. NONLINEAR POLYNOMIAL FIBER CLOSURE THEOREM

The common closure rule survives nonlinear native motion. The sum and hidden-coordinate readouts factor exactly through one two-coordinate effective polynomial map even though the unresolved difference coordinate receives a quadratic hidden contribution and is not autonomous.

Executable root: `_nonlinear_polynomial_fiber_closure_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.451 (226. NONLINEAR POLYNOMIAL FIBER CLOSURE THEOREM)**.** *Prove exact nonlinear polynomial factorization through a many-to-one projection and exhibit a nonclosed readout.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.452. NONLINEAR POLYNOMIAL FIBER CLOSURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 227. CONTINUOUS TIME INTERTWINING SEMIGROUP THEOREM

A continuous effective law is generated rather than guessed. Because the native generator is nilpotent, its exact flow is a finite quadratic polynomial in time. Generator intertwining then proves projection-after-flow equals effective-flow-after-projection and the semigroup composition law holds exactly.

Executable root: `_continuous_time_intertwining_semigroup_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.453 (227. CONTINUOUS TIME INTERTWINING SEMIGROUP THEOREM)**.** *Construct an exact rational continuous-time semigroup whose native and effective flows intertwine at every rational time.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.454. CONTINUOUS TIME INTERTWINING SEMIGROUP THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 228. FINITE MEMORY KERNEL ELIMINATION THEOREM

When the visible variables are not closed, the correct effective equation is not guessed. The hidden native coordinates are eliminated exactly. The projected update becomes an instantaneous visible term plus memory kernels acting on past visible states plus a hidden-force term determined by the initial unresolved state.

Executable root: `_finite_memory_kernel_elimination_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.455 (228. FINITE MEMORY KERNEL ELIMINATION THEOREM)**.** *Eliminate hidden native coordinates exactly and recover the projected trajectory through finite memory kernels.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.456. FINITE MEMORY KERNEL ELIMINATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 229. MINIMAL OBSERVABLE COMPLETION THEOREM

The effective variables are not selected by resemblance to a standard equation. Their row space is enlarged only when native evolution creates a new independent observable. The first invariant span is the unique minimal linear completion of the proposed readout algebra.

Executable root: `_minimal_observable_completion_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.457 (229. MINIMAL OBSERVABLE COMPLETION THEOREM)**.** *Construct the minimal generator-invariant observable completion and prove minimality by exact rank growth.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.458. MINIMAL OBSERVABLE COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 230. NONLINEAR OBSERVABLE ALGEBRA COMPLETION THEOREM

The correct nonlinear effective variables are not selected by resemblance to textbook equations. Starting from constant, sum, hidden coordinate, and difference readouts, repeated native pullback forces a seven-dimensional degree-two observable algebra and no proper generated prefix containing the declared initial observables is invariant.

Executable root: `_nonlinear_observable_algebra_completion_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.459 (230. NONLINEAR OBSERVABLE ALGEBRA COMPLETION THEOREM)**.** *Generate the minimal finite polynomial observable algebra closed under native pullback.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.460*.* NONLINEAR OBSERVABLE ALGEBRA COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 231. JS SH FORWARD REVERSE DYNAMICAL THEOREM

The JS-SH forward equation and reverse equation are not two unrelated formulas. One native source evolves, several individually noninjective shell readouts are produced, their compatible family reconstructs the source, and the inverse native update recovers the complete earlier trajectory.

Executable root: `_js_sh_forward_reverse_dynamical_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.461 (231. JS SH FORWARD REVERSE DYNAMICAL THEOREM)**.** *Join JS-shell closure, multi-projection reconstruction, forward evolution, and exact reverse source recovery.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.462*.* JS SH FORWARD REVERSE DYNAMICAL THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 232. JS SH ADJOINT RECURRENCE THEOREM

The forward shell wave, its reverse equation, and measurement reconstruction are one object. The exact rational orthogonal update preserves norm, every component satisfies the same determinant-one second-order recurrence, the transpose reconstructs the full past, and two rank-one shell readouts jointly recover each state.

Executable root: `_js_sh_adjoint_recurrence_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.463 (232. JS SH ADJOINT RECURRENCE THEOREM)**.** *Derive a norm-preserving JS-SH forward recurrence, its exact adjoint reverse rule, and complete two-readout reconstruction.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.464. JS SH ADJOINT RECURRENCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 233. FOUR FORCE OBSERVABLE ALGEBRA THEOREM

Gravity, electromagnetic, weak, and strong laws are four exact observable algebras of one seventeen-direction residual parent. Their autonomous equations arise because their projection kernels are preserved. An arbitrary mixed partial readout need not be autonomous and is then governed by the exact memory law until its observable algebra is completed.

Executable root: `_force_sector_observable_algebra_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.465 (233. FOUR FORCE OBSERVABLE ALGEBRA THEOREM)**.** *Classify the generated force projections as exact closed algebras and show how incomplete mixed readouts acquire memory.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.466. FOUR FORCE OBSERVABLE ALGEBRA THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 234. FOUR FORCE NONLINEAR PARENT INTERACTION THEOREM

The four effective forces need not be four unrelated native laws. One common-carrier polynomial update leaves electromagnetic and strong source readouts autonomous, while gravity-like and weak-like responses become exact autonomous algebras only after their common and source variables are retained.

Executable root: `_four_force_nonlinear_parent_interaction_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.467 (234. FOUR FORCE NONLINEAR PARENT INTERACTION THEOREM)**.** *Construct one nonlinear parent polynomial map with exact completed force-sector observable algebras.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.468. FOUR FORCE NONLINEAR PARENT INTERACTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 235. FORWARD EFFECTIVE RATIO DECODER THEOREM

Native variables are never matched directly to standard measured values. A native state and a projection state first generate a readout; only a sealed dimensionless readout ratio reaches the comparison boundary.

Executable root: `_nsp_forward_effective_ratio_decoder_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.469 (235. FORWARD EFFECTIVE RATIO DECODER THEOREM)**.** *Prove that comparison with effective physics is forward, projection-mediated, and ratio-only before any external scale is supplied.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.470. FORWARD EFFECTIVE RATIO DECODER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 236. FUNDAMENTAL OBSERVER DYNAMICS SEPARATION THEOREM

Separate intrinsic carrier motion from observer-frame motion and prove that the observable transfer is their relative dynamics rather than either motion alone.

Executable root: `_u876_fundamental_observer_dynamics_separation_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.471 (236. FUNDAMENTAL OBSERVER DYNAMICS SEPARATION THEOREM)**.** *Separate intrinsic carrier motion from observer-frame motion and prove that the observable transfer is their relative dynamics rather than either motion alone.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.472. FUNDAMENTAL OBSERVER DYNAMICS SEPARATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 237. LATENT MASS INERTIA RECONSTRUCTION THEOREM

Prove that latent quadratic mass and inertia are preserved by intrinsic motion, cannot be inferred from an isolated projection, and are reconstructed by the complete force frame with observer calibration.

Executable root: `_u876_latent_mass_inertia_reconstruction_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.473 (237. LATENT MASS INERTIA RECONSTRUCTION THEOREM)**.** *Prove that latent quadratic mass and inertia are preserved by intrinsic motion, cannot be inferred from an isolated projection, and are reconstructed by the complete force frame with observer calibration.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.474. LATENT MASS INERTIA RECONSTRUCTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 238. FULL FRAME OPERATOR EQUIVALENCE THEOREM

Transport latent dynamics, action, unitarity, and a positive spectral gap through the complete four-force frame without inserting an observable equation independently.

Executable root: `_u876_full_frame_operator_equivalence_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.475 (238. FULL FRAME OPERATOR EQUIVALENCE THEOREM)**.** *Transport latent dynamics, action, unitarity, and a positive spectral gap through the complete four-force frame without inserting an observable equation independently.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.476. FULL FRAME OPERATOR EQUIVALENCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 239. FUNDAMENTAL DYNAMICS TOMOGRAPHY THEOREM

Reconstruct the complete latent relative operator and the calibrated intrinsic operator from four independent projected basis evolutions.

Executable root: `_u876_fundamental_dynamics_tomography_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.477 (239. FUNDAMENTAL DYNAMICS TOMOGRAPHY THEOREM)**.** *Reconstruct the complete latent relative operator and the calibrated intrinsic operator from four independent projected basis evolutions.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.478. FUNDAMENTAL DYNAMICS TOMOGRAPHY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 240. PROJECTION ORIGIN FOUR FORCE COMPLETION THEOREM

Integrate latent motion, observer motion, four-force tomography, invariant reconstruction, operator transport, and bidirectional dynamics reconstruction into one dependency-ordered projection-origin theorem.

Executable root: `_u876_projection_origin_four_force_completion_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.479 (240. PROJECTION ORIGIN FOUR FORCE COMPLETION THEOREM)**.** *Integrate latent motion, observer motion, four-force tomography, invariant reconstruction, operator transport, and bidirectional dynamics reconstruction into one dependency-ordered projection-origin theorem.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.480. PROJECTION ORIGIN FOUR FORCE COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 241. MINIMAL LATENT CHARACTER FRAME THEOREM

Derive the four-force projection frame as the complete character table of the primitive two-bit relation group and prove minimal latent dimension four.

Executable root: `_u877_minimal_latent_character_frame_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.481 (241. MINIMAL LATENT CHARACTER FRAME THEOREM)**.** *Derive the four-force projection frame as the complete character table of the primitive two-bit relation group and prove minimal latent dimension four.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.482. MINIMAL LATENT CHARACTER FRAME THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 242. NATIVE OBSERVER AXIS DERIVATION THEOREM

Derive the observer frame from a native positive anchor with a simple ordered spectrum instead of treating observer motion as an external calibration.

Executable root: `_u877_native_observer_axis_derivation_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.483 (242. NATIVE OBSERVER AXIS DERIVATION THEOREM)**.** *Derive the observer frame from a native positive anchor with a simple ordered spectrum instead of treating observer motion as an external calibration.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.484. NATIVE OBSERVER AXIS DERIVATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 243. FOUR FORCE CARRIER SECTOR EQUIVALENCE THEOREM

Join gravity, electromagnetism, weak interaction, and strong interaction to spin-two, U1, SU2, and SU3 carriers in the physically ordered dictionary.

Executable root: `_u877_force_carrier_sector_equivalence_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.485 (243. FOUR FORCE CARRIER SECTOR EQUIVALENCE THEOREM)**.** *Join gravity, electromagnetism, weak interaction, and strong interaction to spin-two, U1, SU2, and SU3 carriers in the physically ordered dictionary.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.486. FOUR FORCE CARRIER SECTOR EQUIVALENCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 244. LATENT MASS SOURCE PROJECTION THEOREM

Generate mass, inertia, and source tensors in the latent layer and prove exact reconstruction from the complete four-force projection layer.

Executable root: `_u877_latent_mass_source_projection_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.487 (244. LATENT MASS SOURCE PROJECTION THEOREM)**.** *Generate mass, inertia, and source tensors in the latent layer and prove exact reconstruction from the complete four-force projection layer.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.488. LATENT MASS SOURCE PROJECTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 245. PROJECTION GENERATED OBSERVABLE ACTION THEOREM

Prove that observable force dynamics and its action arise by projection from intrinsic motion and the internally derived observer frame.

Executable root: `_u877_projection_generated_action_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.489 (245. PROJECTION GENERATED OBSERVABLE ACTION THEOREM)**.** *Prove that observable force dynamics and its action arise by projection from intrinsic motion and the internally derived observer frame.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.490. PROJECTION GENERATED OBSERVABLE ACTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 246. ORIGIN PROJECTION REALITY COMPLETION THEOREM

Integrate the minimal latent carrier, internally derived observer frame, exact force sectors, mass-source reconstruction, and projection-generated action into one ordered theorem.

Executable root: `_u877_origin_projection_reality_completion_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.491 (246. ORIGIN PROJECTION REALITY COMPLETION THEOREM)**.** *Integrate the minimal latent carrier, internally derived observer frame, exact force sectors, mass-source reconstruction, and projection-generated action into one ordered theorem.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.492. ORIGIN PROJECTION REALITY COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 247. ORIGIN SHADOW SCALE SEMIGROUP THEOREM

Derive a continuous multiplicative rational-scale semigroup in the latent layer and prove exact generation and inversion of its four observable shadows.

Executable root: `_u878_origin_shadow_scale_semigroup_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.493 (247. ORIGIN SHADOW SCALE SEMIGROUP THEOREM)**.** *Derive a continuous multiplicative rational-scale semigroup in the latent layer and prove exact generation and inversion of its four observable shadows.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.494. ORIGIN SHADOW SCALE SEMIGROUP THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 248. ALL ORDER SHADOW AMPLITUDE RECURRENCE THEOREM

Prove that every integer-order observable amplitude is uniquely generated by a finite exact recurrence fixed by native poles and residues.

Executable root: `_u878_all_order_shadow_amplitude_recurrence_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.495 (248. ALL ORDER SHADOW AMPLITUDE RECURRENCE THEOREM)**.** *Prove that every integer-order observable amplitude is uniquely generated by a finite exact recurrence fixed by native poles and residues.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.496. ALL ORDER SHADOW AMPLITUDE RECURRENCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 249. OBSERVER APPARATUS COUPLING THEOREM

Derive the observer apparatus as the unique ordered, sign-fixed, orientation-preserving zero-action coupling between the native anchor and its primitive level operator.

Executable root: `_u878_observer_apparatus_coupling_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.497 (249. OBSERVER APPARATUS COUPLING THEOREM)**.** *Derive the observer apparatus as the unique ordered, sign-fixed, orientation-preserving zero-action coupling between the native anchor and its primitive level operator.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.498. OBSERVER APPARATUS COUPLING THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 250. PREDECLARED REALITY RATIO PACKET THEOREM

Generate and seal a dimensionless mass, mixing, sector, scattering, strong-gap, and gravity-ratio packet before any external comparison.

Executable root: `_u878_predeclared_reality_ratio_packet_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.499 (250. PREDECLARED REALITY RATIO PACKET THEOREM)**.** *Generate and seal a dimensionless mass, mixing, sector, scattering, strong-gap, and gravity-ratio packet before any external comparison.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.500. PREDECLARED REALITY RATIO PACKET THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 251. STRONG GAP PROJECTION TRANSFER THEOREM

Transfer the native gauge-invariant SU3 center gap through the physical strong-sector projector and prove that orthogonal projection preserves the declared color gap exactly.

Executable root: `_u878_strong_gap_projection_transfer_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.501 (251. STRONG GAP PROJECTION TRANSFER THEOREM)**.** *Transfer the native gauge-invariant SU3 center gap through the physical strong-sector projector and prove that orthogonal projection preserves the declared color gap exactly.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.502. STRONG GAP PROJECTION TRANSFER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 252. SMOOTH LORENTZIAN PULLBACK COVARIANCE THEOREM

Prove exact Lorentzian pullback covariance and composition for a nonlinear polynomial coframe under a rational affine atlas, with an exact two-polarization and null-constraint carrier.

Executable root: `_u878_smooth_lorentzian_pullback_covariance_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.503 (252. SMOOTH LORENTZIAN PULLBACK COVARIANCE THEOREM)**.** *Prove exact Lorentzian pullback covariance and composition for a nonlinear polynomial coframe under a rational affine atlas, with an exact two-polarization and null-constraint carrier.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.504. SMOOTH LORENTZIAN PULLBACK COVARIANCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 253. ORIGIN SHADOW REALITY FRONTIER THEOREM

Integrate the physical carrier dictionary and six origin-to-reality theorems, closing every internal prerequisite while preserving the exact boundary to unrestricted physics and external validation.

Executable root: `_u878_origin_shadow_reality_frontier_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.505 (253. ORIGIN SHADOW REALITY FRONTIER THEOREM)**.** *Integrate the physical carrier dictionary and six origin-to-reality theorems, closing every internal prerequisite while preserving the exact boundary to unrestricted physics and external validation.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.506. ORIGIN SHADOW REALITY FRONTIER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 254. COMPLETE REALITY IDENTIFIABILITY THEOREM

Prove unique latent-frame identification inside the complete signed-permutation competitor class from the native anchor, ordered apparatus rule, force frame, and mass operator.

Executable root: `_u879_complete_reality_identifiability_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.507 (254. COMPLETE REALITY IDENTIFIABILITY THEOREM)**.** *Prove unique latent-frame identification inside the complete signed-permutation competitor class from the native anchor, ordered apparatus rule, force frame, and mass operator.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.508. COMPLETE REALITY IDENTIFIABILITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 255. ALL ORDER SOURCE MOMENT THEOREM

Generate every integer-order source moment from four native poles and prove exact projection covariance and unique recurrence without order-by-order parameter insertion.

Executable root: `_u879_all_order_source_moment_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.509 (255. ALL ORDER SOURCE MOMENT THEOREM)**.** *Generate every integer-order source moment from four native poles and prove exact projection covariance and unique recurrence without order-by-order parameter insertion.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.510. ALL ORDER SOURCE MOMENT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 256. EXACT POLE AMPUTATION THEOREM

Prove exact pole attachment, observable projection, bilateral amputation, and latent-vertex reconstruction for every declared channel pair.

Executable root: `_u879_exact_pole_amputation_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.511 (256. EXACT POLE AMPUTATION THEOREM)**.** *Prove exact pole attachment, observable projection, bilateral amputation, and latent-vertex reconstruction for every declared channel pair.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.512. EXACT POLE AMPUTATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 257. REFLECTION POSITIVE TRANSFER RECONSTRUCTION THEOREM

Construct a positive transfer carrier, prove reflection-form positivity, unique vacuum separation, and exact Hilbert reconstruction in the declared finite spectral class.

Executable root: `_u879_reflection_positive_transfer_reconstruction_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.513 (257. REFLECTION POSITIVE TRANSFER RECONSTRUCTION THEOREM)**.** *Construct a positive transfer carrier, prove reflection-form positivity, unique vacuum separation, and exact Hilbert reconstruction in the declared finite spectral class.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.514. REFLECTION POSITIVE TRANSFER RECONSTRUCTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 258. SU3 PROJECTIVE REFINEMENT GAP THEOREM

Prove projective consistency and a scale-independent positive SU3 center-loop gap under arbitrary declared edge subdivision.

Executable root: `_u879_su3_projective_refinement_gap_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.515 (258. SU3 PROJECTIVE REFINEMENT GAP THEOREM)**.** *Prove projective consistency and a scale-independent positive SU3 center-loop gap under arbitrary declared edge subdivision.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.516. SU3 PROJECTIVE REFINEMENT GAP THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 259. LORENTZIAN ACTION INVERSE LIMIT THEOREM

Prove exact affine change-of-variable covariance and convergence of a two-polarization Lorentzian action along the declared projective field refinement.

Executable root: `_u879_lorentzian_action_inverse_limit_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.517 (259. LORENTZIAN ACTION INVERSE LIMIT THEOREM)**.** *Prove exact affine change-of-variable covariance and convergence of a two-polarization Lorentzian action along the declared projective field refinement.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.518. LORENTZIAN ACTION INVERSE LIMIT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 260. EXTERNAL REPLICATION CERTIFICATE THEOREM

Generate one immutable ratio certificate and prove deterministic verification rules that cannot report external success without an independently supplied signed result.

Executable root: `_u879_external_replication_certificate_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.519 (260. EXTERNAL REPLICATION CERTIFICATE THEOREM)**.** *Generate one immutable ratio certificate and prove deterministic verification rules that cannot report external success without an independently supplied signed result.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.520. EXTERNAL REPLICATION CERTIFICATE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 261. ORIGIN REALITY CONTINUUM COMPLETION THEOREM

Integrate exact latent identifiability, all-order moments, pole amputation, positive transfer reconstruction, projective SU3 gap, Lorentzian action convergence, and fail-closed external certification.

Executable root: `_u879_origin_reality_continuum_completion_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.521 (261. ORIGIN REALITY CONTINUUM COMPLETION THEOREM)**.** *Integrate exact latent identifiability, all-order moments, pole amputation, positive transfer reconstruction, projective SU3 gap, Lorentzian action convergence, and fail-closed external certification.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.522. ORIGIN REALITY CONTINUUM COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 262. SOURCE PROJECTION RECONSTRUCTION THEOREM

The complete tensor monomial family on the eighteen-state ledger has an exact inverse evaluation matrix. Every source state, field, probability measure, and finite observable is therefore reconstructed from its compatible projection moments.

Executable root: `theorem_source_projection_reconstruction`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.523 (262. SOURCE PROJECTION RECONSTRUCTION THEOREM)**.** *Prove that a complete family of relation projections separates and reconstructs every function and measure on the eighteen-state source ledger.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.524. SOURCE PROJECTION RECONSTRUCTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 263. SOURCE CHARACTER PARSEVAL COMPLETION THEOREM

The eighteen signed-occupancy states carry an explicit tensor character basis whose Gram matrix is positive and diagonal. Exact Parseval reconstruction at one seat propagates to every finite depth by tensor induction and then extends uniquely to the Cauchy completion.

Executable root: `theorem_source_character_parseval_completion`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.525 (263. SOURCE CHARACTER PARSEVAL COMPLETION THEOREM)**.** *Prove an exact rational Fourier-Parseval transform on the eighteen-state source and its arbitrary-depth tensor completion.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.526. SOURCE CHARACTER PARSEVAL COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 264. INDUCTIVE LIMIT SOURCE SHADOW ISOMORPHISM THEOREM

Refinement repeats a parent field over all children while conditional expectation averages the children back to the parent. Because products and norms are preserved, the compatible finite transforms define one global source-shadow algebra isomorphism.

Executable root: `theorem_inductive_limit_source_shadow_isomorphism`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.527 (264. INDUCTIVE LIMIT SOURCE SHADOW ISOMORPHISM THEOREM)**.** *Prove compatible embeddings, conditional expectations, algebra products, and a unique isometric source-shadow map on the inductive-limit completion.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.528. INDUCTIVE LIMIT SOURCE SHADOW ISOMORPHISM THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 265. SOURCE SHADOW DYNAMICAL EQUIVALENCE THEOREM

Two shadow observers may write different evolution matrices while both remain exact conjugate images of one source operator. The intertwiner preserves complete trajectories and spectral invariants, so physical equivalence does not require identical equations inside each shadow.

Executable root: `theorem_source_shadow_dynamical_equivalence`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.529 (265. SOURCE SHADOW DYNAMICAL EQUIVALENCE THEOREM)**.** *Show that distinct shadow equations can encode one source dynamics through exact intertwining, without requiring identical shadow matrices.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.530. SOURCE SHADOW DYNAMICAL EQUIVALENCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 266. PROJECTIVE SHADOW ANALYTIC UNIQUENESS THEOREM

The general projective coefficients are solved before any numerical scale is sampled. Endpoint preservation and odds scaling leave one normalized coefficient ray for every positive rational scale.

Executable root: `theorem_projective_shadow_analytic_uniqueness`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.531 (266. PROJECTIVE SHADOW ANALYTIC UNIQUENESS THEOREM)**.** *Solve the endpoint and odds coefficient equations for every positive rational projective scale.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.532. PROJECTIVE SHADOW ANALYTIC UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 267. BOUNDED SHADOW MOBIUS SEMIGROUP THEOREM

The endpoint-fixing equation T_a(x)=x/[a+(1-a)x] maps the unit interval to itself and fixes both limiting shadow sizes. Its composition multiplies source scales, its inverse uses the reciprocal scale, and its odds coordinate reproduces the source ratio exactly.

Executable root: `theorem_bounded_shadow_mobius_semigroup`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.533 (267. BOUNDED SHADOW MOBIUS SEMIGROUP THEOREM)**.** *Prove analytic uniqueness, endpoint preservation, composition, inverse and odds scaling.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.534. BOUNDED SHADOW MOBIUS SEMIGROUP THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 268. BOUNDARY ADDITIVE DEPTH DECODER THEOREM

The bounded coordinate contains an exact discrete depth because each native step multiplies its odds by the same primitive factor. Zero at the balanced state and one unit per native step uniquely recover the integer depth without evaluating a logarithm.

Executable root: `theorem_boundary_additive_depth_decoder`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.535 (268. BOUNDARY ADDITIVE DEPTH DECODER THEOREM)**.** *Prove the unique additive depth coordinate on the exact discrete bounded-shadow orbit without evaluating a transcendental function.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.536. BOUNDARY ADDITIVE DEPTH DECODER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 269. AXIS PROJECTION CONJUGACY THEOREM

Retaining only the depth axis shows straight translation, whereas retaining the nonlinear bounded axis shows a curved saturating orbit. The apparent laws differ because the readout axis differs; they remain conjugate representations of one source step.

Executable root: `theorem_axis_projection_conjugacy`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.537 (269. AXIS PROJECTION CONJUGACY THEOREM)**.** *Prove that one source step appears as linear translation, bounded curvature, mirrored saturation, or multiplicative scaling according to the retained readout axis.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.538. AXIS PROJECTION CONJUGACY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 270. PROJECTIVE DEPTH ONE FORM INVARIANCE THEOREM

The projective shadow coordinate changes nonlinearly, but the depth one-form dx divided by x times one minus x is exactly unchanged by every positive endpoint-fixing compression. This supplies the logarithmic-coordinate structure without evaluating a logarithm or importing a continuum scale.

Executable root: `theorem_projective_depth_one_form_invariance`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.539 (270. PROJECTIVE DEPTH ONE FORM INVARIANCE THEOREM)**.** *Prove that the canonical depth one-form is invariant under every positive projective boundary compression without evaluating a logarithm.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.540. PROJECTIVE DEPTH ONE FORM INVARIANCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 271. OBSERVATION AXIS CONJUGATE MASTER THEOREM

One source depth is read simultaneously as bounded position, multiplicative length, area, volume and native state count. Different displayed laws therefore arise from retaining different nonlinear readout channels, not from inserting unrelated dynamics.

Executable root: `theorem_observation_axis_conjugate_master`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.541 (271. OBSERVATION AXIS CONJUGATE MASTER THEOREM)**.** *Prove that length, area, volume, bounded position and state count are simultaneous readouts of one source-depth orbit.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.542*.* OBSERVATION AXIS CONJUGATE MASTER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 272. POLYNOMIAL LIE CLOSURE CRITERION THEOREM

A continuous projected law is exact when the Lie derivative of every chosen readout factors through the same projection. The total-and-hidden projection closes exactly, while the unresolved difference readout has equal current values with unequal derivatives and therefore cannot define an autonomous continuous equation.

Executable root: `_polynomial_lie_closure_criterion_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.543 (272. POLYNOMIAL LIE CLOSURE CRITERION THEOREM)**.** *Prove the exact polynomial Lie-closure criterion for a nonlinear continuous native field and one many-to-one projection.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.544*.* POLYNOMIAL LIE CLOSURE CRITERION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 273. NONLINEAR CONTINUOUS FLOW SEMIGROUP THEOREM

The nonlinear native field admits an exact cubic rational flow. The total readout is conserved, the hidden readout advances linearly, the transfer between the two visible coordinates is the exact integral of the squared hidden coordinate, and the complete flow intertwines with the effective world for every tested rational state and time.

Executable root: `_nonlinear_continuous_flow_semigroup_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.545 (273. NONLINEAR CONTINUOUS FLOW SEMIGROUP THEOREM)**.** *Construct and verify the exact global rational flow of the generated nonlinear polynomial field.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.546. NONLINEAR CONTINUOUS FLOW SEMIGROUP THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 274. NONLINEAR FINITE MEMORY COMPLETION THEOREM

A nonlinear shadow that is not autonomous still has an exact effective law. Eliminating the two hidden shift seats gives current visible value plus the square of the value two steps earlier, with the first two updates supplied by the initial hidden state.

Executable root: `_nonlinear_finite_memory_completion_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.547 (274. NONLINEAR FINITE MEMORY COMPLETION THEOREM)**.** *Eliminate a nonlinear hidden shift register and derive its exact finite-memory visible recurrence.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.548. NONLINEAR FINITE MEMORY COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 275. JS SH PROJECTIVE UNITARY FLOW THEOREM

The JS-SH carrier has an exact rational projective clock. Every clock parameter gives a norm-preserving forward map, the negative parameter gives the reverse map, clock parameters compose by one fractional law, and complementary total and difference shell readouts reconstruct the evolved carrier.

Executable root: `_js_sh_projective_unitary_flow_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.549 (275. JS SH PROJECTIVE UNITARY FLOW THEOREM)**.** *Construct a rational projective JS-SH unitary family with exact forward, inverse, composition, recurrence, and reconstruction laws.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.550. JS SH PROJECTIVE UNITARY FLOW THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 276. BOUNDED JET CLOSURE DECIDABILITY THEOREM

Local smooth closure is not inferred from sampled trajectories. The complete order-six jet is inspected coefficient by coefficient. Terms supported only on the two projected coordinates generate an autonomous effective jet, while every nonzero coefficient containing the hidden coordinate is an explicit obstruction.

Executable root: `_bounded_jet_closure_decidability_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.551 (276. BOUNDED JET CLOSURE DECIDABILITY THEOREM)**.** *Decide order-six local smooth closure coefficientwise and exhibit one exact hidden-jet obstruction.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.552. BOUNDED JET CLOSURE DECIDABILITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 277. TRIANGULAR POLYNOMIAL FLOW FAMILY THEOREM

The earlier cubic example belongs to a complete rational triangular family. For any transfer polynomial through degree five, a finite binomial integral gives the exact global flow, its inverse, its differential equation, and its semigroup law.

Executable root: `_triangular_polynomial_flow_family_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.553 (277. TRIANGULAR POLYNOMIAL FLOW FAMILY THEOREM)**.** *Prove the exact semigroup and differential law for every rational transfer polynomial through degree five.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.554. TRIANGULAR POLYNOMIAL FLOW FAMILY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 278. ARBITRARY FINITE MEMORY DEPTH THEOREM

Every finite memory depth has one explicit native realization. A depth-m hidden shift register yields next visible equals current visible plus a declared function of the value m steps earlier. Two states differing only in the oldest hidden seat prove that no shorter current-state description can be exact.

Executable root: `_arbitrary_finite_memory_depth_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.555 (278. ARBITRARY FINITE MEMORY DEPTH THEOREM)**.** *Construct exact nonlinear delay laws for hidden shift registers of every declared depth and prove depth minimality.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.556. ARBITRARY FINITE MEMORY DEPTH THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 279. JS SH CONTINUOUS GENERATOR THEOREM

The projective JS-SH family is not only a collection of norm-preserving matrices. It solves an exact first-order equation whose skew generator is multiplied by one rational clock rate. Norm preservation, identity, and reverse motion follow without trigonometric evaluation.

Executable root: `_js_sh_continuous_generator_compatibility_theorem`

Class and stage: Primary; 6. NATIVE PROJECTION AND EFFECTIVE DYNAMICS

Theorem A.557 (279. JS SH CONTINUOUS GENERATOR THEOREM)**.** *Derive the exact rational differential generator of the JS-SH projective unitary family.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.558. JS SH CONTINUOUS GENERATOR THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 280. FINITE COVARIANT TWENTY TWO TARGET CLOSURE THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_finite_covariant_twenty_two_target_closure_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.559 (280. FINITE COVARIANT TWENTY TWO TARGET CLOSURE THEOREM)**.** *Execute the exact certificate for FINITE COVARIANT TWENTY TWO TARGET CLOSURE THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.560. FINITE COVARIANT TWENTY TWO TARGET CLOSURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 281. NONLINEAR COVARIANT PARENT FIELD THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_nonlinear_covariant_parent_field_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.561 (281. NONLINEAR COVARIANT PARENT FIELD THEOREM)**.** *Execute the exact certificate for NONLINEAR COVARIANT PARENT FIELD THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.562. NONLINEAR COVARIANT PARENT FIELD THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 282. PROJECTIVE DIRECT LIMIT PARENT FIELD THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_projective_direct_limit_parent_field_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.563 (282. PROJECTIVE DIRECT LIMIT PARENT FIELD THEOREM)**.** *Execute the exact certificate for PROJECTIVE DIRECT LIMIT PARENT FIELD THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.564. PROJECTIVE DIRECT LIMIT PARENT FIELD THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 283. COMPLETE DIRECT SYSTEM TARGET THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_complete_direct_system_target_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.565 (283. COMPLETE DIRECT SYSTEM TARGET THEOREM)**.** *Execute the exact certificate for COMPLETE DIRECT SYSTEM TARGET THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.566. COMPLETE DIRECT SYSTEM TARGET THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 284. UNIVERSAL COVARIANT ENERGY COMPLETION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_universal_covariant_energy_completion_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.567 (284. UNIVERSAL COVARIANT ENERGY COMPLETION THEOREM)**.** *Execute the exact certificate for UNIVERSAL COVARIANT ENERGY COMPLETION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.568. UNIVERSAL COVARIANT ENERGY COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 285. COMPACT SMOOTH COVARIANT REPRESENTATION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_compact_smooth_covariant_representation_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.569 (285. COMPACT SMOOTH COVARIANT REPRESENTATION THEOREM)**.** *Execute the exact certificate for COMPACT SMOOTH COVARIANT REPRESENTATION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.570. COMPACT SMOOTH COVARIANT REPRESENTATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 286. FINITE GOOD COVER COVARIANT DESCENT THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_finite_good_cover_covariant_descent_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.571 (286. FINITE GOOD COVER COVARIANT DESCENT THEOREM)**.** *Execute the exact certificate for FINITE GOOD COVER COVARIANT DESCENT THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.572. FINITE GOOD COVER COVARIANT DESCENT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 287. LOCALLY FINITE COVARIANT EXHAUSTION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_locally_finite_covariant_exhaustion_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.573 (287. LOCALLY FINITE COVARIANT EXHAUSTION THEOREM)**.** *Execute the exact certificate for LOCALLY FINITE COVARIANT EXHAUSTION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.574. LOCALLY FINITE COVARIANT EXHAUSTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 288. GLOBAL COMPLETION FRONTIER THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_global_completion_frontier_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.575 (288. GLOBAL COMPLETION FRONTIER THEOREM)**.** *Execute the exact certificate for GLOBAL COMPLETION FRONTIER THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.576. GLOBAL COMPLETION FRONTIER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 289. UNIVERSAL NATIVE FRONTIER CLOSURE THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_universal_native_frontier_closure_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.577 (289. UNIVERSAL NATIVE FRONTIER CLOSURE THEOREM)**.** *Execute the exact certificate for UNIVERSAL NATIVE FRONTIER CLOSURE THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.578. UNIVERSAL NATIVE FRONTIER CLOSURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 290. COUNTABLE HILBERT NATIVE CLOSURE THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_countable_hilbert_native_closure_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.579 (290. COUNTABLE HILBERT NATIVE CLOSURE THEOREM)**.** *Execute the exact certificate for COUNTABLE HILBERT NATIVE CLOSURE THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.580. COUNTABLE HILBERT NATIVE CLOSURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 291. SEPARABLE COVARIANT UNIVERSALITY THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_separable_covariant_universality_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.581 (291. SEPARABLE COVARIANT UNIVERSALITY THEOREM)**.** *Execute the exact certificate for SEPARABLE COVARIANT UNIVERSALITY THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.582. SEPARABLE COVARIANT UNIVERSALITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 292. PARAMETRIC UNIVERSAL NATIVE UNIFICATION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_parametric_universal_native_unification_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.583 (292. PARAMETRIC UNIVERSAL NATIVE UNIFICATION THEOREM)**.** *Execute the exact certificate for PARAMETRIC UNIVERSAL NATIVE UNIFICATION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.584. PARAMETRIC UNIVERSAL NATIVE UNIFICATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 293. SMOOTH COVARIANT DENSITY THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_smooth_covariant_density_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.585 (293. SMOOTH COVARIANT DENSITY THEOREM)**.** *Execute the exact certificate for SMOOTH COVARIANT DENSITY THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.586. SMOOTH COVARIANT DENSITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 294. GLOBAL FOUR FORCE RESOLUTION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_global_four_force_resolution_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.587 (294. GLOBAL FOUR FORCE RESOLUTION THEOREM)**.** *Execute the exact certificate for GLOBAL FOUR FORCE RESOLUTION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.588. GLOBAL FOUR FORCE RESOLUTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 295. GAUGE QUOTIENT OBSERVABLE COMPLETION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_gauge_quotient_observable_completion_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.589 (295. GAUGE QUOTIENT OBSERVABLE COMPLETION THEOREM)**.** *Execute the exact certificate for GAUGE QUOTIENT OBSERVABLE COMPLETION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.590. GAUGE QUOTIENT OBSERVABLE COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 296. FOUR FORCE OPERATOR IDENTIFICATION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_four_force_operator_identification_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.591 (296. FOUR FORCE OPERATOR IDENTIFICATION THEOREM)**.** *Execute the exact certificate for FOUR FORCE OPERATOR IDENTIFICATION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.592. FOUR FORCE OPERATOR IDENTIFICATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 297. EFFECTIVE OPERATOR COEFFICIENT CLOSURE THEOREM

The generated five-one-three-eight sectors are connected to variational operator identities only after projection. No measured coefficient is inserted into the native parent action.

Executable root: `_x893_effective_operator_coefficient_closure_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.593 (297. EFFECTIVE OPERATOR COEFFICIENT CLOSURE THEOREM)**.** *Join the four generated sectors to exact variational operator certificates without direct measured-value identification.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.594. EFFECTIVE OPERATOR COEFFICIENT CLOSURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 298. CLOSED RECURSIVE MODE SPECTRUM THEOREM

A string or loop is not assumed as the fundamental ontology. When the generated relation flow closes into a four-site cycle, the closed carrier has an exact rational mode spectrum. This supplies a string-like effective readout only at the layer where the closure is resolved as a loop.

Executable root: `_e895_closed_recursive_mode_spectrum_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.595 (298. CLOSED RECURSIVE MODE SPECTRUM THEOREM)**.** *Generate an exact closed-loop vibration spectrum from closure alone as the minimal string-like effective example.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.596. CLOSED RECURSIVE MODE SPECTRUM THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 299. EFFECTIVE EQUATION FAMILY GENERATION THEOREM

Standard equations are treated as effective laws on projected variables. Their empirical accuracy is not denied. The stronger task is to regenerate their operator families from the native ledger and prove that the native-first and effective-only calculations agree on the projected result.

Executable root: `_e895_effective_equation_family_generation_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.597 (299. EFFECTIVE EQUATION FAMILY GENERATION THEOREM)**.** *Generate and classify the principal effective equation families from native certificates without using those effective equations as native inputs.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.598. EFFECTIVE EQUATION FAMILY GENERATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 300. NATIVE OPERATOR GRAMMAR THEOREM

The main effective equations are not unrelated formulas. Boundary, coboundary, metric pairing, skew evolution, projection, commutator, and variation form one operator grammar. Conservation, gauge closure, wave propagation, unitary quantum readout, chiral propagation, nonabelian curvature, metric dynamics, and scale flow are different projections of that grammar.

Executable root: `_e896_native_operator_grammar_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.599 (300. NATIVE OPERATOR GRAMMAR THEOREM)**.** *Generate conservation, gauge closure, wave, Dirac-square, unitary probability, nonabelian, metric, closed-mode, and scale-flow operator structures from one exact incidence and variational grammar.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.600. NATIVE OPERATOR GRAMMAR THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 301. STANDARD EQUATION DICTIONARY THEOREM

The dictionary does not copy thirty observed equations into the origin. It records which generated native operation produces each effective operator form and which exact certificate supports that form. Equality of all measured constants and all textbook solutions remains an external phenomenological question.

Executable root: `_e896_standard_equation_dictionary_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.601 (301. STANDARD EQUATION DICTIONARY THEOREM)**.** *Connect thirty principal standard-physics equation forms to native ingredients and exact generated certificates without using the standard forms as native inputs.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.602. STANDARD EQUATION DICTIONARY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 302. STANDARD AND CANDIDATE DYNAMICAL BRIDGE THEOREM

The same compiler connects standard equations and bounded unification-framework structures. It asks whether the selected observables are exactly closed, require an exact memory extension, survive only as a scale limit, or propagate as constraints. No standard or candidate equation is used to choose the native generator.

Executable root: `_standard_and_candidate_dynamical_bridge_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.603 (302. STANDARD AND CANDIDATE DYNAMICAL BRIDGE THEOREM)**.** *Place all thirty standard forms and six bounded framework connections under the universal closure-memory-scale compiler.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.604. STANDARD AND CANDIDATE DYNAMICAL BRIDGE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 303. STANDARD EQUATION TRAJECTORY COMPILER THEOREM

The thirty equation forms are no longer connected only by visual resemblance. Each is assigned to a proved trajectory mechanism: nonlinear fiber factorization, continuous generator intertwining, nonlinear observable closure, JS-SH adjoint recurrence, one nonlinear four-force parent algebra, exact memory completion, or an independently registered covariant certificate.

Executable root: `_standard_equation_trajectory_compiler_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.605 (303. STANDARD EQUATION TRAJECTORY COMPILER THEOREM)**.** *Classify all thirty generated standard-equation forms by exact trajectory closure mechanisms.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.606. STANDARD EQUATION TRAJECTORY COMPILER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 304. CANDIDATE FRAMEWORK COMMON ORIGIN THEOREM

The major unification candidates are treated as distinct effective observable architectures of one deeper native ledger: equation algebras, parent symmetry decomposition, closed spectra, projective networks, scale fixed points, and quotient-world information. Structural overlap is proved without repeating every independently executed heavy continuum certificate inside this root.

Executable root: `_candidate_framework_common_origin_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.607 (304. CANDIDATE FRAMEWORK COMMON ORIGIN THEOREM)**.** *Place six major effective framework architectures under one bounded common-origin closure table.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.608. CANDIDATE FRAMEWORK COMMON ORIGIN THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 305. CLOSURE DEFECT PREDICTION THEOREM

The universal compiler yields a new quantity rather than only rephrasing standard equations. In the minimal nonclosed native block, the first delayed memory-kernel strength is exactly one eighth of the instantaneous kernel strength and every later kernel vanishes.

Executable root: `_closure_defect_prediction_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.609 (305. CLOSURE DEFECT PREDICTION THEOREM)**.** *Seal the first nonzero memory-kernel ratio and the exact two-step closure-defect recurrence as native predictions.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.610. CLOSURE DEFECT PREDICTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 306. SEMANTIC RATIO DECODER SELECTION THEOREM

A blind comparison fails if the physical target is chosen after its value is known. This theorem fixes the admissible target class before data: same-scale quadratic sector ratios and one same-channel memory-kernel ratio, with one common scale removed and no post-seal channel relabeling.

Executable root: `_semantic_ratio_decoder_selection_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.611 (306. SEMANTIC RATIO DECODER SELECTION THEOREM)**.** *Freeze a value-independent semantic decoder for native coefficient and closure-defect ratios.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.612. SEMANTIC RATIO DECODER SELECTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 307. PROJECTION INVARIANT CONSTANT GENERATION THEOREM

The internal theory generates exact dimensionless counts, ratios, ranks, closure periods, and scale-covariant conversion ratios. These are candidate origins of effective constants, but no measured constant is declared identical until a sealed projection decoder and blind ratio comparison are supplied.

Executable root: `_e897_projection_invariant_constant_generation_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.613 (307. PROJECTION INVARIANT CONSTANT GENERATION THEOREM)**.** *Generate a sealed dimensionless invariant ledger from relation counts, ranks, closure ratios, sector proportions, and causal scale covariance.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.614. PROJECTION INVARIANT CONSTANT GENERATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 308. NATIVE COEFFICIENT DESCENT THEOREM

The coefficients of the effective quadratic sector action are generated as basis-independent mean-square eigenvalues of the native scale operator. Their ratios are computed before comparison and therefore cannot be adjusted to observed couplings.

Executable root: `_e898_native_coefficient_descent_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.615 (308. NATIVE COEFFICIENT DESCENT THEOREM)**.** *Derive exact projected quadratic-action coefficient ratios from the native scale spectrum without observed inputs.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.616. NATIVE COEFFICIENT DESCENT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 309. EFFECTIVE WORLD EQUATION CLOSURE THEOREM

The thirty generated equation forms act inside the three-coordinate quotient world and remain exact on projection equivalence classes. Existing frameworks are connected as bounded effective structures, while native selection and empirical coefficients are never inferred backward from those frameworks.

Executable root: `_e897_effective_world_equation_closure_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.617 (309. EFFECTIVE WORLD EQUATION CLOSURE THEOREM)**.** *Join the three-dimensional quotient world to thirty native-generated equation-form certificates without recomputing the large downstream roots.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.618. EFFECTIVE WORLD EQUATION CLOSURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 310. EXISTING UNIFICATION CANDIDATE CONNECTION THEOREM

Existing theories are not dismissed as incorrect shadows. They are compared as mathematically precise effective descriptions. Structural overlap means that the native system generates the same class of projected operator, not that every external theory has been proved identical to this native origin.

Executable root: `_e895_existing_unification_candidate_connection_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.619 (310. EXISTING UNIFICATION CANDIDATE CONNECTION THEOREM)**.** *Place major existing physics and unification frameworks in the effective-readout layer by exact structural overlap, without claiming full equivalence.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.620. EXISTING UNIFICATION CANDIDATE CONNECTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 311. INFINITE VOLUME LOCAL UNITARY FIELD THEOREM

A rational orthogonal local coin and a bijective conditional shift define a norm-preserving operator on the finitely supported dense core of the countable infinite lattice. Finite propagation and norm preservation give a unique unitary extension to its Hilbert completion.

Executable root: `_x893_infinite_volume_local_unitary_field_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.621 (311. INFINITE VOLUME LOCAL UNITARY FIELD THEOREM)**.** *Construct an exact infinite-volume, finite-propagation, norm-preserving field dynamics and its dyadic continuum extension.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.622. INFINITE VOLUME LOCAL UNITARY FIELD THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 312. NONPERTURBATIVE FIELD MEASURE PREDICTION SEAL THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_nonperturbative_field_measure_prediction_seal_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.623 (312. NONPERTURBATIVE FIELD MEASURE PREDICTION SEAL THEOREM)**.** *Execute the exact certificate for NONPERTURBATIVE FIELD MEASURE PREDICTION SEAL THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.624. NONPERTURBATIVE FIELD MEASURE PREDICTION SEAL THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 313. BLIND RATIO FALSIFICATION COMPLETION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_blind_ratio_falsification_completion_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.625 (313. BLIND RATIO FALSIFICATION COMPLETION THEOREM)**.** *Execute the exact certificate for BLIND RATIO FALSIFICATION COMPLETION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.626. BLIND RATIO FALSIFICATION COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 314. RATIO OBSERVATION CONTRACT THEOREM

The final observational layer compares only frozen dimensionless ratios. A direct fractional construction and an independent common-denominator integer construction agree exactly; common scale cancels, and every tested post-seal mutation is rejected.

Executable root: `_e898_ratio_observation_contract_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.627 (314. RATIO OBSERVATION CONTRACT THEOREM)**.** *Create an immutable no-fitting ratio prediction and deterministic external-comparison contract.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.628. RATIO OBSERVATION CONTRACT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 315. NATIVE WORLD STANDARD OBSERVATION HIERARCHY THEOREM

The hierarchy is no longer a verbal analogy. Each arrow is an executable interface: native update to a local three-dimensional quotient, quotient to effective equation forms and coefficients, equations to scale-free predictions, and predictions to a deterministic external verdict.

Executable root: `_e898_native_world_standard_observation_hierarchy_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.629 (315. NATIVE WORLD STANDARD OBSERVATION HIERARCHY THEOREM)**.** *Conjoin the native ledger, effective world, standard-equation, and observation layers into one exact hierarchy theorem.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.630. NATIVE WORLD STANDARD OBSERVATION HIERARCHY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 316. NATIVE ENERGY RATIO DECODER THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_native_energy_ratio_decoder_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.631 (316. NATIVE ENERGY RATIO DECODER THEOREM)**.** *Execute the exact certificate for NATIVE ENERGY RATIO DECODER THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.632. NATIVE ENERGY RATIO DECODER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 317. CONTINUUM SCHUR BETA UNIVERSALITY THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_continuum_schur_beta_universality_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.633 (317. CONTINUUM SCHUR BETA UNIVERSALITY THEOREM)**.** *Execute the exact certificate for CONTINUUM SCHUR BETA UNIVERSALITY THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.634. CONTINUUM SCHUR BETA UNIVERSALITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 318. TIME FIRST SINGLE ANCHOR COMPLETION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_time_first_single_anchor_completion_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.635 (318. TIME FIRST SINGLE ANCHOR COMPLETION THEOREM)**.** *Execute the exact certificate for TIME FIRST SINGLE ANCHOR COMPLETION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.636. TIME FIRST SINGLE ANCHOR COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 319. FOUR FORCE COMMON CARRIER SCHUR ACTION THEOREM

The four sectors share one common carrier rather than four unrelated actions. Exact stationary elimination of that carrier produces a positive Schur action, generated cross-sector couplings, and the same sector variation as the parent action.

Executable root: `_four_force_common_carrier_schur_action_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.637 (319. FOUR FORCE COMMON CARRIER SCHUR ACTION THEOREM)**.** *Derive a positive four-sector effective action by exact stationary elimination of one common carrier.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.638. FOUR FORCE COMMON CARRIER SCHUR ACTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 320. FORCE DECODER IDENTIFIABILITY THEOREM

A physical decoder is not selected by numerical proximity. The generated sector rank, closure type, and operator role form distinct semantic signatures, and exhaustive permutation leaves exactly one admissible four-sector identification.

Executable root: `_force_decoder_identifiability_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.639 (320. FORCE DECODER IDENTIFIABILITY THEOREM)**.** *Prove unique four-sector semantic decoding under generated rank, closure, and operator signatures.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.640. FORCE DECODER IDENTIFIABILITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 321. STANDARD EQUATION COMMUTING DIAGRAM THEOREM

Each standard equation form now belongs to one executable commuting diagram: one native mechanism, one generated world readout, one effective trajectory law, and one ratio-only observation interface. The equation form is not used backward to construct its native source.

Executable root: `_standard_equation_commuting_diagram_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.641 (321. STANDARD EQUATION COMMUTING DIAGRAM THEOREM)**.** *Close one commuting native-world-equation-ratio diagram for every generated standard equation form.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.642. STANDARD EQUATION COMMUTING DIAGRAM THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 322. SEALED MULTIRATIO PREDICTION THEOREM

The prediction layer is not a single fitted number. Native coefficient ratios, the first memory-strength ratio, six common-carrier pair fractions, and four force-rank fractions are generated independently, normalized without an absolute unit, and sealed before external data.

Executable root: `_sealed_multiratio_prediction_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.643 (322. SEALED MULTIRATIO PREDICTION THEOREM)**.** *Generate and seal a scale-free multi-ratio prediction packet from independent native mechanisms.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.644. SEALED MULTIRATIO PREDICTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 323. FOUR FORCE ACTION SCALE INVARIANCE THEOREM

A recursive layer change may alter the overall action unit but cannot alter the generated relative force content. The complete Schur matrix scales uniformly, all principal minors scale with their degree, positivity survives, and the normalized action matrix is identical at every declared layer.

Executable root: `_four_force_action_scale_invariance_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.645 (323. FOUR FORCE ACTION SCALE INVARIANCE THEOREM)**.** *Prove common-layer scale covariance and normalized-ratio invariance of the generated Schur action.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.646. FOUR FORCE ACTION SCALE INVARIANCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 324. STANDARD MECHANISM DECODER UNIQUENESS THEOREM

The thirty effective equations are not attached to mechanisms by numerical resemblance. Time type, algebra type, closure type, physical role, and equation multiplicity form seven distinct signatures. Exhaustive permutation leaves exactly one mechanism decoder.

Executable root: `_standard_mechanism_decoder_uniqueness_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.647 (324. STANDARD MECHANISM DECODER UNIQUENESS THEOREM)**.** *Prove unique semantic identification of the seven mechanisms used by the thirty-equation trajectory compiler.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.648. STANDARD MECHANISM DECODER UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 325. BLIND RATIO CAPSULE VERDICT THEOREM

The comparison operation is finished without importing observations into the proof. A capsule can only provide labeled dimensionless intervals. Exact protocol data pass, a one-component displacement fails, a missing label fails, and no verdict can alter the frozen prediction.

Executable root: `_blind_ratio_capsule_verdict_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.649 (325. BLIND RATIO CAPSULE VERDICT THEOREM)**.** *Complete and self-test the deterministic ratio-capsule verdict operation without fabricating external evidence.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.650. BLIND RATIO CAPSULE VERDICT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 326. CONTINUUM REFINEMENT THEOREM

An induction identity proves the exact quadratic energy at every positive integer refinement. Dyadic and triadic refinements share the same continuum value and cutoff coefficient.

Executable root: `_u905_continuum_refinement_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.651 (326. CONTINUUM REFINEMENT THEOREM)**.** *Execute the exact certificate for CONTINUUM REFINEMENT THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.652. CONTINUUM REFINEMENT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 327. NATIVE CONTINUUM CYLINDER RECONSTRUCTION THEOREM

Sixteen children per four-dimensional refinement preserve parent weight exactly at every depth. The symmetric transfer kernel is reflection positive and every affine-quadratic local observable is reconstructed exactly by finite differences.

Executable root: `theorem_native_continuum_cylinder_reconstruction`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.653 (327. NATIVE CONTINUUM CYLINDER RECONSTRUCTION THEOREM)**.** *Construct a four-dimensional projective cylinder measure, positive transfer form, and exact local quadratic observable reconstruction.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.654. NATIVE CONTINUUM CYLINDER RECONSTRUCTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 328. SOURCE CYLINDER ALL CORRELATION RECONSTRUCTION THEOREM

A strictly positive eighteen-state transition kernel generates normalized and projectively consistent finite cylinders. Complete indicator correlations equal the cylinder probabilities and reconstruct every finite source correlation by induction.

Executable root: `theorem_source_cylinder_all_correlation_reconstruction`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.655 (328. SOURCE CYLINDER ALL CORRELATION RECONSTRUCTION THEOREM). *Construct a positive eighteen-state source cylinder measure and prove that its complete indicator correlations reconstruct every finite cylinder probability.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.656. SOURCE CYLINDER ALL CORRELATION RECONSTRUCTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 329. COMPLETED GENERATOR SEMIGROUP INTERTWINING THEOREM

One identity-retention channel and one common redistribution channel select a symmetric positive transfer kernel with a closed power formula at every scale. The exact character transform intertwines the transfer and generator on the dense cylinder core and therefore on their closures.

Executable root: `theorem_completed_generator_semigroup_intertwining`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.657 (329. COMPLETED GENERATOR SEMIGROUP INTERTWINING THEOREM). *Construct the source transfer semigroup, prove positivity and an all-scale closed form, and extend exact intertwining to the cylinder completion.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.658. COMPLETED GENERATOR SEMIGROUP INTERTWINING THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 330. UNIQUE REFLECTION POSITIVE SOURCE MEASURE THEOREM

Strict positivity and a single residual contraction make the uniform stationary source measure unique. The transfer quadratic form is a sum of two rational squares and every residual correlation obeys one exact clustering power law.

Executable root: `theorem_unique_reflection_positive_source_measure`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.659 (330. UNIQUE REFLECTION POSITIVE SOURCE MEASURE THEOREM)**.** *Prove uniqueness, reflection positivity, clustering, and all-cylinder consistency of the source transfer measure selected by the native generator.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.660. UNIQUE REFLECTION POSITIVE SOURCE MEASURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 331. ALL CORRELATION FUNCTIONAL EQUIVALENCE THEOREM

Every finite correlation word is a product of multiplication and transfer operators between an initial measure and the terminal constant state. Transform and inverse-transform factors cancel at every internal boundary, proving equality of all words and their bounded completion.

Executable root: `theorem_all_correlation_functional_equivalence`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.661 (331. ALL CORRELATION FUNCTIONAL EQUIVALENCE THEOREM)**.** *Prove equality of every finite source correlation word and its complete-shadow image, then extend the bounded functional to the completion.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.662. ALL CORRELATION FUNCTIONAL EQUIVALENCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 332. ARBITRARY REFINEMENT POLYNOMIAL BRIDGE THEOREM

Parent-child cylinder weights close by an algebraic induction at arbitrary depth. The binomial expansion shows that central first- and second-difference errors contain an h-squared factor for every finite rational polynomial degree, while mixed coordinate operators commute.

Executable root: `theorem_arbitrary_refinement_polynomial_bridge`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.663 (332. ARBITRARY REFINEMENT POLYNOMIAL BRIDGE THEOREM)**.** *Prove arbitrary-depth cylinder consistency and coefficientwise continuum convergence for every finite rational polynomial field.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.664. ARBITRARY REFINEMENT POLYNOMIAL BRIDGE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 333. NATIVE DYADIC CYLINDER ALGEBRA ISOMORPHISM THEOREM

Each native sixteen-way digit is bijective with four dyadic bits; Cartesian powers preserve this bijection at every finite depth. Pointwise algebra, refinement prefixes, and uniform cylinder measures are preserved, inducing an inverse-limit cylinder-algebra isomorphism.

Executable root: `theorem_native_dyadic_cylinder_algebra_isomorphism`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.665 (333. NATIVE DYADIC CYLINDER ALGEBRA ISOMORPHISM THEOREM)**.** *Construct an exact depth-preserving algebra and measure isomorphism between native sixteen-branch cylinders and dyadic boxes in four coordinates.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.666. NATIVE DYADIC CYLINDER ALGEBRA ISOMORPHISM THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 334. NATIVE RENORMALIZATION FLOW THEOREM

The temporal curvature-to-signed ratio generates the scale contraction. Orthogonal fixed and residual projectors prove the exact semigroup and beta law.

Executable root: `_u905_native_renormalization_flow_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.667 (334. NATIVE RENORMALIZATION FLOW THEOREM)**.** *Execute the exact certificate for NATIVE RENORMALIZATION FLOW THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.668. NATIVE RENORMALIZATION FLOW THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 335. NATIVE FLUCTUATION RG BRIDGE THEOREM

The temporal contraction multiplies the native gauge-closure, Weyl-orientation, and scalar-common multiplicities to generate the three fluctuation weights. The resulting group and matter indices drive exact rational scale flows with the strong and weak channels decreasing and the Abelian channel increasing.

Executable root: `theorem_native_fluctuation_rg_bridge`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.669 (335. NATIVE FLUCTUATION RG BRIDGE THEOREM)**.** *Derive fluctuation multiplicities from the primitive ledger and join group indices to an exact scale-dependent rational flow.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.670. NATIVE FLUCTUATION RG BRIDGE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 336. NATIVE RG CLOSED FORM INDUCTION THEOREM

Orthogonal fixed and residual projectors reduce every blocking power to one closed expression valid at all nonnegative integer scales. The beta increment and every channel coupling product then follow by an exact one-step induction without logarithms or fitted coefficients.

Executable root: `theorem_native_rg_closed_form_induction`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.671 (336. NATIVE RG CLOSED FORM INDUCTION THEOREM)**.** *Prove the native blocking semigroup and beta increment for every nonnegative scale by exact projector induction.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.672. NATIVE RG CLOSED FORM INDUCTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 337. NATIVE MULTICHANNEL SCATTERING UNITARITY THEOREM

The generated ranks 5, 1, 3, and 8 determine one rational skew interaction without an external coupling value. Its Cayley transform has determinant one, preserves every incoming norm, and acts only on channels whose Green residues are positive.

Executable root: `theorem_native_multichannel_scattering_unitarity`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.673 (337. NATIVE MULTICHANNEL SCATTERING UNITARITY THEOREM)**.** *Construct a parameter-free rational four-channel Cayley scattering operator from the generated carrier ranks and prove exact unitarity and positive channel probabilities.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.674. NATIVE MULTICHANNEL SCATTERING UNITARITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 338. RG SCATTERING SCALE BINDING THEOREM

The temporal contraction transports the skew interaction through every integer blocking scale. The Cayley identity proves a unitary scattering operator at every scale without logarithms or fitted parameters.

Executable root: `theorem_rg_scattering_scale_binding`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.675 (338. RG SCATTERING SCALE BINDING THEOREM)**.** *Bind the exact native blocking flow to a unitary four-channel scattering family at every nonnegative integer scale without logarithms.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.676. RG SCATTERING SCALE BINDING THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 339. NATIVE SCALE MOMENTUM RATIO DECODER UNIQUENESS THEOREM

The native one-step contraction fixes the inverse-length momentum ratio, and multiplicativity fixes every later depth uniquely. Only one overall dimensional calibration remains external; all interscale ratios are exact and parameter free.

Executable root: `theorem_native_scale_momentum_ratio_decoder_uniqueness`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.677 (339. NATIVE SCALE MOMENTUM RATIO DECODER UNIQUENESS THEOREM)**.** *Derive the unique multiplicative momentum and length ratios from the native blocking contraction.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.678. NATIVE SCALE MOMENTUM RATIO DECODER UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 340. NATIVE ALL ORDER RESOLVENT CLOSURE THEOREM

The interaction generator satisfies its exact characteristic polynomial and every higher power reduces to the finite basis I, A and A squared. Every declared resolvent and Cayley response is therefore a closed rational operator rather than an uncontrolled infinite list of independent terms.

Executable root: `theorem_native_all_order_resolvent_closure`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.679 (340. NATIVE ALL ORDER RESOLVENT CLOSURE THEOREM)**.** *Reduce every power and rational resolvent of the native interaction generator to its finite exact operator algebra.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.680. NATIVE ALL ORDER RESOLVENT CLOSURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 341. NATIVE DISCRETE SCALE FLOW IDENTITY THEOREM

The unique momentum-ratio coordinate turns native blocking into an exact finite-difference beta identity. The corresponding Cayley scattering difference is derived as an exact resolvent identity at every native scale.

Executable root: `theorem_native_discrete_scale_flow_identity`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.681 (341. NATIVE DISCRETE SCALE FLOW IDENTITY THEOREM)**.** *Prove exact finite-difference beta and Cayley-flow identities in the uniquely decoded momentum-ratio coordinate.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.682. NATIVE DISCRETE SCALE FLOW IDENTITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 342. RATIONAL SEMIGROUP TANGENT UNIQUENESS THEOREM

The scale increment is treated as a formal variable in an exact polynomial identity. The first increment coefficient uniquely gives the continuum generator without logarithms.

Executable root: `theorem_rational_semigroup_tangent_uniqueness`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.683 (342. RATIONAL SEMIGROUP TANGENT UNIQUENESS THEOREM)**.** *Derive the continuum generator from a formal bivariate polynomial identity.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.684. RATIONAL SEMIGROUP TANGENT UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 343. DISCRETE CONTINUUM GENERATOR BRIDGE THEOREM

The rational semigroup has an exact difference quotient whose zero-increment tangent is V(x)=-x(1-x). This derives the continuum generator from the discrete source law rather than assuming an independent smooth equation.

Executable root: `theorem_discrete_continuum_generator_bridge`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.685 (343. DISCRETE CONTINUUM GENERATOR BRIDGE THEOREM)**.** *Derive the continuum boundary generator from the formal rational tangent identity.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.686. DISCRETE CONTINUUM GENERATOR BRIDGE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 344. PROJECTIVE RESOLVENT ANALYTIC COMPLETION THEOREM

The unique bounded projective equation is approximated by exact rational polynomials with a closed geometric remainder. This supplies a constructive analytic continuum class beyond finite polynomials without evaluating a logarithm or exponential.

Executable root: `theorem_projective_resolvent_analytic_completion`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.687 (344. PROJECTIVE RESOLVENT ANALYTIC COMPLETION THEOREM)**.** *Place the bounded projective shadow map in an exact polynomial Cauchy completion on every compact native interior chart.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.688. PROJECTIVE RESOLVENT ANALYTIC COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 345. ADMISSIBLE CONTINUUM BILATERAL EQUIVALENCE THEOREM

Compatible source families and their completed shadow fields are inverse descriptions on the full declared finite-action projective class, including gauge orbits, action variation, correlations, metric and connection.

Executable root: `theorem_admissible_continuum_bilateral_equivalence`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.689 (345. ADMISSIBLE CONTINUUM BILATERAL EQUIVALENCE THEOREM)**.** *Close a two-sided source-shadow equivalence on the complete admissible finite-action projective continuum class, including action, variation, gauge quotient, correlations, and metric gravity.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.690. ADMISSIBLE CONTINUUM BILATERAL EQUIVALENCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 346. CONDITIONAL STANDARD MODEL GR SOLUTION LIFT THEOREM

Prove a universal lift for every continuous gauge-gravity solution satisfying the generated parent-action, finite-action, topology, and completion conditions.

Executable root: `theorem_conditional_standard_model_gr_solution_lift`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.691 (346. CONDITIONAL STANDARD MODEL GR SOLUTION LIFT THEOREM)**.** *Prove a universal lift for every continuous gauge-gravity solution satisfying the generated parent-action, finite-action, topology, and completion conditions.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.692. CONDITIONAL STANDARD MODEL GR SOLUTION LIFT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 347. GLOBAL SCALE CENTER CLASSIFICATION THEOREM

All relative four-sector constraints leave exactly one common central scale direction. Boundary exchange fixes the native depth origin and the temporal spectrum fixes the native unit step.

Executable root: `theorem_global_scale_center_classification`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.693 (347. GLOBAL SCALE CENTER CLASSIFICATION THEOREM)**.** *Classify the complete common unit freedom and prove that the relative four-sector scale data leave exactly one central global scale direction.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.694. GLOBAL SCALE CENTER CLASSIFICATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 348. SINGLE TIME ANCHOR UNIT DECODER UNIQUENESS THEOREM

Because the residual unit freedom is one-dimensional and the causal dependency graph begins with time, one time anchor fixes every generated length, energy, mass and force representative. A second fitted dimensional parameter is neither needed nor allowed.

Executable root: `theorem_single_anchor_unit_decoder_uniqueness`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.695 (348. SINGLE TIME ANCHOR UNIT DECODER UNIQUENESS THEOREM)**.** *Prove that one external time anchor fixes the sole central unit orbit and determines every generated dimensionful decoder with no second fit parameter.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.696. SINGLE TIME ANCHOR UNIT DECODER UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 349. TIME ANCHOR DECODER PRIORITY THEOREM

Time is the only source node of the dimensional decoder. Length follows from the causal speed unit, energy from action divided by time, mass from energy, and force from energy divided by length.

Executable root: `theorem_time_anchor_decoder_priority`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.697 (349. TIME ANCHOR DECODER PRIORITY THEOREM)**.** *Prove that time is the unique source node of the generated dimensional decoder and that length, energy, mass and force cannot be independent external anchors.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.698. TIME ANCHOR DECODER PRIORITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 350. SECTOR RESOLVED THRESHOLD RG FLOW THEOREM

The parent-scale ratios are not frozen physical couplings at every energy. Native mass gaps activate the three temporal generations at distinct depth thresholds. The generated field indices then give different electromagnetic, weak, strong, and gravitational scale increments.

Executable root: `theorem_sector_resolved_threshold_rg_flow`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.699 (350. SECTOR RESOLVED THRESHOLD RG FLOW THEOREM)**.** *Generate distinct electromagnetic, weak, strong, and gravitational scale trajectories with exact native generation thresholds and no logarithm, fitted beta coefficient, or observed coupling value.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.700*.* SECTOR RESOLVED THRESHOLD RG FLOW THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 351. RG PROJECTIVE CROSS RATIO SCOPE AND CURVE THEOREM

The sealed parent cross ratio is a structural invariant under one common projective compression. It is not asserted to equal a fixed measured running coupling. Sector-dependent RG motion instead generates a scale-indexed cross-ratio curve, and each point of that curve remains invariant under a common coordinate compression.

Executable root: `theorem_rg_projective_cross_ratio_scope_and_curve`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.701 (351. RG PROJECTIVE CROSS RATIO SCOPE AND CURVE THEOREM)**.** *Distinguish the fixed parent-scale projective invariant from the scale-dependent cross-ratio curve generated by sector-resolved running and prove common-compression invariance at every scale.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.702*.* RG PROJECTIVE CROSS RATIO SCOPE AND CURVE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 352. RG HORIZON COMMON COMPRESSION GENERATOR THEOREM

The RG residual and the horizon residual are the same primitive contraction power at every depth. Their difference is the physical readout label: momentum suppression in one channel and boundary-distance compression in another.

Executable root: `theorem_rg_horizon_common_compression_generator`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.703 (352. RG HORIZON COMMON COMPRESSION GENERATOR THEOREM)**.** *Prove that native RG blocking and zero-boundary horizon compression are the same multiplicative residual semigroup under different readout names.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.704. RG HORIZON COMMON COMPRESSION GENERATOR THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 353. NATIVE THRESHOLD SCHEME COVARIANCE THEOREM

Native mass ratios determine exact generation-threshold windows without an absolute unit. Common positive unit changes preserve every active set, while positive affine inverse-coordinate schemes conjugate the orbit without changing its ordering or fixed-point content.

Executable root: `theorem_native_threshold_scheme_covariance`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.705 (353. NATIVE THRESHOLD SCHEME COVARIANCE THEOREM)**.** *Derive generation-threshold windows and exact covariance under common scale and positive affine inverse-coordinate schemes.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.706. NATIVE THRESHOLD SCHEME COVARIANCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 354. NATIVE REGULATOR PATH UNIVERSALITY THEOREM

All six positive permutations of the three residual transfer eigenvalues share one fixed projector and commute on the common character projectors. Every regulator path contracts residual components by at most one half per step and therefore reaches the same orientation-independent limit.

Executable root: `theorem_native_regulator_path_universality`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.707 (354. NATIVE REGULATOR PATH UNIVERSALITY THEOREM)**.** *Prove a unique continuum fixed projector for every path through the complete native residual-spectrum regulator family.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.708. NATIVE REGULATOR PATH UNIVERSALITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 355. NATIVE ASYMPTOTIC STATE THEOREM

At every RG scale the Cayley operator maps a complete orthonormal in basis to a complete orthonormal out basis. The exact native contraction drives the interaction to the free fixed point with rational tolerance witnesses and no logarithm.

Executable root: `theorem_native_asymptotic_state`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.709 (355. NATIVE ASYMPTOTIC STATE THEOREM)**.** *Construct complete orthonormal in and out bases for every native RG scale and prove exact convergence of the interacting Cayley family to the free fixed point.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.710. NATIVE ASYMPTOTIC STATE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 356. EMERGENT POINCARE INTERNAL FACTORIZATION THEOREM

The rational Lorentz and translation generators act on a kinematic factor, while the exact internal S-matrix centralizer acts on a separate force-channel factor. Their tensor representations commute, both preserve the total scattering operator, and their generator spans form an exact direct sum.

Executable root: `theorem_emergent_poincare_internal_factorization`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.711 (356. EMERGENT POINCARE INTERNAL FACTORIZATION THEOREM)**.** *Construct the rational Poincare algebra and the exact internal S-matrix centralizer on separate tensor factors.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.712. EMERGENT POINCARE INTERNAL FACTORIZATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 357. COLEMAN MANDULA DIRECT COMPATIBILITY THEOREM

A common native origin does not require nontrivial mixing of emergent spacetime and internal symmetry generators. The declared bosonic asymptotic symmetry satisfies the direct-product conclusion with zero mixed commutators instead of claiming an unproved no-go evasion.

Executable root: `theorem_coleman_mandula_direct_compatibility`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.713 (357. COLEMAN MANDULA DIRECT COMPATIBILITY THEOREM)**.** *Meet the bosonic direct-product conclusion on the declared asymptotic class rather than claiming an unproved no-go evasion.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.714. COLEMAN MANDULA DIRECT COMPATIBILITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 358. NATIVE FLUX AND PHASE SPACE UNIQUENESS THEOREM

Orthogonality fixes current, tensor products fix pair flux, and the signed-permutation group acts transitively on all twelve directions. Transitivity and normalization force the unique angular measure, while amplitude and area gauges cancel from every ratio.

Executable root: `theorem_native_flux_phase_space_uniqueness`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.715 (358. NATIVE FLUX AND PHASE SPACE UNIQUENESS THEOREM)**.** *Derive exact conserved flux and the unique invariant angular phase-space measure on the twelve-direction native orbit, including exchange and unit-gauge cancellation.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.716. NATIVE FLUX AND PHASE SPACE UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 359. DIFFERENTIAL SCATTERING RATIO THEOREM

The all-scale channel probabilities are tensored with the unique native angular measure and carrier multiplicities. Every differential packet is normalized, satisfies optical balance and inverse reversal, and is transported exactly by the RG law.

Executable root: `theorem_differential_scattering_ratio`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.717 (359. DIFFERENTIAL SCATTERING RATIO THEOREM)**.** *Tensor the all-scale channel scattering family with the unique native angular measure and derive exact differential cross-section ratios, optical balance, reversal, and RG compatibility.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.718. DIFFERENTIAL SCATTERING RATIO THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 360. NATIVE OBSERVABLE ALGEBRA THEOREM

Generation-gap, sector-response, mixing, scattering-residue, causal-decay, Higgs-curvature, and Lorentz-defect ratios are generated before empirical naming. Every entry is exact, dimensionless, nonnegative where required, and invariant under a common scale multiplier.

Executable root: `theorem_native_observable_algebra`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.719 (360. NATIVE OBSERVABLE ALGEBRA THEOREM)**.** *Execute the exact certificate for NATIVE OBSERVABLE ALGEBRA THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.720. NATIVE OBSERVABLE ALGEBRA THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 361. CANONICAL SHADOW OBSERVABLE FUNCTOR THEOREM

The complete character map transports the source algebra, generator, gauge morphisms, correlations, metric-curvature data, scattering norm, and native ratios. Agreement on the eighteen generators and continuity make this transport unique on the completed admissible shadow algebra.

Executable root: `theorem_canonical_shadow_observable_functor`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.721 (361. CANONICAL SHADOW OBSERVABLE FUNCTOR THEOREM)**.** *Execute the exact certificate for CANONICAL SHADOW OBSERVABLE FUNCTOR THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.722. CANONICAL SHADOW OBSERVABLE FUNCTOR THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 362. NATIVE SHADOW EQUATION DESCENT THEOREM

Causal balance, parent-action response, gauge transport, metric variation, and transfer evolution are generated in the source theory. Their shadow equations follow by intertwining; no textbook equation or measured coefficient is used as a premise.

Executable root: `theorem_native_shadow_equation_descent`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.723 (362. NATIVE SHADOW EQUATION DESCENT THEOREM)**.** *Execute the exact certificate for NATIVE SHADOW EQUATION DESCENT THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.724. NATIVE SHADOW EQUATION DESCENT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 363. NATIVE OBSERVABLE DECODER UNIQUENESS THEOREM

The mass-like gap, response-like coupling, mixing-like overlap, and complete shadow transport are selected in explicit low-degree grammars. Observed particle names and measured constants do not participate in the selection.

Executable root: `theorem_native_observable_decoder_uniqueness`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.725 (363. NATIVE OBSERVABLE DECODER UNIQUENESS THEOREM)**.** *Execute the exact certificate for NATIVE OBSERVABLE DECODER UNIQUENESS THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.726. NATIVE OBSERVABLE DECODER UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 364. NATIVE MULTIRATIO BLIND CAPSULE THEOREM

Every native observable family is serialized into one immutable ratio packet. A mutation in any single family changes the packet digest, preventing isolated post-comparison adjustment.

Executable root: `theorem_native_multiratio_blind_capsule`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.727 (364. NATIVE MULTIRATIO BLIND CAPSULE THEOREM)**.** *Execute the exact certificate for NATIVE MULTIRATIO BLIND CAPSULE THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.728. NATIVE MULTIRATIO BLIND CAPSULE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 365. NATIVE TARGET CHANNEL FORCING THEOREM

Only after the anonymous fingerprints are complete is a conventional chiral role schema admitted for comparison. Exhausting all role assignments leaves one structural mapping and the temporal mass ordering forces first, second and third generation target channels.

Executable root: `theorem_native_target_channel_forcing`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.729 (365. NATIVE TARGET CHANNEL FORCING THEOREM)**.** *Generate target roles only after the anonymous fingerprints are complete and prove one exact structural assignment to the conventional chiral role schema without using masses or measured rates.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.730. NATIVE TARGET CHANNEL FORCING THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 366. EMPIRICAL TARGET PRE REGISTRATION CONTRACT THEOREM

Labels, ratio definitions, scales, angle bins, evidence tiers and the pass rule are frozen before external numerical data are admitted. Only dimensionless ratios are registered and post-unblinding target changes are forbidden.

Executable root: `theorem_empirical_target_preregistration_contract`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.731 (366. EMPIRICAL TARGET PRE REGISTRATION CONTRACT THEOREM)**.** *Freeze ratio-only target channels, conventions, scales, and verdict rules before any external numerical record is admitted.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.732. EMPIRICAL TARGET PRE REGISTRATION CONTRACT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 367. IMMUTABLE COMPARISON CAPSULE THEOREM

Forced targets and mass, branching, width and differential scattering packets are sealed together. Changing any one family changes the digest, preventing isolated post-comparison repair.

Executable root: `theorem_immutable_comparison_capsule`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.733 (367. IMMUTABLE COMPARISON CAPSULE THEOREM)**.** *Seal the forced targets and all quantitative native predictions together so that neither labels nor any prediction family can be altered independently after comparison begins.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.734. IMMUTABLE COMPARISON CAPSULE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 368. EXTERNAL EMPIRICAL VERDICT GATE THEOREM

The internal program supplies a complete sealed capsule but cannot manufacture an independent custodian or experiment. With no signed external record the exact verdict is NOT EVALUATED rather than PASS.

Executable root: `theorem_external_empirical_verdict_gate`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.735 (368. EXTERNAL EMPIRICAL VERDICT GATE THEOREM)**.** *Prove a fail-closed empirical gate: the internal theory supplies a sealed comparison capsule, while an absent independent record yields NOT EVALUATED rather than a fabricated pass.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.736. EXTERNAL EMPIRICAL VERDICT GATE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 369. MULTISCALE HOLDOUT PREREGISTRATION CAPSULE THEOREM

The symmetry, momentum-ratio, running, threshold, operator-closure and regulator-universality packets are frozen together before external data are admitted. Changing any one packet changes the digest, so no isolated post-comparison repair is possible.

Executable root: `theorem_multiscale_holdout_preregistration_capsule`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.737 (369. MULTISCALE HOLDOUT PREREGISTRATION CAPSULE THEOREM)**.** *Seal the new symmetry, scale, threshold, all-order closure, and regulator-universality predictions before external comparison.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.738. MULTISCALE HOLDOUT PREREGISTRATION CAPSULE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 370. EXTERNAL UNIVERSALITY VERDICT GATE THEOREM

The native theorem chain closes its own asymptotic, scale and regulator classes but cannot create independent analytic, experimental or software evidence. Without a signed external record the unrestricted universality verdict remains NOT EVALUATED rather than being promoted to an internal pass.

Executable root: `theorem_external_universality_verdict_gate`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.739 (370. EXTERNAL UNIVERSALITY VERDICT GATE THEOREM)**.** *Fail closed on unrestricted analytic S-matrix, all-order external RG, smooth-regulator universality, and empirical realization.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.740. EXTERNAL UNIVERSALITY VERDICT GATE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 371. EXTERNAL REALITY VERDICT FIREWALL THEOREM

The internal proof terminates after generating the decoder, shadow equations, and sealed multifamily ratio packet. Only independent measurement and independently written reproduction can decide whether nature realizes that packet.

Executable root: `theorem_external_reality_verdict_firewall`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.741 (371. EXTERNAL REALITY VERDICT FIREWALL THEOREM)**.** *Execute the exact certificate for EXTERNAL REALITY VERDICT FIREWALL THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.742. EXTERNAL REALITY VERDICT FIREWALL THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 372. NATIVE PROJECTIVE BRIDGE PACKET THEOREM

The all-rational Lorentz identity, temporal generation uniqueness, arbitrary refinement, all-scale RG, and native-dyadic cylinder isomorphism are joined into one bridge certificate. The proved domain is stated exactly as the native and dyadic cylinder-polynomial field algebra, avoiding an unsupported identification with every external smooth quantum field measure.

Executable root: `theorem_native_projective_bridge_packet`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.743 (372. NATIVE PROJECTIVE BRIDGE PACKET THEOREM)**.** *Join the all-rational Lorentz, three-generation, arbitrary-refinement, RG-induction, and dyadic-cylinder isomorphism theorems.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.744. NATIVE PROJECTIVE BRIDGE PACKET THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 373. SCATTERING RATIO THEOREM

Two native scales generate a four-channel Cayley scattering matrix. Unitarity, the optical identity, and normalized transition ratios are exact.

Executable root: `_u905_scattering_ratio_theorem`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.745 (373. SCATTERING RATIO THEOREM)**.** *Execute the exact certificate for SCATTERING RATIO THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.746. SCATTERING RATIO THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 374. BLIND RATIO IDENTIFICATION FALSIFICATION CONTRACT THEOREM

The sealed native ratio signature is matched by exhaustive permutation using ratios only, without an absolute standard value. A one-component negative control is rejected, making the external validation protocol both identifying and falsifiable.

Executable root: `theorem_blind_ratio_identification_falsification_contract`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.747 (374. BLIND RATIO IDENTIFICATION FALSIFICATION CONTRACT THEOREM)**.** *Seal a ratio-only signature and prove an exhaustive blind permutation matcher with a negative-control falsification outcome.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.748. BLIND RATIO IDENTIFICATION FALSIFICATION CONTRACT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 375. RATIO ONLY OBSERVABLE PROTOCOL COMPLETION THEOREM

The internal theory generates and seals dimensionless ratio packets before external evidence arrives. External comparison may accept or reject the packet but cannot alter the generating equations.

Executable root: `theorem_ratio_only_observable_protocol_completion`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.749 (375. RATIO ONLY OBSERVABLE PROTOCOL COMPLETION THEOREM)**.** *Prove that the prospective observable interface is sealed as dimensionless ratios before external comparison and cannot rewrite the internal theory.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.750. RATIO ONLY OBSERVABLE PROTOCOL COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 376. SOURCE SHADOW COMPLETION BRIDGE THEOREM

Source selection, complete projection reconstruction, conjugate dynamics, variational pushforward, correlation reconstruction, metric gravity, and blind ratio identification form one bridge. The bridge replaces the demand that all observed shadows use the same equations with the stronger requirement that they reconstruct and commute with one common source.

Executable root: `theorem_source_shadow_completion_bridge`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.751 (376. SOURCE SHADOW COMPLETION BRIDGE THEOREM)**.** *Join source selection, projection reconstruction, shadow dynamics, variational pushforward, correlations, gravity, and blind ratio identification.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.752. SOURCE SHADOW COMPLETION BRIDGE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 377. EFFECTIVE EQUATION COMMUTING DIAGRAM COMPLETION THEOREM

No standard-looking equation is counted merely because its final syntax is familiar. Each of the thirty rows begins at the primitive grammar, crosses the covariant residual middle gate, and receives its incidence, metric, matter, holonomy or shell companion before the effective form is read.

Executable root: `theorem_effective_equation_commuting_diagram_completion`

Class and stage: Primary; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.753 (377. EFFECTIVE EQUATION COMMUTING DIAGRAM COMPLETION THEOREM)**.** *Prove that every declared effective equation passes through the same residual middle gate and an explicitly generated companion structure.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.754. EFFECTIVE EQUATION COMMUTING DIAGRAM COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 378. COMPACT GAUGE CHIRAL MATTER QUANTUM GRAVITY CLOSURE

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_compact_gauge_chiral_matter_quantum_gravity`

Class and stage: Narrative; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.755 (378. COMPACT GAUGE CHIRAL MATTER QUANTUM GRAVITY CLOSURE)**.** *Execute the exact certificate for COMPACT GAUGE CHIRAL MATTER QUANTUM GRAVITY CLOSURE on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.756. COMPACT GAUGE CHIRAL MATTER QUANTUM GRAVITY CLOSURE holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 379. UNIT SCALE QUOTIENT CHIRAL CONTENT AND NONLINEAR GRAVITY

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_unit_scale_chiral_content_nonlinear_gravity`

Class and stage: Narrative; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.757 (379. UNIT SCALE QUOTIENT CHIRAL CONTENT AND NONLINEAR GRAVITY)**.** *Execute the exact certificate for UNIT SCALE QUOTIENT CHIRAL CONTENT AND NONLINEAR GRAVITY on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.758. UNIT SCALE QUOTIENT CHIRAL CONTENT AND NONLINEAR GRAVITY holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 380. GENERAL METRIC VARIATION PROJECTIVE MEASURE AND IDENTIFIABILITY

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_general_metric_measure_identifiability`

Class and stage: Narrative; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.759 (380. GENERAL METRIC VARIATION PROJECTIVE MEASURE AND IDENTIFIABILITY)**.** *Execute the exact certificate for GENERAL METRIC VARIATION PROJECTIVE MEASURE AND IDENTIFIABILITY on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.760. GENERAL METRIC VARIATION PROJECTIVE MEASURE AND IDENTIFIABILITY holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 381. SHELL CONVERSE ACTION EQUIVALENCE AND HIERARCHICAL QUANTUM MEASURE

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_shell_converse_action_quantum_measure`

Class and stage: Narrative; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.761 (381. SHELL CONVERSE ACTION EQUIVALENCE AND HIERARCHICAL QUANTUM MEASURE)**.** *Execute the exact certificate for SHELL CONVERSE ACTION EQUIVALENCE AND HIERARCHICAL QUANTUM MEASURE on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.762. SHELL CONVERSE ACTION EQUIVALENCE AND HIERARCHICAL QUANTUM MEASURE holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 382. PRIMITIVE PROVENANCE ACTION RATIOS AND GAUGE INVARIANTS

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_primitive_provenance_action_gauge_invariants`

Class and stage: Narrative; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.763 (382. PRIMITIVE PROVENANCE ACTION RATIOS AND GAUGE INVARIANTS)**.** *Execute the exact certificate for PRIMITIVE PROVENANCE ACTION RATIOS AND GAUGE INVARIANTS on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.764. PRIMITIVE PROVENANCE ACTION RATIOS AND GAUGE INVARIANTS holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 383. PARENT KERNEL RESOLVENT RENORMALIZATION AND SCATTERING

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_parent_kernel_renormalization_scattering`

Class and stage: Narrative; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.765 (383. PARENT KERNEL RESOLVENT RENORMALIZATION AND SCATTERING)**.** *Execute the exact certificate for PARENT KERNEL RESOLVENT RENORMALIZATION AND SCATTERING on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.766. PARENT KERNEL RESOLVENT RENORMALIZATION AND SCATTERING holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 384. CONNECTED GAUGE COMPLETION SPECTRAL SCATTERING AND LINEARIZED GRAVITY

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_connected_gauge_scattering_linearized_gravity`

Class and stage: Narrative; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.767 (384. CONNECTED GAUGE COMPLETION SPECTRAL SCATTERING AND LINEARIZED GRAVITY)**.** *Execute the exact certificate for CONNECTED GAUGE COMPLETION SPECTRAL SCATTERING AND LINEARIZED GRAVITY on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.768. CONNECTED GAUGE COMPLETION SPECTRAL SCATTERING AND LINEARIZED GRAVITY holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 385. PROJECTIVE MEASURE INTERTWINER AND POSITIVE QUANTUM GEOMETRY

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_projective_intertwiner_positive_geometry`

Class and stage: Narrative; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.769 (385. PROJECTIVE MEASURE INTERTWINER AND POSITIVE QUANTUM GEOMETRY)**.** *Execute the exact certificate for PROJECTIVE MEASURE INTERTWINER AND POSITIVE QUANTUM GEOMETRY on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.770. PROJECTIVE MEASURE INTERTWINER AND POSITIVE QUANTUM GEOMETRY holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 386. CURVATURE SENSITIVE MEASURE AND NONLINEAR FIXED POINT

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_curvature_measure_nonlinear_fixed_point`

Class and stage: Narrative; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.771 (386. CURVATURE SENSITIVE MEASURE AND NONLINEAR FIXED POINT)**.** *Execute the exact certificate for CURVATURE SENSITIVE MEASURE AND NONLINEAR FIXED POINT on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.772. CURVATURE SENSITIVE MEASURE AND NONLINEAR FIXED POINT holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 387. TRACE CLASS DETERMINANT AND TOPOLOGY CHANGING MEASURE

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_trace_class_determinant_topology_measure`

Class and stage: Narrative; 7. COVARIANT CONTINUUM FIELD CONSTRUCTION

Theorem A.773 (387. TRACE CLASS DETERMINANT AND TOPOLOGY CHANGING MEASURE)**.** *Execute the exact certificate for TRACE CLASS DETERMINANT AND TOPOLOGY CHANGING MEASURE on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.774. TRACE CLASS DETERMINANT AND TOPOLOGY CHANGING MEASURE holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 388. U862 MULTI FRONTIER ADVANCEMENT THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u862_multi_frontier_advancement_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.775 (388. U862 MULTI FRONTIER ADVANCEMENT THEOREM)**.** *Execute the exact certificate for U862 MULTI FRONTIER ADVANCEMENT THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.776. U862 MULTI FRONTIER ADVANCEMENT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 389. U863 FINAL CONTINUUM QUANTUM STRUCTURE THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u863_final_continuum_quantum_structure_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.777 (389. U863 FINAL CONTINUUM QUANTUM STRUCTURE THEOREM)**.** *Execute the exact certificate for U863 FINAL CONTINUUM QUANTUM STRUCTURE THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.778. U863 FINAL CONTINUUM QUANTUM STRUCTURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 390. U864 NONLINEAR CHIRAL COLOR COMPLETION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u864_nonlinear_chiral_color_completion_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.779 (390. U864 NONLINEAR CHIRAL COLOR COMPLETION THEOREM)**.** *Execute the exact certificate for U864 NONLINEAR CHIRAL COLOR COMPLETION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.780. U864 NONLINEAR CHIRAL COLOR COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 391. U865 CHIRAL QUANTUM COLOR GEOMETRY COMPLETION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u865_chiral_quantum_color_geometry_completion_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.781 (391. U865 CHIRAL QUANTUM COLOR GEOMETRY COMPLETION THEOREM)**.** *Execute the exact certificate for U865 CHIRAL QUANTUM COLOR GEOMETRY COMPLETION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.782. U865 CHIRAL QUANTUM COLOR GEOMETRY COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 392. FOUR FORCE COMPLETE PROOF

Place the complete native proof in a dependency order readable without hidden jumps.

Executable root: `_complete_dependency_ordered_four_force_proof_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.783 (392. FOUR FORCE COMPLETE PROOF)**.** *Place the complete native proof in a dependency order readable without hidden jumps.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.784. FOUR FORCE COMPLETE PROOF holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 393. U867 QUANTUM RECONSTRUCTION FRONTIER THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_u867_quantum_reconstruction_frontier_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.785 (393. U867 QUANTUM RECONSTRUCTION FRONTIER THEOREM)**.** *Execute the exact certificate for U867 QUANTUM RECONSTRUCTION FRONTIER THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.786. U867 QUANTUM RECONSTRUCTION FRONTIER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 394. FOUR FORCE COMPLETE PROOF

This theorem supplies the generated object required by the next causal stage.

Executable root: `_complete_quantum_reconstruction_four_force_proof_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.787 (394. FOUR FORCE COMPLETE PROOF)**.** *Execute the exact certificate for FOUR FORCE COMPLETE PROOF on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.788. FOUR FORCE COMPLETE PROOF holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 395. FOUR FORCE COMPLETE PROOF

Verify that every displayed conclusion follows from earlier proved inputs without a hidden dependency jump.

Executable root: `_u868_no_jump_dependency_proof_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.789 (395. FOUR FORCE COMPLETE PROOF)**.** *Verify that every displayed conclusion follows from earlier proved inputs without a hidden dependency jump.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.790. FOUR FORCE COMPLETE PROOF holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 396. FINITE CERTIFICATE NONIMPLICATION THEOREM

Construct exact pairs that agree on every declared finite certificate but differ on an unrestricted target.

Executable root: `_u869_finite_certificate_nonimplication_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.791 (396. FINITE CERTIFICATE NONIMPLICATION THEOREM)**.** *Construct exact pairs that agree on every declared finite certificate but differ on an unrestricted target.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.792. FINITE CERTIFICATE NONIMPLICATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 397. MINIMAL UNRESTRICTED COMPLETION THEOREM

Prove the minimal six-condition implication structure for an unrestricted physical completion.

Executable root: `_u869_minimal_unrestricted_completion_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.793 (397. MINIMAL UNRESTRICTED COMPLETION THEOREM)**.** *Prove the minimal six-condition implication structure for an unrestricted physical completion.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.794. MINIMAL UNRESTRICTED COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 398. FOUR FORCE COMPLETE PROOF

Extend the no-jump proof by one final chapter that separates proved conclusions from independent completion obligations.

Executable root: `_u869_complete_boundary_ordered_four_force_proof_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.795 (398. FOUR FORCE COMPLETE PROOF)**.** *Extend the no-jump proof by one final chapter that separates proved conclusions from independent completion obligations.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.796. FOUR FORCE COMPLETE PROOF holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 399. MINIMAL CUMULATIVE RECYCLE CONSERVATION THEOREM

Derive variable self-similar scale growth from the full prior structural history while conserving one physical total.

Executable root: `_u870_minimal_cumulative_scale_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.797 (399. MINIMAL CUMULATIVE RECYCLE CONSERVATION THEOREM)**.** *Derive variable self-similar scale growth from the full prior structural history while conserving one physical total.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.798. MINIMAL CUMULATIVE RECYCLE CONSERVATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 400. FOUR FORCE CUMULATIVE RECYCLE PROJECTION THEOREM

Project one conservative cumulative recycle carrier into signed, transition, locking, and geometric four-force readouts.

Executable root: `_u870_four_force_recycle_projection_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.799 (400. FOUR FORCE CUMULATIVE RECYCLE PROJECTION THEOREM)**.** *Project one conservative cumulative recycle carrier into signed, transition, locking, and geometric four-force readouts.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.800. FOUR FORCE CUMULATIVE RECYCLE PROJECTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 401. CUMULATIVE RECYCLE UNBOUNDED EXTENSION THEOREM

Prove four exact recursive-class extensions from cumulative recycle growth while keeping unrestricted hypotheses separate.

Executable root: `_u870_recursive_unbounded_extension_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.801 (401. CUMULATIVE RECYCLE UNBOUNDED EXTENSION THEOREM)**.** *Prove four exact recursive-class extensions from cumulative recycle growth while keeping unrestricted hypotheses separate.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.802. CUMULATIVE RECYCLE UNBOUNDED EXTENSION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 402. FOUR FORCE COMPLETE PROOF

Append the cumulative recycle law to the dependency-ordered four-force proof without converting restricted bridges into unrestricted claims.

Executable root: `_u870_cumulative_recycle_ordered_unification_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.803 (402. FOUR FORCE COMPLETE PROOF)**.** *Append the cumulative recycle law to the dependency-ordered four-force proof without converting restricted bridges into unrestricted claims.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.804. FOUR FORCE COMPLETE PROOF holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 403. PROJECTIVE CONTINUUM AMPLITUDE UNIQUENESS THEOREM

Construct the unique positive nonfactorizing cylinder amplitude functional on the infinite recursive path class.

Executable root: `_u871_projective_continuum_amplitude_uniqueness_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.805 (403. PROJECTIVE CONTINUUM AMPLITUDE UNIQUENESS THEOREM)**.** *Construct the unique positive nonfactorizing cylinder amplitude functional on the infinite recursive path class.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.806. PROJECTIVE CONTINUUM AMPLITUDE UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 404. SU3 DIRECT LIMIT UNIFORM GAP THEOREM

Prove a uniform positive color-Casimir gap on the countable finite-support SU3 direct-limit carrier.

Executable root: `_u871_su3_direct_limit_uniform_gap_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.807 (404. SU3 DIRECT LIMIT UNIFORM GAP THEOREM)**.** *Prove a uniform positive color-Casimir gap on the countable finite-support SU3 direct-limit carrier.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.808. SU3 DIRECT LIMIT UNIFORM GAP THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 405. UNIQUE STRUCTURAL RATIO AND MIXING DECODER THEOREM

Derive a unique scale-covariant ratio and mixing decoder inside the declared structural-signature class.

Executable root: `_u871_unique_ratio_mixing_decoder_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.809 (405. UNIQUE STRUCTURAL RATIO AND MIXING DECODER THEOREM)**.** *Derive a unique scale-covariant ratio and mixing decoder inside the declared structural-signature class.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.810. UNIQUE STRUCTURAL RATIO AND MIXING DECODER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 406. TOPOLOGY PROJECTIVE NONLINEAR QUANTUM GEOMETRY THEOREM

Combine Pachner invariance, recursive projective conservation, nonlinear classical geometry, and physical graviton cohomology.

Executable root: `_u871_topology_projective_nonlinear_quantum_geometry_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.811 (406. TOPOLOGY PROJECTIVE NONLINEAR QUANTUM GEOMETRY THEOREM)**.** *Combine Pachner invariance, recursive projective conservation, nonlinear classical geometry, and physical graviton cohomology.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.812. TOPOLOGY PROJECTIVE NONLINEAR QUANTUM GEOMETRY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 407. FOUR FORCE COMPLETE PROOF

Integrate four exact frontier theorems into the dependency-ordered four-force proof.

Executable root: `_u871_four_frontier_recursive_completion_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.813 (407. FOUR FORCE COMPLETE PROOF)**.** *Integrate four exact frontier theorems into the dependency-ordered four-force proof.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.814. FOUR FORCE COMPLETE PROOF holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 408. SPECTRAL CYLINDER RESIDUE EQUIVALENCE THEOREM

Prove an exact one-to-one moment, resolvent, pole-residue, and unitary-scattering bridge in the declared three-mode class.

Executable root: `_u872_lsz_cylinder_residue_equivalence_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.815 (408. SPECTRAL CYLINDER RESIDUE EQUIVALENCE THEOREM)**.** *Prove an exact one-to-one moment, resolvent, pole-residue, and unitary-scattering bridge in the declared three-mode class.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.816. SPECTRAL CYLINDER RESIDUE EQUIVALENCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 409. SU3 HILBERT COMPLETION GAP THEOREM

Extend the exact SU3 Casimir gap from finite support to the declared weighted Hilbert completion by a coercive quadratic-form identity.

Executable root: `_u872_su3_hilbert_completion_gap_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.817 (409. SU3 HILBERT COMPLETION GAP THEOREM)**.** *Extend the exact SU3 Casimir gap from finite support to the declared weighted Hilbert completion by a coercive quadratic-form identity.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.818. SU3 HILBERT COMPLETION GAP THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 410. PREDECLARED STRUCTURAL RATIO DECODER SEAL THEOREM

Generate and seal a unique internal ratio, charge, mixing, color, and gravity packet before any external comparison.

Executable root: `_u872_predeclared_ratio_decoder_seal_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.819 (410. PREDECLARED STRUCTURAL RATIO DECODER SEAL THEOREM)**.** *Generate and seal a unique internal ratio, charge, mixing, color, and gravity packet before any external comparison.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.820. PREDECLARED STRUCTURAL RATIO DECODER SEAL THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 411. SMOOTH LORENTZIAN COFINAL REFINEMENT THEOREM

Prove exact cofinal PL refinement and smooth polynomial Lorentzian-volume convergence on the declared compact coframe class.

Executable root: `_u872_smooth_lorentzian_refinement_equivalence_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.821 (411. SMOOTH LORENTZIAN COFINAL REFINEMENT THEOREM)**.** *Prove exact cofinal PL refinement and smooth polynomial Lorentzian-volume convergence on the declared compact coframe class.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.822. SMOOTH LORENTZIAN COFINAL REFINEMENT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 412. EXTERNAL REPRODUCTION AND BLIND PROTOCOL THEOREM

Create an immutable independent-reproduction and blind-comparison protocol without claiming an external event has occurred.

Executable root: `_u872_external_reproduction_and_blind_protocol_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.823 (412. EXTERNAL REPRODUCTION AND BLIND PROTOCOL THEOREM)**.** *Create an immutable independent-reproduction and blind-comparison protocol without claiming an external event has occurred.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.824. EXTERNAL REPRODUCTION AND BLIND PROTOCOL THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 413. FOUR FORCE COMPLETE PROOF

Integrate four strengthened mathematical bridges and two immutable external protocols into the ordered four-force proof.

Executable root: `_u872_six_frontier_integrated_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.825 (413. FOUR FORCE COMPLETE PROOF)**.** *Integrate four strengthened mathematical bridges and two immutable external protocols into the ordered four-force proof.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.826. FOUR FORCE COMPLETE PROOF holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 414. RECURSIVE BLOCK TRANSFER EQUIVALENCE THEOREM

Prove exact variable-scale block transfer consistency, positivity, interaction persistence, and a uniform spectral gap.

Executable root: `_u873_recursive_block_transfer_equivalence_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.827 (414. RECURSIVE BLOCK TRANSFER EQUIVALENCE THEOREM)**.** *Prove exact variable-scale block transfer consistency, positivity, interaction persistence, and a uniform spectral gap.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.828. RECURSIVE BLOCK TRANSFER EQUIVALENCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 415. ALL PARTICLE CLUSTER SCATTERING THEOREM

Lift an exact rational one-particle scattering family to a norm-preserving tensor carrier and prove cluster factorization.

Executable root: `_u873_all_particle_cluster_scattering_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.829 (415. ALL PARTICLE CLUSTER SCATTERING THEOREM)**.** *Lift an exact rational one-particle scattering family to a norm-preserving tensor carrier and prove cluster factorization.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.830. ALL PARTICLE CLUSTER SCATTERING THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 416. SU3 GAUGE INVARIANT NETWORK GAP THEOREM

Prove a uniform positive gap on an exact gauge-invariant meson, baryon, and glue-network Hilbert completion.

Executable root: `_u873_su3_gauge_invariant_network_gap_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.831 (416. SU3 GAUGE INVARIANT NETWORK GAP THEOREM)**.** *Prove a uniform positive gap on an exact gauge-invariant meson, baryon, and glue-network Hilbert completion.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.832. SU3 GAUGE INVARIANT NETWORK GAP THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 417. HIERARCHICAL BOUND STATE DECODER THEOREM

Generate a unique scale and shell spectral fingerprint from native thresholds and shell capacities without observed inputs.

Executable root: `_u873_hierarchical_bound_state_decoder_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.833 (417. HIERARCHICAL BOUND STATE DECODER THEOREM)**.** *Generate a unique scale and shell spectral fingerprint from native thresholds and shell capacities without observed inputs.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.834. HIERARCHICAL BOUND STATE DECODER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 418. NONLINEAR SCALE COVARIANT GRAVITY THEOREM

Prove an exact nonlinear Einstein-Maxwell recursive scaling family with invariant dimensionless geometry and conserved integrated burden.

Executable root: `_u873_nonlinear_scale_covariant_gravity_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.835 (418. NONLINEAR SCALE COVARIANT GRAVITY THEOREM)**.** *Prove an exact nonlinear Einstein-Maxwell recursive scaling family with invariant dimensionless geometry and conserved integrated burden.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.836. NONLINEAR SCALE COVARIANT GRAVITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 419. NONLINEAR GRAVITY BLACK HOLE COSMOLOGY THEOREM

The gravity sector is tested on exact nonlinear families rather than only on a linearized shadow: charged spherical horizons, Ricci-flat Schwarzschild limit, rational Kasner vacuum, perfect-fluid FLRW evolution, Lorentzian constraint propagation, and the two-polarization inverse-limit action.

Executable root: `_x893_nonlinear_gravity_blackhole_cosmology_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.837 (419. NONLINEAR GRAVITY BLACK HOLE COSMOLOGY THEOREM)**.** *Combine exact nonlinear vacuum, charged spherical, anisotropic, cosmological, constraint, and inverse-limit gravity families.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.838. NONLINEAR GRAVITY BLACK HOLE COSMOLOGY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 420. SOURCE GENERATED METRIC GRAVITY COVARIANCE THEOREM

The metric is a Gram shadow of the source coframe and is invariant under exact source-frame transformations. Coordinate congruence preserves scalar curvature and density weight while the native Palatini selector and nonlinear metric variation supply the gravitational dynamics.

Executable root: `theorem_source_generated_metric_gravity_covariance`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.839 (420. SOURCE GENERATED METRIC GRAVITY COVARIANCE THEOREM)**.** *Generate a metric from a source coframe using the generated Lorentz signature class and prove exact covariance.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.840. SOURCE GENERATED METRIC GRAVITY COVARIANCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 421. DIMENSION FLUX POTENTIAL HIERARCHY THEOREM

One conserved radial flux produces a field proportional to the reciprocal shell measure r^(d-1). The two-dimensional primitive belongs to the logarithmic class, while generated three-dimensional space gives inverse-radius potential and inverse-square field.

Executable root: `theorem_dimension_flux_potential_hierarchy`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.841 (421. DIMENSION FLUX POTENTIAL HIERARCHY THEOREM)**.** *Derive inverse-power shell ratios from generated rank, derived mass signatures and fixed-flux action minimization.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.842. DIMENSION FLUX POTENTIAL HIERARCHY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 422. DIMENSION GREEN FAMILY BRIDGE THEOREM

One conserved radial flux gives dimension-dependent fields through the shell measure r^(d-1). The two-dimensional additive-depth potential and the three-dimensional inverse-radius potential are therefore one Green-family law.

Executable root: `theorem_dimension_green_family_bridge`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.843 (422. DIMENSION GREEN FAMILY BRIDGE THEOREM)**.** *Derive the logarithmic two-dimensional and inverse-power three-dimensional Green families from one conserved radial flux law.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.844. DIMENSION GREEN FAMILY BRIDGE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 423. PRIMITIVE MASS VALUATION UNIVERSALITY THEOREM

A composite body is a finite disjoint union of occupied primitive seats, so one normalized additive valuation fixes its mass ratio. Because one parent matter action is extensive over those same seats, kinetic and source-coupling terms receive one common coefficient rather than two fitted masses.

Executable root: `theorem_primitive_mass_valuation_universality`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.845 (423. PRIMITIVE MASS VALUATION UNIVERSALITY THEOREM)**.** *Derive the common inertial-source coefficient from finite additivity and generated parent-action coefficients.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.846. PRIMITIVE MASS VALUATION UNIVERSALITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 424. NEWTON EQUIVALENCE RATIO THEOREM

The common gravity-sector response multiplies source and test mass and is diluted by the generated three-dimensional shell measure. Dividing force by the same native inertial mass cancels the test body exactly, leaving universal acceleration ratios.

Executable root: `theorem_newton_equivalence_ratio`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.847 (424. NEWTON EQUIVALENCE RATIO THEOREM)**.** *Derive test-body-independent acceleration from the additive one-parent worldline action.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.848. NEWTON EQUIVALENCE RATIO THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 425. HORIZON ZERO BOUNDARY CONJUGACY THEOREM

A regular source depth produces a residual that contracts by an exact primitive ratio and a bounded coordinate that approaches zero. The finite shadow boundary and the unbounded additive depth are inverse descriptions of the same orbit, not separate source laws.

Executable root: `theorem_horizon_zero_boundary_conjugacy`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.849 (425. HORIZON ZERO BOUNDARY CONJUGACY THEOREM)**.** *Prove that a regular unit-depth source orbit is compressed toward the zero shadow boundary while the inverse additive depth remains unbounded.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.850. HORIZON ZERO BOUNDARY CONJUGACY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 426. HORIZON BOUNDARY RATE RATIO THEOREM

The zero-boundary generator has a unique unit-normalized inward rate in the boundary limit and a primitive one-step contraction ratio. The gravity-sector coefficient weights this rate while no absolute surface-gravity unit is inserted.

Executable root: `theorem_horizon_boundary_rate_ratio`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.851 (426. HORIZON BOUNDARY RATE RATIO THEOREM)**.** *Derive a unique dimensionless near-zero boundary rate and its gravity-sector weighting without an absolute surface-gravity calibration.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.852. HORIZON BOUNDARY RATE RATIO THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 427. NATIVE BOUNDARY ENTROPY AREA DEPTH THEOREM

Independent native boundary seats multiply hidden state counts. The additive base-depth decoder converts this product into an information quantity exactly proportional to boundary seat number.

Executable root: `theorem_native_boundary_entropy_area_depth`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.853 (427. NATIVE BOUNDARY ENTROPY AREA DEPTH THEOREM)**.** *Derive an additive boundary-information depth from multiplicative native microstate counts on the generated shell seats.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.854. NATIVE BOUNDARY ENTROPY AREA DEPTH THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 428. SOURCE SHADOW INFORMATION CONSERVATION THEOREM

Visible and hidden fibers multiply to the full source count for every partition of the eighteen-state carrier. Projection may hide information from one shadow algebra, but the complete source-shadow reconstruction does not destroy it.

Executable root: `theorem_source_shadow_information_conservation`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.855 (428. SOURCE SHADOW INFORMATION CONSERVATION THEOREM)**.** *Prove exact visible-hidden information balance for every partition of the eighteen-state source carrier.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.856. SOURCE SHADOW INFORMATION CONSERVATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 429. PROJECTION SINGULARITY CLASSIFIER THEOREM

The forward projection generator remains finite and vanishes at the boundary, while its inverse decoder Jacobian grows without bound along the exact orbit. Constant source trace and determinant classify this as a projection-coordinate singularity; a genuine source singularity requires source-invariant failure.

Executable root: `theorem_projection_singularity_classifier`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.857 (429. PROJECTION SINGULARITY CLASSIFIER THEOREM)**.** *Separate inverse-decoder divergence caused by a bounded projection from divergence of source invariants.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.858. PROJECTION SINGULARITY CLASSIFIER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 430. DEPTH COFRAME LORENTZ METRIC COMPLETION THEOREM

One temporal and three residual depth one-forms form a canonical coframe. Their Lorentz Gram tensor is unchanged when each bounded shadow axis is projectively compressed.

Executable root: `theorem_depth_coframe_lorentz_metric_completion`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.859 (430. DEPTH COFRAME LORENTZ METRIC COMPLETION THEOREM)**.** *Use the invariant depth one-forms as a canonical coframe and recover the generated Lorentz metric under projective shadow changes.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.860*.* DEPTH COFRAME LORENTZ METRIC COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 431. EXACT NONLINEAR COVARIANT BIANCHI LIFT THEOREM

A variable temporal-depth coframe produces a nonconstant metric, a nonzero Levi-Civita connection, nonzero Ricci curvature and a full nonlinear Einstein tensor. Its ordinary divergence is balanced exactly by the connection correction, leaving the covariant divergence identically zero.

Executable root: `theorem_exact_nonlinear_covariant_bianchi_lift`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.861 (431. EXACT NONLINEAR COVARIANT BIANCHI LIFT THEOREM)**.** *Compute an exact nonconstant four-dimensional metric from a variable coframe and prove that its full Einstein tensor has zero covariant divergence through cancellation of the ordinary-divergence and connection terms.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.862*.* EXACT NONLINEAR COVARIANT BIANCHI LIFT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 432. DYNAMIC COFRAME CARRIER NONLINEAR COMPATIBILITY THEOREM

The eighteen-state grammar and JS incidence determine the finite fiber and its five-one-three-eight sector decomposition. The coframe is a field over the generated base; its coefficients vary, its metric determinant changes, and its curvature is nonzero while the internal carrier dimensions remain exact.

Executable root: `theorem_dynamic_coframe_carrier_nonlinear_compatibility`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.863 (432. DYNAMIC COFRAME CARRIER NONLINEAR COMPATIBILITY THEOREM)**.** *Prove that the fixed finite carrier supplies fiber incidence and sector projectors while the coframe coefficients vary over the generated base and create nonzero nonlinear curvature.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.864. DYNAMIC COFRAME CARRIER NONLINEAR COMPATIBILITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 433. LOCAL SCALE CONNECTION CURVATURE COMPLETION THEOREM

A common central scale shift commutes with every sector and is therefore a flat unit change. Local changes that conjugate the sector compression operator fail to commute and create a nonzero exact curvature.

Executable root: `theorem_local_scale_connection_curvature_completion`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.865 (433. LOCAL SCALE CONNECTION CURVATURE COMPLETION THEOREM)**.** *Separate the central global unit shift from local sector-changing compression and derive a nonzero exact internal curvature from one operator.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.866. LOCAL SCALE CONNECTION CURVATURE COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 434. TEMPORAL COMPRESSION GRAVITY ORIGIN THEOREM

A spatially varying temporal-depth compression changes the lapse coframe, creates a Lorentz metric response, conserves radial flux, and gives probe-independent acceleration ratios.

Executable root: `theorem_temporal_compression_gravity_origin`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.867 (434. TEMPORAL COMPRESSION GRAVITY ORIGIN THEOREM)**.** *Generate the spherical gravity readout from a spatially nonuniform temporal-depth compression and prove its inverse-square flux, metric lapse and probe-independent acceleration ratios.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.868. TEMPORAL COMPRESSION GRAVITY ORIGIN THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 435. TEMPORAL SECOND RESIDUAL SCALE UNIQUENESS THEOREM

The local scale contains the unchanged source, one direct temporal residual and one residual-of-residual. These are the complete independent orders before the generated spatial coframe is formed. Unit normalization of the three generated orders fixes the scale polynomial without a coefficient fit.

Executable root: `theorem_temporal_second_residual_scale_uniqueness`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.869 (435. TEMPORAL SECOND RESIDUAL SCALE UNIQUENESS THEOREM)**.** *Derive the polynomial one plus temporal depth plus temporal depth squared as the unique normalized scale in the generated second-residual class.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.870. TEMPORAL SECOND RESIDUAL SCALE UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 436. TEMPORAL SCALE EDGE RATIO UNIQUENESS THEOREM

An edge compares two locally generated clock scales. Common unit changes must cancel, reversing the edge must invert the factor, and path factors must compose. Inside the typed endpoint-monomial class these requirements uniquely give target scale divided by source scale.

Executable root: `theorem_temporal_scale_edge_ratio_uniqueness`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.871 (436. TEMPORAL SCALE EDGE RATIO UNIQUENESS THEOREM)**.** *Derive the target-over-source temporal scale ratio as the unique typed monomial edge compression compatible with identity, common-scale invariance, orientation, and path composition.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.872. TEMPORAL SCALE EDGE RATIO UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 437. LOG FREE GRAVITY RESIDUAL CORRELATION THEOREM

Gravity is correlated by finite clock-compression ratios rather than by a logarithm. The source and target residuals cross-multiply with their generated temporal scales; a local gravitational frame transport may rotate the five-component metric response without changing the correlation invariant.

Executable root: `theorem_log_free_gravity_residual_correlation`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.873 (437. LOG FREE GRAVITY RESIDUAL CORRELATION THEOREM)**.** *Derive one finite, exact, logarithm-free gravitational residual correlation equation from the generated temporal scale and the gravity projector.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.874. LOG FREE GRAVITY RESIDUAL CORRELATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 438. LOG FREE GRAVITY CONNECTION LIMIT THEOREM

The one-form dN/N is not introduced by evaluating a logarithm. It is the exact first-order coefficient of the finite ratio N(q)/N(p). The covariant residual derivative equals the frame commutator plus this finite-ratio scale rate times the residual.

Executable root: `theorem_log_free_gravity_connection_limit`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.875 (438. LOG FREE GRAVITY CONNECTION LIMIT THEOREM)**.** *Recover the finite-ratio gravitational connection dN/N and its covariant residual derivative as the exact first-order coefficient of the logarithm-free correlation equation.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.876. LOG FREE GRAVITY CONNECTION LIMIT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 439. LOG FREE GRAVITY FINITE DIFFERENCE IDENTITY THEOREM

Before a differential symbol is introduced, the finite gravitational edge already satisfies an exact quotient identity. The transported residual changes by the target-over-source clock ratio, and division by the finite scale increment gives the exact coefficient R divided by N. The continuum dN/N form is only the local notation for this finite identity.

Executable root: `theorem_log_free_gravity_finite_difference_identity`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.877 (439. LOG FREE GRAVITY FINITE DIFFERENCE IDENTITY THEOREM)**.** *Derive the gravitational covariant scale-rate equation as an exact finite-difference identity before taking any continuum interpretation.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.878. LOG FREE GRAVITY FINITE DIFFERENCE IDENTITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 440. FINITE GRAVITY METRIC BRIDGE COMPLETION THEOREM

The primitive gravitational statement is the finite target-over-source temporal-scale ratio. Its local coefficient changes the temporal coframe; the coframe generates a nonconstant Lorentz metric, curvature and an exact covariant Bianchi identity. No evaluated logarithm is required.

Executable root: `theorem_finite_gravity_metric_bridge_completion`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.879 (440. FINITE GRAVITY METRIC BRIDGE COMPLETION THEOREM)**.** *Close the finite gravity ratio, local coefficient, temporal coframe, nonlinear metric curvature, and exact Bianchi chain without evaluating a logarithm.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.880. FINITE GRAVITY METRIC BRIDGE COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 441. RESIDUAL CONSISTENCY FUNCTIONAL SEPARATION THEOREM

The squared residual defect answers only whether a proposed edge state satisfies the kinematic middle equation. It is not varied to create an additional force. The sole physical action is the parent curvature-matter action, whose one variation yields the four field equations and the matter equation.

Executable root: `theorem_residual_consistency_functional_separation`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.881 (441. RESIDUAL CONSISTENCY FUNCTIONAL SEPARATION THEOREM)**.** *Prove that the nonnegative residual defect functional is only an equation-consistency certificate and that the unique physical variational action remains the parent curvature-matter action.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.882. RESIDUAL CONSISTENCY FUNCTIONAL SEPARATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 442. PARENT ACTION VARIATION COEFFICIENT DECOMPOSITION THEOREM

Insert F plus epsilon H into the positive parent curvature response and collect the exact constant, linear and quadratic coefficients. Orthogonal projector products remove every cross-sector term. The parent linear coefficient is exactly the sum of the four sector linear coefficients, so one parent Euler-Lagrange equation is equivalent to its four projected field equations.

Executable root: `theorem_parent_action_variation_coefficient_decomposition`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.883 (442. PARENT ACTION VARIATION COEFFICIENT DECOMPOSITION THEOREM)**.** *Expand the positive parent quadratic curvature response under an exact rational variation and prove coefficient-by-coefficient decomposition into the four orthogonal sector equations.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.884. PARENT ACTION VARIATION COEFFICIENT DECOMPOSITION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 443. SINGLE PARENT ACTION NO FIFTH FORCE COMPLETION THEOREM

The nonnegative residual consistency functional tests whether the middle equation holds and is never varied as a fifth interaction. The sole physical parent action has one variation whose orthogonal projections yield the four field equations, with an independent matter variation.

Executable root: `theorem_single_parent_action_no_fifth_force_completion`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.885 (443. SINGLE PARENT ACTION NO FIFTH FORCE COMPLETION THEOREM)**.** *Prove that residual consistency is a certificate, not a second physical action, and that one variation gives the four field projections and matter equation.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.886. SINGLE PARENT ACTION NO FIFTH FORCE COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 444. COVARIANT RESIDUAL CURVATURE DESCENT THEOREM

The master residual equation separates force roles without separating their source law. Compression eigenvalues encode finite temporal or internal burden, transports encode local frames, and the closed-path defect is the common curvature object used by the parent action.

Executable root: `theorem_covariant_residual_curvature_descent`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.887 (444. COVARIANT RESIDUAL CURVATURE DESCENT THEOREM)**.** *Join finite residual holonomy, local temporal compression, Abelian phase curvature, compact non-Abelian curvature, and nonlinear metric curvature into one parent-curvature descent.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.888. COVARIANT RESIDUAL CURVATURE DESCENT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 445. COVARIANT RESIDUAL PARENT ACTION BRIDGE THEOREM

The residual field equation is the kinematic middle gate between temporal-dimensional genesis and the parent action. Its nonnegative consistency functional vanishes on the generated transport and becomes positive under a one-entry mutation. This functional is not a second physical action. The same transport supplies the parent curvature, and one parent variation supplies the four field equations and matter equation.

Executable root: `theorem_covariant_residual_parent_action_bridge`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.889 (445. COVARIANT RESIDUAL PARENT ACTION BRIDGE THEOREM)**.** *Place the covariant residual field equation immediately before the single parent curvature action and prove its exact zero-defect variational bridge.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.890. COVARIANT RESIDUAL PARENT ACTION BRIDGE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 446. GRAVITY TEMPORAL ROLE UNIQUENESS THEOREM

Only the five-component spin-two role changes the universal temporal-depth coframe and metric. The one-, three- and eight-component roles act on internal gauge carriers.

Executable root: `theorem_gravity_temporal_role_uniqueness`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.891 (446. GRAVITY TEMPORAL ROLE UNIQUENESS THEOREM)**.** *Prove that only the generated five-component spin-two sector acts universally on the temporal-depth coframe and metric, while the one-, three- and eight-component sectors remain internal gauge curvatures.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.892. GRAVITY TEMPORAL ROLE UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 447. NONLINEAR PARENT MODE LOCAL WELL-POSEDNESS THEOREM

The generated parent Hessian and a zero-sum skew transport define a nonlinear constrained mode. The common constraint propagates exactly, the parent energy decreases by the fourth-power residual burden, and one rational Banach interval gives existence and uniqueness uniformly through every declared refinement level.

Executable root: `theorem_nonlinear_parent_mode_local_wellposedness`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.893 (447. NONLINEAR PARENT MODE LOCAL WELLPOSEDNESS THEOREM)**.** *Construct a nonlinear constrained four-sector parent mode, prove exact energy dissipation and constraint propagation, and give a uniform rational contraction interval for every native refinement level.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.894. NONLINEAR PARENT MODE LOCAL WELLPOSEDNESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 448. NONLINEAR PARENT LATTICE GLOBAL ENERGY THEOREM

On every finite periodic relation lattice the parent field follows the projected gradient of a positive edge-plus-quartic energy. The local zero-sum constraint propagates, the energy derivative is minus the exact projected-gradient norm, and the quartic term gives a coercive continuation bound.

Executable root: `theorem_nonlinear_parent_lattice_global_energy`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.895 (448. NONLINEAR PARENT LATTICE GLOBAL ENERGY THEOREM)**.** *Construct the constrained nonlinear parent field on every finite periodic relation lattice, prove exact energy decay, constraint propagation and a coercive continuation bound.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.896. NONLINEAR PARENT LATTICE GLOBAL ENERGY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 449. NONLINEAR GRAVITY PARENT FIELD COMPLETION THEOREM

The temporal-depth coframe is lifted to a full variable metric with exact nonlinear curvature and covariant Bianchi balance. The same generated carrier supports a constrained nonlinear parent field, one parent variation, exact gravitational solution families and sector-resolved scale flow.

Executable root: `theorem_nonlinear_gravity_parent_field_completion`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.897 (449. NONLINEAR GRAVITY PARENT FIELD COMPLETION THEOREM)**.** *Join the exact nonlinear metric, covariant Bianchi balance, variable coframe, nonlinear parent lattice, exact black-hole and cosmological families, parent variation and sector-resolved running into one dependency-ordered field theorem.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.898*.* NONLINEAR GRAVITY PARENT FIELD COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 450. SU3 CENTER FLUX AREA LAW COMPLETION THEOREM

The uniform singlet gap prevents arbitrarily soft colored excitations, while the positive flux-tube coercivity supplies one transfer penalty for every enclosed center-flux plaquette. Rectangular loop amplitudes therefore multiply by area rather than perimeter in the declared native lattice class.

Executable root: `theorem_su3_center_flux_area_law_completion`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.899 (450. SU3 CENTER FLUX AREA LAW COMPLETION THEOREM)**.** *Combine the generated uniform color gap and positive flux-tube coercivity into an exact rectangular area-law transfer family on the declared gauge-invariant center-flux system.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.900*.* SU3 CENTER FLUX AREA LAW COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 451. GRAVITATIONAL PROJECTION METRIC LIGHT PATH THEOREM

A total closure burden can alter the effective metric decoder while the native adjacency graph remains fixed. The minimum-cost causal path and arrival time then change, which is the exact structural content required before comparing with gravitational delay or lensing.

Executable root: `_e897_gravitational_projection_metric_light_path_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.901 (451. GRAVITATIONAL PROJECTION METRIC LIGHT PATH THEOREM)**.** *Show exactly that a closure-burden-dependent effective metric changes light-candidate travel time and geodesic readout on a fixed native graph.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.902. GRAVITATIONAL PROJECTION METRIC LIGHT PATH THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 452. FOUR FORCE COMPLETE PROOF

Integrate five independently registered theorem roots into the ordered four-force dependency chain.

Executable root: `_u873_five_frontier_integrated_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.903 (452. FOUR FORCE COMPLETE PROOF)**.** *Integrate five independently registered theorem roots into the ordered four-force dependency chain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.904. FOUR FORCE COMPLETE PROOF holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 453. INTERACTING FINITE FOCK SCATTERING THEOREM

Construct exact norm-preserving interacting circuits for every finite particle number and prove separated-cluster factorization.

Executable root: `_u874_interacting_fock_scattering_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.905 (453. INTERACTING FINITE FOCK SCATTERING THEOREM)**.** *Construct exact norm-preserving interacting circuits for every finite particle number and prove separated-cluster factorization.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.906. INTERACTING FINITE FOCK SCATTERING THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 454. SU3 FLUX TUBE COERCIVITY THEOREM

Prove a uniform gauge-invariant flux-tube coercive gap on the recursive direct-limit carrier and its square-summable completion.

Executable root: `_u874_su3_flux_tube_coercivity_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.907 (454. SU3 FLUX TUBE COERCIVITY THEOREM)**.** *Prove a uniform gauge-invariant flux-tube coercive gap on the recursive direct-limit carrier and its square-summable completion.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.908. SU3 FLUX TUBE COERCIVITY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 455. RATIO TRANSPORT UNIQUENESS THEOREM

Prove that the complete internal generation-shell ratio graph is scale invariant and has only the identity structure-preserving decoder.

Executable root: `_u874_ratio_transport_uniqueness_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.909 (455. RATIO TRANSPORT UNIQUENESS THEOREM)**.** *Prove that the complete internal generation-shell ratio graph is scale invariant and has only the identity structure-preserving decoder.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.910. RATIO TRANSPORT UNIQUENESS THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 456. LORENTZIAN CONSTRAINT PROPAGATION THEOREM

Prove exact nonlinear-scale Lorentzian constraint propagation by a rational canonical boost across every recursive level.

Executable root: `_u874_lorentzian_constraint_propagation_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.911 (456. LORENTZIAN CONSTRAINT PROPAGATION THEOREM)**.** *Prove exact nonlinear-scale Lorentzian constraint propagation by a rational canonical boost across every recursive level.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.912. LORENTZIAN CONSTRAINT PROPAGATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 457. CONSERVED GROWTH FOUR FORCE LIMIT THEOREM

Prove by exact recurrence that structural support grows while the parent four-force total and every projected sector total remain conserved at every recursive depth.

Executable root: `_u874_conserved_growth_four_force_limit_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.913 (457. CONSERVED GROWTH FOUR FORCE LIMIT THEOREM)**.** *Prove by exact recurrence that structural support grows while the parent four-force total and every projected sector total remain conserved at every recursive depth.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.914. CONSERVED GROWTH FOUR FORCE LIMIT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 458. FOUR FORCE COMPLETE PROOF

Place five new exact roots after the preceding proof chain and separate their native completion from unrestricted empirical equivalence.

Executable root: `_u874_five_frontier_completion_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.915 (458. FOUR FORCE COMPLETE PROOF)**.** *Place five new exact roots after the preceding proof chain and separate their native completion from unrestricted empirical equivalence.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.916. FOUR FORCE COMPLETE PROOF holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 459. CROSSING UNITARY SCATTERING THEOREM

Construct an exact rational multichannel scattering family with crossing symmetry, optical consistency, and reversible Cayley reconstruction at every recursive scale.

Executable root: `_u875_crossing_unitary_scattering_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.917 (459. CROSSING UNITARY SCATTERING THEOREM)**.** *Construct an exact rational multichannel scattering family with crossing symmetry, optical consistency, and reversible Cayley reconstruction at every recursive scale.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.918. CROSSING UNITARY SCATTERING THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 460. SU3 CENTER GRAPH GAP THEOREM

Prove a graph-size-independent positive gap for every nonzero divergence-free SU3 center-triality flux on an arbitrary finite simple graph.

Executable root: `_u875_su3_center_graph_gap_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.919 (460. SU3 CENTER GRAPH GAP THEOREM). *Prove a graph-size-independent positive gap for every nonzero divergence-free SU3 center-triality flux on an arbitrary finite simple graph.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.920. SU3 CENTER GRAPH GAP THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 461. LEAVE ONE OUT RATIO PREDICTION THEOREM

Prove that every internal generation-shell state is uniquely recovered after its row is withheld, using only dimensionless relations to the remaining states.

Executable root: `_u875_leave_one_out_ratio_prediction_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.921 (461. LEAVE ONE OUT RATIO PREDICTION THEOREM). *Prove that every internal generation-shell state is uniquely recovered after its row is withheld, using only dimensionless relations to the remaining states.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.922. LEAVE ONE OUT RATIO PREDICTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 462. TWO POLARIZATION PROJECTIVE FIELD THEOREM

Construct a normalized projective continuum field with exact martingale consistency, rational variance control, and two positive physical graviton polarizations over a nonlinear recursive background.

Executable root: `_u875_two_polarization_projective_field_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.923 (462. TWO POLARIZATION PROJECTIVE FIELD THEOREM). *Construct a normalized projective continuum field with exact martingale consistency, rational variance control, and two positive physical graviton polarizations over a nonlinear recursive background.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.924. TWO POLARIZATION PROJECTIVE FIELD THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 463. RECURSIVE CONTINUUM FIXED POINT THEOREM

Prove exact convergence of the interacting block-transfer hierarchy to its unique vacuum projector while preserving nonfactorization at every finite scale.

Executable root: `_u875_recursive_continuum_fixed_point_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.925 (463. RECURSIVE CONTINUUM FIXED POINT THEOREM)**.** *Prove exact convergence of the interacting block-transfer hierarchy to its unique vacuum projector while preserving nonfactorization at every finite scale.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.926. RECURSIVE CONTINUUM FIXED POINT THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 464. FIVE FRONTIER STRENGTHENING THEOREM

Register five new exact frontier theorems after the complete conserved-growth parent proof and preserve the separation between exact native closure and unrestricted external completion.

Executable root: `_u875_five_frontier_strengthening_theorem`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.927 (464. FIVE FRONTIER STRENGTHENING THEOREM)**.** *Register five new exact frontier theorems after the complete conserved-growth parent proof and preserve the separation between exact native closure and unrestricted external completion.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.928. FIVE FRONTIER STRENGTHENING THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 465. HOLONOMY PARENT CURVATURE COMPLETION THEOREM

The constant plaquette coefficient is the identity, the linear coefficient cancels, and the quadratic coefficient is the generator commutator. Link variation supplies the exterior-difference term, producing one parent curvature that enters one parent action.

Executable root: `theorem_holonomy_parent_curvature_completion`

Class and stage: Primary; 8. RECURSIVE QUANTUM AND GRAVITY

Theorem A.929 (465. HOLONOMY PARENT CURVATURE COMPLETION THEOREM)**.** *Prove the coefficientwise descent from exact plaquette transport to parent curvature and the sole physical parent action.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.930. HOLONOMY PARENT CURVATURE COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 466. GLOBAL TARGET WEB THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_global_target_web_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.931 (466. GLOBAL TARGET WEB THEOREM)**.** *Execute the exact certificate for GLOBAL TARGET WEB THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.932. GLOBAL TARGET WEB THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 467. DOMAIN CONSISTENCY THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_domain_consistency_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.933 (467. DOMAIN CONSISTENCY THEOREM)**.** *Execute the exact certificate for DOMAIN CONSISTENCY THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.934. DOMAIN CONSISTENCY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 468. GLOBAL CLAUSE DEPENDENCY THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_global_clause_dependency_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.935 (468. GLOBAL CLAUSE DEPENDENCY THEOREM)**.** *Execute the exact certificate for GLOBAL CLAUSE DEPENDENCY THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.936. GLOBAL CLAUSE DEPENDENCY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 469. STRUCTURE FILTER RECONSTRUCTION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_structure_filter_reconstruction_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.937 (469. STRUCTURE FILTER RECONSTRUCTION THEOREM)**.** *Execute the exact certificate for STRUCTURE FILTER RECONSTRUCTION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.938. STRUCTURE FILTER RECONSTRUCTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 470. UNIVERSAL DESCENT REFINEMENT COMPLETION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_universal_descent_refinement_completion_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.939 (470. UNIVERSAL DESCENT REFINEMENT COMPLETION THEOREM)**.** *Execute the exact certificate for UNIVERSAL DESCENT REFINEMENT COMPLETION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.940. UNIVERSAL DESCENT REFINEMENT COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 471. UNIVERSAL SELECTOR OBSTRUCTION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_universal_selector_obstruction_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.941 (471. UNIVERSAL SELECTOR OBSTRUCTION THEOREM)**.** *Execute the exact certificate for UNIVERSAL SELECTOR OBSTRUCTION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.942. UNIVERSAL SELECTOR OBSTRUCTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 472. MASTER FRONTIER COMPLETION THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_master_frontier_completion_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.943 (472. MASTER FRONTIER COMPLETION THEOREM)**.** *Execute the exact certificate for MASTER FRONTIER COMPLETION THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.944. MASTER FRONTIER COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 473. FINAL FIELD FRONTIER THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_final_field_frontier_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.945 (473. FINAL FIELD FRONTIER THEOREM)**.** *Execute the exact certificate for FINAL FIELD FRONTIER THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.946. FINAL FIELD FRONTIER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 474. ULTIMATE UNIFICATION FRONTIER THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_ultimate_unification_frontier_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.947 (474. ULTIMATE UNIFICATION FRONTIER THEOREM). *Execute the exact certificate for ULTIMATE UNIFICATION FRONTIER THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.948. ULTIMATE UNIFICATION FRONTIER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 475. COMPLETE BRIDGE FRONTIER THEOREM

This theorem supplies the generated object required by the next causal stage.

Executable root: `_complete_bridge_frontier_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.949 (475. COMPLETE BRIDGE FRONTIER THEOREM). *Execute the exact certificate for COMPLETE BRIDGE FRONTIER THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.950. COMPLETE BRIDGE FRONTIER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 476. ORIGIN NESTED RESIDUAL COMPLETION THEOREM

Integrate time residuals, nested signs, deterministic readout, twelve JS signatures, the one-hundred-forty-four board, the sixteen-state carrier, generation separation, and the parent force action.

Executable root: `_origin_nested_residual_completion_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.951 (476. ORIGIN NESTED RESIDUAL COMPLETION THEOREM). *Integrate time residuals, nested signs, deterministic readout, twelve JS signatures, the one-hundred-forty-four board, the sixteen-state carrier, generation separation, and the parent force action.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.952. ORIGIN NESTED RESIDUAL COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 477. DUAL IMPLEMENTATION REPRODUCTION THEOREM

The core numbers are recomputed by two algorithmically different methods: exhaustive enumeration and factorized local counting. Agreement is checked before any external comparison.

Executable root: `_x893_dual_implementation_reproduction_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.953 (477. DUAL IMPLEMENTATION REPRODUCTION THEOREM)**.** *Recompute the core counts by direct enumeration and by an independent factorized algorithm.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.954. DUAL IMPLEMENTATION REPRODUCTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 478. FRACTAL READER EXPOSITION COMPLETION THEOREM

Mathematical motivation and derivation direction are stated beside the exact proof that uses them, so every theorem is readable from its declared premises and calculations.

Executable root: `_fractal_reader_exposition_completion_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.955 (478. FRACTAL READER EXPOSITION COMPLETION THEOREM)**.** *Verify that the multiscale conceptual chain is both computed and explained before its dependent proof roots execute.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.956. FRACTAL READER EXPOSITION COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 479. NATIVE EFFECTIVE READER EXPOSITION THEOREM

The explanatory chain states every mathematically relevant reason, distinction, admissible example and derivation direction before the dependent proof.

Executable root: `_e895_native_effective_reader_exposition_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.957 (479. NATIVE EFFECTIVE READER EXPOSITION THEOREM)**.** *Preserve the full conceptual reason for the native-to-effective derivation beside the mathematical certificates that implement it.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.958. NATIVE EFFECTIVE READER EXPOSITION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 480. COMPLETE CONCEPT EXPOSITION THEOREM

Every mathematically relevant reason, admissible example, generation step and interpretation is stated before the dependent certificate.

Executable root: `_e896_complete_concept_exposition_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.959 (480. COMPLETE CONCEPT EXPOSITION THEOREM)**.** *Preserve every mathematically relevant concept from the structured-zero, fractal, role, generalized-shadow, and native-first discussions immediately beside the completed proof chain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.960. COMPLETE CONCEPT EXPOSITION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 481. COMPLETE WORLD LIGHT READER EXPOSITION THEOREM

The three-dimensional-world and light construction is stated as a sequence of generated geometric and causal propositions before its proof certificates.

Executable root: `_e897_complete_world_light_reader_exposition_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.961 (481. COMPLETE WORLD LIGHT READER EXPOSITION THEOREM)**.** *Verify complete standalone exposition for the three-dimensional-world projection and light-role chain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.962. COMPLETE WORLD LIGHT READER EXPOSITION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 482. COMPLETE HIERARCHY READER EXPOSITION THEOREM

Every hierarchy premise is explained before its calculation, and the output records the motivation, exact result and dependency needed for independent reading.

Executable root: `_e898_complete_hierarchy_reader_exposition_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.963 (482. COMPLETE HIERARCHY READER EXPOSITION THEOREM)**.** *Verify complete premise-before-proof exposition for the native-world-standard-observation hierarchy.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.964. COMPLETE HIERARCHY READER EXPOSITION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 483. DYNAMICAL CLOSURE READER EXPOSITION THEOREM

Every conceptual step needed to follow the universal dynamical rule is stated before the mathematical certificates, together with its assumptions and consequences.

Executable root: `_dynamical_closure_reader_exposition_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.965 (483. DYNAMICAL CLOSURE READER EXPOSITION THEOREM)**.** *Preserve the complete reader-facing explanation of static agreement, trajectory agreement, closure, memory, and reverse reconstruction.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.966. DYNAMICAL CLOSURE READER EXPOSITION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 484. NONLINEAR CONTINUOUS READER EXPOSITION THEOREM

Every new premise is explained before its exact calculation: nonlinear factorization, continuous generator flow, observable completion, JS-SH recurrence, nonlinear force coupling, standard-equation trajectory classes, candidate-framework overlap, and semantic blind decoding.

Executable root: `_nonlinear_continuous_reader_exposition_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.967 (484. NONLINEAR CONTINUOUS READER EXPOSITION THEOREM)**.** *Verify premise-before-proof exposition for every nonlinear, continuous, trajectory, framework, and decoder theorem.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.968. NONLINEAR CONTINUOUS READER EXPOSITION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 485. NATIVE TO EFFECTIVE PHYSICS COMPLETION THEOREM

All available native proof material is used in one direction: origin first, effective physics second. The effective layer is generated, not fitted, and structural agreement with existing theories is evaluated only after the native prediction packet has been sealed.

Executable root: `_e895_native_to_effective_physics_completion_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.969 (485. NATIVE TO EFFECTIVE PHYSICS COMPLETION THEOREM)**.** *Complete the native-first generation of effective physics and its bounded connection to existing equation families.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.970. NATIVE TO EFFECTIVE PHYSICS COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 486. BLIND RATIO REPRODUCTION PACKAGE THEOREM

The program completes everything it can control: source identity, frozen native prediction, equation dictionary, two internal algorithms, and a no-fitting scoring rule. Hidden data and third-party identity remain externally signed fields and therefore fail closed rather than being falsely marked complete.

Executable root: `_e896_blind_ratio_reproduction_package_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.971 (486. BLIND RATIO REPRODUCTION PACKAGE THEOREM)**.** *Create a fail-closed blind-comparison and independent-reproduction package whose internal components are complete while external signatures remain impossible to fabricate in-file.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.972. BLIND RATIO REPRODUCTION PACKAGE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 487. COMPLETE NATIVE EFFECTIVE UNIFICATION THEOREM

The complete in-file result now runs from structured zero through fractal rebasing, three-coordinate continuum decoding, temporal causality, one-plus-seventeen coarse graining, native quotient dynamics, role-dependent projection, one operator grammar, thirty effective equation forms, and a sealed external challenge. External empirical signatures remain separate by logical necessity.

Executable root: `_e896_complete_native_effective_unification_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.973 (487. COMPLETE NATIVE EFFECTIVE UNIFICATION THEOREM)**.** *Conjoin the complete symbolic continuum, scale quotient, role, operator, equation dictionary, exposition, and fail-closed external package into one native-first unification theorem.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.974. COMPLETE NATIVE EFFECTIVE UNIFICATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 488. COMPLETE THREE DIMENSIONAL WORLD UNIFICATION THEOREM

The declared exact class now contains one causal chain from structured zero and residual recursion to a three-coordinate quotient world, a layer-invariant causal speed, a separated electromagnetic light candidate, wave-particle projection duality, closure-burden metric paths, projection-invariant constants, and thirty effective equation forms.

Executable root: `_e897_complete_three_dimensional_world_unification_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.975 (488. COMPLETE THREE DIMENSIONAL WORLD UNIFICATION THEOREM)**.** *Conjoin the complete three-dimensional quotient-world, causal-light, metric, constant, equation, and exposition proofs.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.976. COMPLETE THREE DIMENSIONAL WORLD UNIFICATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 489. EXTERNAL FRONTIER INTEGRATION THEOREM

Every frontier that can be closed by mathematics and deterministic computation inside one file is closed here. Actions that logically require an independent person, hidden data custodian, or resolution of a broader open problem are converted into signed fail-closed protocols rather than fabricated PASS values.

Executable root: `_x893_external_frontier_integration_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.977 (489. EXTERNAL FRONTIER INTEGRATION THEOREM)**.** *Integrate the eight formerly weak frontiers and expose the exact remaining external-only actions.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.978. EXTERNAL FRONTIER INTEGRATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 490. COMPLETE HIERARCHICAL UNIFICATION THEOREM

The complete declared-class result now follows one uninterrupted hierarchy from structured zero and temporal residuals to a minimal local three-dimensional world, exact causal and no-signalling quantum structure, generated effective equations and coefficients, and a frozen ratio-only observational verdict.

Executable root: `_e898_complete_hierarchical_unification_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.979 (490. COMPLETE HIERARCHICAL UNIFICATION THEOREM)**.** *Conjoin all new rank, causality, quantum, scale, coefficient, observation, hierarchy, and exposition gates.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.980. COMPLETE HIERARCHICAL UNIFICATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 491. COMPLETE DYNAMICAL CLOSURE UNIFICATION THEOREM

One native generator now connects static readouts, complete trajectories, exact force-sector quotient laws, finite memory when closure fails, minimal observable completion, JS-SH forward and reverse reconstruction, standard equations, bounded framework connections, and a sealed closure-defect prediction.

Executable root: `_complete_dynamical_closure_unification_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.981 (491. COMPLETE DYNAMICAL CLOSURE UNIFICATION THEOREM)**.** *Conjoin the universal closure, memory, observable completion, JS-SH, four-force, standard-framework, prediction, and exposition theorems.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.982. COMPLETE DYNAMICAL CLOSURE UNIFICATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 492. COMPLETE NONLINEAR CONTINUOUS UNIFICATION THEOREM

The universal rule now spans exact linear fibers, nonlinear polynomial factorization, continuous-time semigroups, finite memory, minimal nonlinear observables, JS-SH forward and reverse recurrences, one nonlinear four-force parent, thirty standard equation trajectory mechanisms, bounded framework connections, and a value-independent semantic ratio decoder.

Executable root: `_complete_nonlinear_continuous_unification_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.983 (492. COMPLETE NONLINEAR CONTINUOUS UNIFICATION THEOREM)**.** *Conjoin every nonlinear, continuous, JS-SH, four-force, standard-equation, framework, and semantic-decoder gate.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.984. COMPLETE NONLINEAR CONTINUOUS UNIFICATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 493. GENERALIZED DYNAMICAL READER EXPOSITION THEOREM

The generalized closure program explains the premise before every calculation: Lie closure, exact nonlinear flow, nonlinear memory, JS-SH projective time, common-carrier action, decoder identifiability, thirty commuting diagrams, and the sealed multi-ratio packet.

Executable root: `_generalized_dynamical_reader_exposition_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.985 (493. GENERALIZED DYNAMICAL READER EXPOSITION THEOREM)**.** *Verify premise-before-proof exposition for every generalized flow, memory, action, decoder, diagram, and prediction theorem.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.986. GENERALIZED DYNAMICAL READER EXPOSITION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 494. COMPLETE GENERALIZED DYNAMICAL UNIFICATION THEOREM

One generalized rule now covers continuous polynomial Lie closure, an exact nonlinear global flow, nonlinear finite memory, JS-SH projective unitary motion, common-carrier four-force action descent, unique semantic force decoding, thirty commuting standard-equation diagrams, and an immutable multi-ratio prediction packet.

Executable root: `_complete_generalized_dynamical_unification_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.987 (494. COMPLETE GENERALIZED DYNAMICAL UNIFICATION THEOREM)**.** *Conjoin every generalized nonlinear-flow, memory, JS-SH, action, decoder, equation, prediction, and exposition gate.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.988. COMPLETE GENERALIZED DYNAMICAL UNIFICATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 495. FINITE SAMPLE UNIVERSALITY BOUNDARY THEOREM

Finite agreement is necessary but not sufficient for an unrestricted law. The exact vanishing polynomial agrees with zero at every tested point and differs elsewhere. Therefore bounded certificates must state their function class, and empirical universality requires independent evidence beyond finite internal replay.

Executable root: `_finite_sample_universality_boundary_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.989 (495. FINITE SAMPLE UNIVERSALITY BOUNDARY THEOREM)**.** *Prove that no finite trajectory sample can certify unrestricted global identity and provide one exact counterexample polynomial.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.990. FINITE SAMPLE UNIVERSALITY BOUNDARY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 496. EXACT FRONTIER READER EXPOSITION THEOREM

Every final generalization states its premise and exact calculation before the certificate, yielding a self-contained proof sequence.

Executable root: `_exact_frontier_reader_exposition_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.991 (496. EXACT FRONTIER READER EXPOSITION THEOREM)**.** *Verify complete reader orientation for the exact jet, flow, memory, JS-SH, action, decoder, verdict, and boundary layer.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.992. EXACT FRONTIER READER EXPOSITION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 497. COMPLETE EXACT INTERNAL FRONTIER THEOREM

The exact internal frontier now includes coefficientwise smooth-jet closure, a parameterized nonlinear flow family, arbitrary finite memory construction, continuous JS-SH generator compatibility, recursive action scale invariance, unique standard-mechanism decoding, a complete blind verdict contract, and a proof of why finite evidence cannot be promoted to unrestricted identity.

Executable root: `_complete_exact_internal_frontier_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.993 (497. COMPLETE EXACT INTERNAL FRONTIER THEOREM)**.** *Conjoin the final exact internal frontier while preserving external and unrestricted logical boundaries.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.994. COMPLETE EXACT INTERNAL FRONTIER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 498. RETAINED CENTRAL PROOF THEOREM

The compact algebra, local parent action, non-Abelian interaction, chiral matter, Lorentz, quantum, color, gravity, continuum, and scattering roots are recomputed. Every retained central root closes through its executable theorem gate.

Executable root: `_u905_retained_central_proof_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.995 (498. RETAINED CENTRAL PROOF THEOREM)**.** *Execute the exact certificate for RETAINED CENTRAL PROOF THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.996. RETAINED CENTRAL PROOF THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 499. RATIO PREDICTION PACKET THEOREM

All final observables are dimensionless ratios generated by the proof chain. Every single-component mutation changes the sealed packet.

Executable root: `_u905_ratio_prediction_packet_theorem`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.997 (499. RATIO PREDICTION PACKET THEOREM)**.** *Execute the exact certificate for RATIO PREDICTION PACKET THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.998. RATIO PREDICTION PACKET THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 500. CLAIM SCOPE CONSISTENCY THEOREM

Earlier theorem-local boundaries are paired with the exact later theorem that resolves each declared native obligation. Unrestricted external equivalence, empirical identification, and independent reproduction remain explicitly separated from internal completion.

Executable root: `theorem_claim_scope_consistency`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.999 (500. CLAIM SCOPE CONSISTENCY THEOREM)**.** *Extend the complete stage-to-resolution ledger through symmetry factorization, physical scale ratios, all-order closure and native regulator universality.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.1000. CLAIM SCOPE CONSISTENCY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 501. INTERNAL UNIFICATION COMPLETION THEOREM

The arbitrary-depth algebra, completed generator, unique measure, gauge quotient, all correlations, gravity, RG, scattering, and blind ratio contract are joined in one theorem. Only an independent empirical verdict remains outside the mathematical source theory.

Executable root: `theorem_internal_unification_completion`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1001 (501. INTERNAL UNIFICATION COMPLETION THEOREM)**.** *Close every nonempirical obligation of the declared source theory through the factorized asymptotic, scale-ratio and native regulator-universality layers.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.1002. INTERNAL UNIFICATION COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 502. STRUCTURAL NECESSITY CHAIN THEOREM

The structural theorem follows one causal order from the unique minimal ledger through time, dimension, Lorentz structure, force carriers, matter, Higgs closure, continuum reconstruction, scale flow, and scattering. Every entry is recalculated from definitions in this file and every theorem gate is true.

Executable root: `theorem_structural_necessity_chain`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1003 (502. STRUCTURAL NECESSITY CHAIN THEOREM)**.** *Join the complete primitive-to-force proof with direct asymptotic factorization, scale decoding, native universality and external verdict separation.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.1004. STRUCTURAL NECESSITY CHAIN THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 503. FINAL CHAIN TARGET INDEPENDENCE THEOREM

The repaired final bridge is inspected at source level to reject a literal generation count, a copied beta-coefficient target, or an enabled external-value flag.

Executable root: `theorem_final_chain_target_independence`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1005 (503. FINAL CHAIN TARGET INDEPENDENCE THEOREM)**.** *Verify that the repaired final field-content and one-loop bridge obtain their inputs from native theorem calls rather than literal external targets.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.1006. FINAL CHAIN TARGET INDEPENDENCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 504. UNIFIED FIELD DEPENDENCY CLOSURE THEOREM

The complete proof is a directed acyclic chain from the eighteen nested source states to time, space, causal signature, four carriers, one action, matter, quantum measure, scale flow, scattering, gravity, and a ratio-only empirical gate.

Executable root: `theorem_unified_field_dependency_closure`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1007 (504. UNIFIED FIELD DEPENDENCY CLOSURE THEOREM)**.** *Close the complete internal dependency chain from the eighteen source states to time, space, Lorentz structure, four forces, matter, quantum measure, RG, scattering, gravity, and ratio-only falsification.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.1008. UNIFIED FIELD DEPENDENCY CLOSURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 505. MOBIUS CROSS RATIO PREDICTION THEOREM

The common projective compression preserves ordered cross ratios. The generated four-sector points therefore supply exact scale- and unit-independent predictions that can be sealed before any external comparison.

Executable root: `theorem_mobius_cross_ratio_prediction`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1009 (505. MOBIUS CROSS RATIO PREDICTION THEOREM)**.** *Generate and seal unit-free four-sector projective invariants that survive every common source-shadow compression.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.1010. MOBIUS CROSS RATIO PREDICTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 506. OPERATOR SPECTRAL PREDICTION CAPSULE THEOREM

Eigenvalue ratios, spectral moments, first crossing depths and projective cross ratios are sealed in one immutable packet without measured target values.

Executable root: `theorem_operator_spectral_prediction_capsule`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1011 (506. OPERATOR SPECTRAL PREDICTION CAPSULE THEOREM)**.** *Seal the scale-free spectral moments, sector threshold depths, eigenvalue ratios, and projective cross ratios of the parent compression operator.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.1012. OPERATOR SPECTRAL PREDICTION CAPSULE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 507. FIVE ITEM COMPLETION RESOLUTION THEOREM

Continuum equivalence on the generated domain, native scale classification, one-anchor unit decoding and new prediction sealing are internal theorems. Only the occurrence of independent measurement and reproduction remains an external event.

Executable root: `theorem_five_item_completion_resolution`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1013 (507. FIVE ITEM COMPLETION RESOLUTION THEOREM)**.** *Join the five completed mathematical constructions into one dependency-ordered closure packet.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.1014. FIVE ITEM COMPLETION RESOLUTION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 508. LOG GRAVITY APPLICATION CLOSURE THEOREM

The source compression law is now applied to continuum completion, RG, dimension-dependent gravity, horizon rate, singularity classification and information balance. These are one acyclic dependency chain inside the declared native class, while absolute calibration and empirical identity remain external tests.

Executable root: `theorem_log_gravity_application_closure`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1015 (508. LOG GRAVITY APPLICATION CLOSURE THEOREM)**.** *Close the exact internal application of one source compression generator to continuum, RG, gravity, horizon and information readouts.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.1016. LOG GRAVITY APPLICATION CLOSURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 509. MASTER SOURCE SHADOW COMPRESSION EQUATION THEOREM

The final equation chain joins regular source motion, bounded projection, discrete-to-continuum generation, mass rates, dimensional gravity, horizon compression, and four force readouts. Every numerical statement is an internally generated exact ratio; absolute calibration and unrestricted external identity remain separate falsifiable layers.

Executable root: `theorem_master_source_shadow_compression_equation`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1017 (509. MASTER SOURCE SHADOW COMPRESSION EQUATION THEOREM). *Join the corrected origin, projective, continuum, metric, gravity, information and four-role certificates.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.1018. MASTER SOURCE SHADOW COMPRESSION EQUATION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 510. COMPLETE INTERNAL FOUR FORCE FIELD CLOSURE THEOREM

The final dependency chain begins with the eighteen signed occupancy states and ends with one operator-generated field theory, one scale gauge, a sealed prediction packet and a fail-closed external verdict boundary.

Executable root: `theorem_complete_internal_four_force_field_closure`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1019 (510. COMPLETE INTERNAL FOUR FORCE FIELD CLOSURE THEOREM). *Close the complete source theory from the eighteen primitive states through operator compression, continuum fields, scale classification and prediction sealing.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.1020. COMPLETE INTERNAL FOUR FORCE FIELD CLOSURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 511. THIRTY EQUATION DECLARED SOLUTION SPACE COMPLETION THEOREM

Each of the thirty effective equation forms is placed after its native generator. The operator origin, trajectory mechanism, world projection, equation evolution, and ratio interface form one commuting diagram. The result is exact on the registered generated solution classes and does not import measured coefficients.

Executable root: `theorem_thirty_equation_declared_solution_space_completion`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1021 (511. THIRTY EQUATION DECLARED SOLUTION SPACE COMPLETION THEOREM)**.** *Join every registered standard equation form to one native origin, one unique mechanism decoder, one commuting trajectory diagram, and the available internally generated coefficient class on the declared solution spaces.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.1022. THIRTY EQUATION DECLARED SOLUTION SPACE COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 512. COVARIANT RESIDUAL EQUATION DESCENT DICTIONARY THEOREM

The parent covariant residual equation is the mandatory middle gate for compression, transport, curvature and action descent. Matter representation, closed-shell spectrum and boundary data remain generated companions rather than being falsely compressed into one recurrence. Every principal equation retains its proved native trajectory certificate.

Executable root: `theorem_covariant_residual_equation_descent_dictionary`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1023 (512. COVARIANT RESIDUAL EQUATION DESCENT DICTIONARY THEOREM)**.** *Map every registered principal effective-equation form to the covariant residual middle gate, its required generated companion, and the already proved trajectory certificate without claiming that one recurrence alone replaces all companion structures.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.1024. COVARIANT RESIDUAL EQUATION DESCENT DICTIONARY THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 513. TIME GRAVITY FOUR FORCE ORIGIN CLOSURE THEOREM

The final chronological chain runs from the eighteen origin states to time, additive depth, space, gravity, three internal gauge curvatures, one parent variation, one time anchor and a sealed external test.

Executable root: `theorem_time_gravity_four_force_origin_closure`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1025 (513. TIME GRAVITY FOUR FORCE ORIGIN CLOSURE THEOREM)**.** *Close the chronological origin proof from the eighteen states through time, additive depth, space, gravity and the three internal gauge forces, with time as the sole dimensional anchor.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.1026. TIME GRAVITY FOUR FORCE ORIGIN CLOSURE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 514. PARENT SHADOW FOUR FORCE COMPLETION THEOREM

One source does not select one of several universes. The primitive grammar generates all canonical shadow blocks at once; admissible relabelings are equivalent descriptions, refinement removes the directional fourth-order defect, and one typed parent response produces the four projected field sectors.

Executable root: `theorem_parent_shadow_four_force_completion`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1027 (514. PARENT SHADOW FOUR FORCE COMPLETION THEOREM)**.** *Join the canonical no-choice shadow decomposition, actual dispersion refinement, typed formula seal, and existing parent action into one strengthened four-force proof chain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.1028. PARENT SHADOW FOUR FORCE COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 515. TIME FORCE DYNAMICAL COMPLETION THEOREM

The proof begins with the complete negative-zero-positive, A-B-C, full-empty grammar; generates time before space; derives the residual dimensions; identifies four anonymous carriers without a label permutation; fixes one parent action and response law analytically; and continues through nonlinear dynamics, running, confinement, matter, and thirty effective equation classes.

Executable root: `theorem_time_force_dynamical_completion`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1029 (515. TIME FORCE DYNAMICAL COMPLETION THEOREM)**.** *Close the strengthened internal chain from structured zero and time-first dimensional genesis through anonymous force identification, analytic parent action, nonlinear dynamics, sector RG, confinement, matter, and thirty effective equations.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.1030. TIME FORCE DYNAMICAL COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 516. COVARIANT RESIDUAL FIELD COMPLETION THEOREM

The proof is one continuous causal chain. Signed occupied and vacant relations generate time; temporal residuals generate space; the complete parent grammar generates the four anonymous shadows; the rational residual equation transports all sectors; its gravity projection gives the finite clock-correlation law; its holonomy gives curvature; and one parent action generates the four field equations and matter dynamics.

Executable root: `theorem_covariant_residual_field_completion`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1031 (516. COVARIANT RESIDUAL FIELD COMPLETION THEOREM)**.** *Close the dependency-ordered proof from the eighteen-state grammar through time, spatial residuals, canonical force shadows, the logarithm-free covariant residual equation, gravity, curvature, one parent action, matter, RG, scattering, and thirty effective equation forms.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.1032. COVARIANT RESIDUAL FIELD COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 517. THIRTY EQUATION NO JUMP DEPENDENCY LEDGER THEOREM

Every effective equation is displayed only after an explicit chain from the primitive grammar to the common residual update and then to the necessary generated companion. No row is certified merely because its final textbook form is recognizable.

Executable root: `theorem_thirty_equation_no_jump_dependency_ledger`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1033 (517. THIRTY EQUATION NO JUMP DEPENDENCY LEDGER THEOREM)**.** *Expand every principal effective-equation row into an explicit dependency path from primitive grammar through the covariant residual middle gate and its generated companion mechanism.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.1034. THIRTY EQUATION NO JUMP DEPENDENCY LEDGER THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 518. COVARIANT RESIDUAL UNIFICATION COMPLETION THEOREM

The covariant residual equation is the central dynamical gate rather than a decorative bridge. Its edge operator is generated, its target is unique, its domain is preserved, its gauge and path laws close, its plaquette coefficient gives curvature, its gravity block gives the finite clock-ratio identity, and the sole parent action yields the four force and matter equations.

Executable root: `theorem_covariant_residual_unification_completion`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1035 (518. COVARIANT RESIDUAL UNIFICATION COMPLETION THEOREM)**.** *Close the strengthened unification chain with the covariant residual equation as the explicit central dynamical gate and with no-jump curvature, gravity, action, equation and registry certificates.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.1036. COVARIANT RESIDUAL UNIFICATION COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 519. AUTHORITATIVE CAUSAL SPINE COMPLETION THEOREM

The proof map used by the executable certificate and the manuscript is uniquely fixed by dependency: primitive relation, time, residual space, shell, canonical carrier, covariant residual dynamics, gravity and holonomy, parent action, matter, observables, continuum and external test interface.

Executable root: `theorem_authoritative_causal_spine_completion`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1037 (519. AUTHORITATIVE CAUSAL SPINE COMPLETION THEOREM)**.** *Prove that the complete theory has one authoritative primitive-to-observable order and no obsolete alternative proof map.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.1038. AUTHORITATIVE CAUSAL SPINE COMPLETION THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 520. RUNTIME SOURCE METADATA COHERENCE THEOREM

The executable certificate does not copy a line count or source digest from an earlier artifact. It reads the exact file being executed and calculates its metadata at runtime.

Executable root: `theorem_runtime_source_metadata_coherence`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1039 (520. RUNTIME SOURCE METADATA COHERENCE THEOREM)**.** *Compute source line count, byte count, and SHA-256 directly from the executed standalone file rather than from copied metadata.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.1040. RUNTIME SOURCE METADATA COHERENCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 521. EXTERNAL SCOPE AND INTERNAL CLAIM FIREWALL THEOREM

An internal exact theorem cannot manufacture evidence about nature or historical priority. Those questions remain signed external inputs even when the generated mathematical class is internally closed.

Executable root: `theorem_external_scope_and_internal_claim_firewall`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1041 (521. EXTERNAL SCOPE AND INTERNAL CLAIM FIREWALL THEOREM)**.** *Separate internal exact closure from empirical identity, unrestricted continuum equivalence, historical priority, and third-party reproduction.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.1042. EXTERNAL SCOPE AND INTERNAL CLAIM FIREWALL THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 522. FINAL STANDALONE RESIDUAL FIELD UNIFICATION CERTIFICATE

One self-contained file recomputes the primitive grammar, time, residual dimensions, canonical force carrier, covariant residual dynamics, finite gravity relation, curvature, single action, matter, ratios, continuum and effective descendants while preserving the boundary between internal theorem and external science.

Executable root: `theorem_final_standalone_residual_field_unification_certificate`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1043 (522. FINAL STANDALONE RESIDUAL FIELD UNIFICATION CERTIFICATE)**.** *Conjoin the complete authoritative chain into one final standalone internal certificate.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.1044. FINAL STANDALONE RESIDUAL FIELD UNIFICATION CERTIFICATE holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 523. PRIMARY REGISTRY STATEMENT COHERENCE THEOREM

The appendix registry is sourced from the same chronological callable list used by execution. Every root has one contiguous ordinal, one unique callable name and one deterministic nonempty statement source. Every newly strengthened root is additionally executed here to verify exact agreement among its callable, proof title, theorem statement and true result.

Executable root: `theorem_primary_registry_statement_coherence`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1045 (523. PRIMARY REGISTRY STATEMENT COHERENCE THEOREM)**.** *Prove coherence of the chronological registry source and verify title-statement-result agreement for every newly strengthened root without recursively executing the registry itself.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.1046*.* PRIMARY REGISTRY STATEMENT COHERENCE THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 524. CHRONOLOGICAL FOUR FORCE UNIFIED FIELD THEOREM

The proof proceeds without a derivation jump from signed occupancy to time, dimension, four carriers, one action, quantum structure, continuum flow, and scattering. All four interaction sectors close in one dependency-ordered executable theorem.

Executable root: `_chronological_origin_to_four_force_completion_theorem_905`

Class and stage: Primary; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1047 (524. CHRONOLOGICAL FOUR FORCE UNIFIED FIELD THEOREM)**.** *Execute the exact certificate for CHRONOLOGICAL FOUR FORCE UNIFIED FIELD THEOREM on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. ☐

Remark A.1048*.* CHRONOLOGICAL FOUR FORCE UNIFIED FIELD THEOREM holds on the declared generated domain, and its root gate is TRUE.

Reader guide before the proof: 525. NATIVE PROOF CONSISTENCY

This theorem supplies the generated object required by the next causal stage.

Executable root: `prove_native_proof_consistency`

Class and stage: Narrative; 9. GLOBAL CLOSURE AND SINGLE PROOF

Theorem A.1049 (525. NATIVE PROOF CONSISTENCY)**.** *Execute the exact certificate for NATIVE PROOF CONSISTENCY on its generated dependency domain.*

Proof. The executable recomputes the theorem from objects generated earlier in the causal order. Its complete rational, matrix, incidence, path, differential, variational, or closure certificate is printed in the deterministic output under the same proof root. □

Remark A.1050. NATIVE PROOF CONSISTENCY holds on the declared generated domain, and its root gate is TRUE.

B Complete explicit derivation ledger: all 544 rows

Each row records generated input, exact operation, generated output, and the mathematical consequence used by later stages. The ledger is stated without implementation-history language.

B.1 Derivation rows 1–68

No.	Generated input	Exact operation	Generated output	Mathematical consequence
1	two primitive channels; four transformation layers	form 2, 4, 8, 10 and the twelve-seat carrier	0: 2; 1: 4; 2: 8; 3: 10; 4: 12	the primitive carrier data are fixed
2	two-fold condition; three-fold condition; four-fold condition	take the least positive common carrier	12	no smaller positive carrier satisfies all three conditions
3	carrier ranks; root sets	add rank and root count	0: 1; 1: 3; 2: 8	the U1, A1 and A2 algebra dimensions are obtained
4	gauge representations; Yukawa invariance equations	solve all linear and cubic anomaly equations together	primitive chiral charge ray	relative chiral charges are fixed
5	chiral charge ray; weak isospin	add the two charge contributions and divide by their common divisor	0: 2; 1: -1; 2: 0; 3: -3	the relative electric charges follow
6	Higgs charge; lower weak-isospin charge	add the two charges	0	the selected vacuum is electrically neutral
7	neutral gauge mass matrix	compute determinant, trace and null vector	0: 0; 1: 8; 2: 1	exactly one neutral gauge direction is massless
8	C12 carrier; gamma-five; Hermitian involution	construct the Ginsparg-Wilson operator and compute rank and index	0: 11; 1: 1	one exact chiral mode remains
9	nilpotent BRST operator	compute kernel, image and quotient dimensions	0: 4; 1: 2; 2: 2	gauge-exact states are removed and positive physical states remain
10	field representations; canonical dimension bound	enumerate all gauge-invariant monomials	kinetic, scalar, Yukawa and neutral Majorana classes	the admissible matter operator basis is fixed
11	signed conserved source; abelian connection	form the abelian curvature and divergence equation	electromagnetic field operator	the electromagnetic sector is identified
12	left-chiral doublets; A1 generators	form the noncommuting covariant derivative and curvature	weak field operator	the weak sector is identified
13	A2 generators; color tensor products; positive local gap	decompose meson and baryon carriers and compute the first excitation	0: 1+8; 1: 1+8+8+10; 2: positive gap	the strong sector is identified

No.	Generated input	Exact operation	Generated output	Mathematical consequence
14	symmetric metric; electromagnetic stress tensor	vary the nonlinear Einstein-Maxwell action	curvature equals matter stress	the gravity sector is identified
15	four operator signatures	compare rank, commutator, chirality and source type	four pairwise distinct signatures	the assignment of the four forces is unique in the declared grammar
16	four sector actions; permutation symmetry; unit normalization	solve the coefficient equations	0: 1/4; 1: 1/4; 2: 1/4; 3: 1/4	one normalized parent action is fixed
17	four sector gradients	differentiate each sector and add the results	parent Euler-Lagrange gradient	the four field equations are projections of one parent equation
18	reflection-positive correlations	form the Gram matrix and quotient null vectors	positive physical Hilbert space	Euclidean data reconstruct physical states
19	positive transfer contraction	construct an exact orthogonal dilation	norm-preserving dilation	the transfer step embeds in reversible quantum evolution
20	Ginsparg-Wilson operator; positive native masses; anomaly sums	compute determinants and anomaly gates	positive anomaly-free determinant packet	the chiral matter measure has no displayed finite obstruction
21	SU3 Dynkin labels; quadratic Casimir	compute the representation contraction per plaquette	strict contraction for every nontrivial representation	each nontrivial color representation has positive area decay
22	single-plaquette contractions	multiply contractions over added plaquettes	exact area semigroup	Wilson ratios obey the declared area law
23	symmetric tensor modes; BRST gauge pairs	take BRST cohomology after transverse and trace constraints	2	only plus and cross graviton polarizations remain
24	parent action; interacting transfer; dimensionless ratio packet	hash all declared holdout ratios before comparison	immutable prediction digest	later numerical retuning is excluded
25	finite native scattering coefficients	construct two exact continuations with the same finite prefix	equal prefix and unequal tails	finite coefficients do not determine all-order amplitudes
26	tested finite-volume color gaps	construct a uniformly gapped extension and a gapless extension	equal tested prefix and different infinite-volume gap property	finite-volume tests do not determine the uniform continuum gap
27	finite local curvature jet	compare zero with a polynomial beginning at the next derivative order	equal finite jet and unequal global profile	finite local geometry leaves multiple admissible continuations for global quantum geometry

No.	Generated input	Exact operation	Generated output	Mathematical consequence
28	internal dimensionless ratios	apply two distinct external label maps preserving the same ratio multiset	identical internal algebra and different physical identification	the projection decoder requires a separate uniqueness theorem
29	sealed prediction packet	compare matching and nonmatching independent datasets while preserving the seal	one pass and one failure from the same internal packet	blind independent confirmation is logically external
30	declared native theorem	evaluate the complete sixty-four-row truth table and remove each obligation once	all six are jointly sufficient and each is necessary in the completion definition	the exact final completion architecture is fixed by the jointly sufficient and individually necessary obligations
31	primitive closure loads	take their greatest common divisor	the least positive closure unit	the recycle threshold is derived rather than inserted
32	all antecedent structural scales	sum the complete history and choose the least larger integer refinement factor	a variable strictly growing factor	the next scale uses the whole prior state rather than copying the preceding scale
33	one conserved total; integer refinement factor	split every parent into equal children and push child weights back to the parent	exact projective conservation at every pair of levels	support growth and total conservation coexist
34	threshold; incoming residue	perform quotient and remainder decomposition	closed nodes plus a strictly subthreshold remainder	recycling loses no residue and retains the excess to zero
35	one cumulative recycle carrier	take signed, transition, color-locking, and area-flux projections	four distinct force readouts with one-quarter parent weights	the four-force split preserves the conserved total
36	all recursive levels	derive interaction-order weights, uniform color gaps, projective geometry measures, and internal ratio spectra	four exact recursive-class extension theorems	four recursive-class extensions are proved on the generated recursive class
37	positive nonfactorizing transfer kernel	extend path weights by exact cylinder refinement	one normalized projective family	every finite cylinder amplitude has one recursively fixed value
38	projective cylinder family; transfer eigencharacters	evaluate every integer-time mode correlation	closed rational amplitude formulas	the continuum path functional and all discrete-time transfer amplitudes are fixed in the declared class
39	all nonnegative SU3 Dynkin labels	minimize the exact quadratic Casimir	minimum four-thirds at the fundamental and antifundamental representations	every nonvacuum finite-support direct-limit state has a uniform positive gap

No.	Generated input	Exact operation	Generated output	Mathematical consequence
40	finite-support color excitations	sum local Casimir energies	energy never below four-thirds	the gap survives the countable direct limit of the declared representation carrier
41	cumulative thresholds; charge-color-chirality signatures	select the first three nontrivial scales and normalize by the first	a scale-covariant generation hierarchy	the ratio decoder contains all ratios arise from the generated hierarchy
42	primitive integer pairs	form and multiply three rational Givens rotations	an orthogonal determinant-one noncommuting mixing matrix	mixing is generated internally in a fixed structural order
43	Pachner-invariant topology amplitude; recursive node measure	multiply and push child weights to every parent	topology-labeled projective conservation	triangulation moves and refinement preserve the declared measure
44	projective topology measure; nonlinear Einstein-Maxwell sector; graviton BRST cohomology	combine topology, nonlinear geometry, and physical propagation gates	one recursive PL quantum-geometry theorem	the geometry sector is dynamically stronger on the generated recursive PL class
45	recursive cylinder moments; positive transfer eigenvalues	construct the exact rational resolvent	three positive poles and residues	moments and pole data determine one another in the declared class
46	pole data; rational Cayley scattering	verify residue positivity and exact unitarity	one spectral scattering packet	the finite spectral LSZ bridge is closed on the generated finite spectral class
47	SU3 Casimir spectrum; square-summable coefficients	subtract four-thirds times the norm	a nonnegative coefficient sum	the gap extends to the declared Hilbert completion
48	structural family signatures; ordered thresholds	enumerate all label permutations	only the identity preserves every signature and order	the internal decoder is unique before physical naming
49	dimensionless ratios; mixing matrix; force ratios	serialize and hash the complete packet	one immutable prediction digest	later comparison leaves unchanged the prediction
50	smooth polynomial Lorentzian coframe	refine by rational PL partitions	preserved Lorentzian signature and decreasing volume error	the declared smooth and PL descriptions are cofinally connected
51	Pachner-projective topology measure; cofinal Lorentzian refinements	combine topology and refinement gates	one declared smoothable PL geometry class	the gravity bridge is stronger on the generated smoothable PL class

No.	Generated input	Exact operation	Generated output	Mathematical consequence
52	sealed packet	reconstruct in an independent key order	the same digest	the reproduction protocol is deterministic
53	sealed packet; hidden independent comparison key	require a separate signed result	the comparison result is determined by independent signed fields	blind success remains an independent verification event
54	six frontier rows	separate exact bridges from unrestricted completions	six exact internal gates and zero fabricated external completions	the final scope is explicit and noncircular
55	positive interacting transfer; variable cumulative block sizes	diagonalize by exact character projectors and coarse-grain every block	a scale-consistent positive nonfactorizing transfer semigroup	the declared interacting transfer is stable under the full recursive scale law
56	one-particle rational scattering	form tensor powers and apply the tensor-product norm identity	unitary finite-particle carriers	norm preservation extends to every finite particle number in the declared carrier
57	separated incoming and outgoing labels	evaluate tensor-product matrix entries	exact product amplitudes	cluster factorization is explicit rather than assumed
58	unique meson and baryon singlets; adjoint glue carrier	assign additive Casimir energy to every finite gauge-invariant network	a minimal nonvacuum network gap of eight-thirds	the declared gauge-invariant network completion is uniformly gapped
59	recursive generation thresholds; native shell capacities	form complete-graph shell spectra	one unique generation-shell fingerprint for every internal state	the no-fit internal decoder gains an exact bound-state spectral layer
60	dimensionless spectral packet	serialize independently in two key orders	one identical digest	the enlarged ratio packet is immutable before independent comparison
61	nonlinear Einstein-Maxwell family; recursive scale sequence	scale mass charge and radius together	invariant lapse and dimensionless ratios with inverse-area density	nonlinear geometry grows without duplicating the integrated burden
62	five independently proved roots; external completion boundary	place every new theorem after the preceding completed chapter and before the integration chapter	one acyclic dependency extension	the proof advances without recomputing or circularly assuming a new theorem inside itself
63	rational one-particle scattering; controlled local coupling	compose local rotations and reversible couplings for arbitrary finite particle number	one exact interacting Fock circuit	unitarity and genuine interaction coexist
64	disconnected particle clusters	remove every coupling edge crossing the partition	tensor-product cluster evolution	cluster factorization follows from the graph rather than being assumed

No.	Generated input	Exact operation	Generated output	Mathematical consequence
65	meson color singlet endpoints; SU3 Casimir gap	add positive nearest-neighbor flux gradients	a coercive flux-tube quadratic form	the direct-limit color carrier has a uniform gauge-invariant lower bound
66	generation thresholds; native shell spectra	form the complete dimensionless ratio graph	one scale-invariant state fingerprint	the internal decoder is unique in its declared structural class
67	nonlinear scale-covariant geometry; rational Lorentz boost	propagate the null Hamiltonian constraint through every scale	constraint-preserving Lorentzian recursion	the growing geometry remains dynamically admissible
68	recursive thresholds; inverse node weights	sum every child weight and every force-sector projection	one conserved parent total at every depth	fractal support growth preserves physical quantity

B.2 Derivation rows 69–136

No.	Generated input	Exact operation	Generated output	Mathematical consequence
69	growing area; inverse-area density	multiply support by local density	constant integrated burden	gravity growth and total conservation are compatible
70	five exact native roots; six unrestricted obligations	separate internal theorem truth from external equivalence	five exact bridges and zero fabricated external completions	the conclusion remains noncircular and falsifiable
71	recursive scales; crossing-symmetric skew generators	apply the rational Cayley transform	one crossing-symmetric optical scattering family	finite-scale interaction and reversible reconstruction coexist
72	SU3 center triality; finite simple graph	apply divergence-free support and cycle containment	one graph-size-independent center-flux gap	gauge-invariant center flux cannot approach zero energy by graph growth
73	twelve internal states; all remaining-state ratios	withhold one row and match every dimensionless relation	one uniquely recovered hidden state	the prediction packet is internally leave-one-out falsifiable
74	binary cylinder refinement; plus-cross physical basis	push measures and conditional means through every refinement	one normalized projective two-polarization field	continuum field consistency follows without duplicating total measure
75	interacting transfer spectrum; cumulative block exponents	compare each blocked kernel with the vacuum projector	one strictly convergent fixed-point hierarchy	the continuum limit is unique in the declared transfer class
76	five exact frontier roots; six unrestricted obligations	separate mathematical strengthening from empirical completion	five exact bridges and zero fabricated external claims	the final statement remains independently testable

No.	Generated input	Exact operation	Generated output	Mathematical consequence
77	latent four-component carrier; four primitive readout signs	construct the complete Hadamard projection frame	four orthogonal force readouts and an exact inverse	all four forces together recover information that every proper subset loses
78	intrinsic motion; observer-frame motion	form the relative transfer before projection	one observable transfer distinct from both underlying motions	observable dynamics is not automatically intrinsic dynamics
79	projected trajectory; two moving factors	split each observable increment exactly	intrinsic and observer contributions	a changing readout can arise from either source or both
80	one force scalar; a hidden orthogonal direction	hold the scalar readout fixed while changing the carrier	different latent quadratic invariants	one force requires the complete native state to determine mass or inertia
81	four force readouts; observer calibration	invert the complete frame and frame motion	latent state, mass, and weighted inertia	the complete force system reconstructs declared latent invariants
82	latent transfer and positive operator; normalized force frame	apply exact orthogonal similarity	observable unitary and positive operators with identical spectrum	observable equations are derived projections rather than inserted targets
83	four independent projected basis evolutions; known observer law	assemble and invert the complete evolution matrix	relative and intrinsic operators	the fundamental dynamics is uniquely reconstructed after observer calibration
84	one projected trajectory; an operator perturbation annihilating that input	compare two compatible transfers	the same trajectory from different operators	complete tomography is necessary for uniqueness
85	five exact projection-origin roots; the existing parent proof chain	order every dependency before its consequence	one bidirectional latent-to-observable theorem	the standard observable layer is an output of the native layer within the declared class
86	exact native projection equivalence; unrestricted empirical obligations	separate proven structural equivalence from external identification	one sealed comparison interface	no external physical identity is fabricated
87	four independent force readouts; primitive two-bit relations	derive the complete character frame	a minimal four-dimensional latent carrier	the latent dimension is selected before any standard force equation
88	native positive anchor; ordered primitive levels	solve the sign-fixed ordered eigentetrad	one observer frame with the native positive anchor	the observer law is part of the native construction
89	complete force scalars; compact carrier algebra	form orthogonal sector projectors of ranks one, three, eight, and five	one parent carrier with four exact force sectors	the force dictionary is generated structurally

No.	Generated input	Exact operation	Generated output	Mathematical consequence
90	native state; native positive mass operator	compute mass, gradient, and source before projection	latent physical invariants independent of the observable shadow	mass is not inferred from one shadow shape
91	complete force frame; latent mass and source	project and invert every invariant	exact observable reconstruction with no information loss	all four views recover what every proper subset loses
92	intrinsic motion; internally generated observer frame	form relative motion and observable transfer	the force-layer dynamics as a derived projection	observable dynamics is not mistaken for intrinsic dynamics
93	latent action; observable action	compare values and Euler gradients	exact action equivalence under the complete projection	standard-layer equations are outputs rather than fitting targets
94	observable transfer; native observer law	invert the force projection and restore the observer motion	the intrinsic transfer	the origin dynamics is recoverable in the declared class
95	five origin-projection roots; the preceding complete proof chain	order every dependency before its consequence	one origin-to-shadow-to-origin theorem	no derivation jump is used
96	exact native reconstruction; independent physical obligations	separate structural proof from nature identification	one sealed comparison interface	future measurements and continuum completion conditions are retained as independently decidable
97	complete latent character frame; rational scale parameter	build orthogonal spectral projectors and compose their kernels	one exact multiplicative origin semigroup	the observable scale law is generated from the latent frame
98	origin scale kernel; complete force projection	transport the kernel to the shadow basis and invert it	four exact scale eigenchannels	shadow dynamics remains a derived image
99	native poles; native positive residues	derive the characteristic recurrence	all integer-order amplitudes	higher orders are not individually fitted
100	first four moments; invertible Vandermonde map	recover every residue exactly	unique amplitude tomography	the all-order packet is identifiable in the declared class
101	native anchor; primitive level operator	minimize the apparatus coupling action	one ordered sign-fixed observer frame	the observer axis is generated rather than freely calibrated
102	latent mass levels; observer mixing; sector dimensions	seal a ratio-only packet	one immutable independent comparison target	generation uses only native ratios
103	strong-SU3 dictionary; native center gap	embed color states through the strong projector	exact gap-energy preservation	the strong shadow inherits the declared native gap

No.	Generated input	Exact operation	Generated output	Mathematical consequence
104	polynomial Lorentzian coframe; rational affine atlas	compose exact pullbacks	smooth-class coordinate covariance	coordinate change is separated from intrinsic geometry
105	cofinal PL refinement; Lorentzian constraint; two polarization field	join the three limits	one declared smoothable quantum-geometry class	no unrestricted DiffM claim is substituted
106	six generated closure dependencies; physical carrier dictionary	order every dependency	one origin-shadow reality frontier theorem	the force sectors follow their generated carrier dimensions
107	internal mathematical prerequisites	count exact completions	six of six internal frontier gates	more than half of the named frontier is strengthened directly
108	unrestricted continuum and independent verification events	keep independent obligations explicit	one independent-comparison condition	open problems and future measurements are not converted into Boolean success
109	native anchor; mass operator; complete force frame	test every signed-permutation competitor	one invariant-preserving latent frame	the declared origin is identifiable within the complete discrete competitor class
110	four native poles; positive native residues	derive one characteristic identity	all integer-order source moments	no order is fitted separately
111	latent vertex; native pole propagators	attach, project, and amputate exactly	one recovered interaction vertex	observable pole factors do not replace the origin interaction
112	positive transfer spectrum; complete force frame	form all reflected quadratic forms	nonnegative reconstruction carrier	the declared Euclidean transfer has a positive Hilbert interpretation
113	SU3 center loop; weighted edge subdivision	refine and push forward every parent edge	one scale-independent positive loop gap	the declared color gap survives projective refinement
114	nonlinear Lorentzian background; two projective polarizations	transport the exact action through affine charts and depth	one rational inverse-limit action	coordinate shadows remain separated from intrinsic action
115	sealed ratio packet; required independent fields	construct a fail-closed verifier	one immutable replication certificate	internal execution requires independent verification for external success
116	seven exact roots; correct physical force-carrier order	place every premise before its consequence	one origin-reality continuum theorem	the strengthened internal bridges are combined without renaming independent verification events as proofs

No.	Generated input	Exact operation	Generated output	Mathematical consequence
117	internal mathematical frontier	count direct completions	seven completed exact bridges	the named mathematical frontier is established in one standalone dependency-ordered proof
118	unrestricted physics and independent events	retain their distinct completion conditions	one explicit independent-comparison condition	physical identification and public open problems remain externally decidable
119	one ordered time depth; binary split count	count each new nonidentity temporal residual	cumulative levels 1,2,5,12	the native level ledger is generated before force observation
120	binary depths	take reciprocal branch scales and normalize cumulative levels	four temporal poles and positive residues	the amplitude seed comes from time refinement rather than observable fitting
121	one scalar moment stream	form successive Hankel matrices	ranks 1,2,3,4,4	four is the exact minimal latent dimension of the declared temporal stream
122	four consecutive times	apply residual orders zero through three	one invertible residual coordinate system	dimension coordinates arise from temporal differences
123	one characteristic recurrence	form its companion map	unique causal four-delay evolution	one stationary and three contracting modes select forward time
124	temporal residual coordinates; complete character frame	project and invert exactly	gravity, electromagnetic, weak, and strong readouts	all four forces are complete shadows of one temporal history
125	temporal Hankel form	test every leading principal minor	one positive mass operator	mass is defined before the force shadows
126	one primary time value; three generated residual orders	assign the causal Lorentzian sign and transform it	one force-coordinate Lorentzian form	time and generated dimensions remain distinct under observation
127	companion target; positive temporal mass form	minimize the exact quadratic step action	one unique temporal evolution	dynamics follows from the origin action rather than a fitted shadow law
128	temporal-to-force transform	transport the action by exact congruence	one equivalent force action	the observed dynamics is generated from the fundamental dynamics
129	seven exact temporal roots; all preceding origin proofs	place time before residuals, dimensions, forces, mass, and action	one time-residual-dimension-force theorem	the complete internal chain has no standard-equation input

No.	Generated input	Exact operation	Generated output	Mathematical consequence
130	unrestricted physics and independent events	retain their separate decision conditions	one explicit independent-comparison condition	the new origin theorem preserves independent verification of empirical or public completion
131	two primitive polarity projectors	form their complete occupancy lattice	empty zero, negative, positive, and full zero	zero in the shadow has two distinct origin preimages
132	nonnegative polarity counts	separate signed residue from closed pairs	T=ABS(R)+2C and exact recovery	balanced content is conserved rather than deleted
133	four primitive states	take the complete character table	one orthogonal four-readout frame	the force frame is the Fourier frame of the origin occupancy group
134	all numeral bases and all state names	change representation while preserving matrices and incidence	one radix-independent invariant structure	numbers label results but do not generate them
135	four primitive modes	remove self-relations and form every directed difference	twelve relation roots in a three-dimensional sum-zero space	three relative directions arise after one common mode is removed
136	twelve relation roots	compute exact supporting facets	twelve vertices, twenty-four edges, eight triangles, six squares	the JS shell follows from relation incidence rather than a preselected solid

B.3 Derivation rows 137–204

No.	Generated input	Exact operation	Generated output	Mathematical consequence
137	oriented edges and faces	form both boundary matrices	boundary of boundary zero and Betti numbers 1,0,1	the generated shell is an exact closed sphere carrier
138	JS adjacency	form its exact graph Laplacian	spectrum 0,2,4,6 with positive gap two	the local carrier has a scale-free positive discrete gap
139	empty/full zero distinction; signed residue; closed pairs	construct total, sign, zero-type, and lock readouts	one invertible physical force frame	gravity, electromagnetic, weak, and strong channels arise from one ledger
140	twelve latent relation roots	transport them through the physical force frame and its induced metric	twelve unique force-transition vectors with preserved incidence	the JS geometry survives observation without loss
141	one temporal companion; residual coordinates; double-zero force frame	conjugate the origin dynamics exactly	one force companion and exact inverse recovery	the observed dynamics is generated from the temporal origin dynamics

No.	Generated input	Exact operation	Generated output	Mathematical consequence
142	triangular and square relation cycles	sum every directed transition around each closure	zero latent and force boundary	local cycle closure is compatible with the four-force observation
143	seven exact roots; all preceding temporal proofs	order every premise before geometry and forces	one time-double-zero-relation-cell-force theorem	the internal chain contains no numeral-base or standard-equation premise
144	unrestricted physics and independent events	retain their separate completion conditions	one explicit independent-comparison condition	the new root-cell theorem preserves independent verification of empirical or public completion
145	negative, closure, positive slots	enumerate the complete Boolean state cube	eight internal states	the signed shadow has two negative, four zero, and two positive preimages
146	six single-polarity states	retain simultaneous negative-positive occupancy	two additional balanced-zero states	the earlier six-state sector is a proper subspace rather than the full origin algebra
147	signed residue; pair depth; closure phase	form a complete invariant key	exact recovery of all three slots	the closure phase cannot be compressed into pair depth without losing particle-wave information
148	arbitrary polarity counts	separate residue and closed-pair depth	T=ABS(R)+2C	balanced content is conserved while the closure phase remains an independent dynamical variable
149	four zero preimages	classify empty, closure-only, open balanced, and closed balanced states	one zero fibre with four internal origins	zero is not a unique physical state
150	open balanced pair	retain symmetric and antisymmetric channel projectors	two orthogonal propagation modes	the open zero supports interference without net signed residue
151	closed balanced pair	apply the symmetric closure projector	one quotient mode	closure produces a localized particle-like manifestation from the same balanced carrier
152	detected and complementary branches	retain both orthogonal outputs	exact reconstruction and quadratic conservation	localized detection retains both complementary branches under exact reconstruction
153	carrier burden; signed residue; closure phase; pair lock	form four physical readouts	complete eight-state tomography	gravity, electromagnetic, weak, and strong channels distinguish states hidden by the signed shadow

No.	Generated input	Exact operation	Generated output	Mathematical consequence
154	open and closed states with equal polarity content	toggle only the closure slot	unchanged burden and charge with exchanged wave-particle readout	particle-wave manifestation is a closure transition rather than two unrelated objects
155	four primitive polarity states	construct all directed non-self relations in the open layer	one twelve-seat JS shell	the original relation-root geometry remains intact
156	the same four polarity states	lift them into the closed layer	a second isomorphic JS shell	closure adds a vertical state direction without changing horizontal root incidence
157	one temporal ordering step	apply the closure involution	open-closed-open period two	time orders the manifestation change while preserving the latent carrier
158	balanced open zero 101	advance one closure step	balanced closed zero 111	the two missing zero states supply the exact wave-capable and particle-capable pair
159	seven exact roots; all preceding time and JS proofs	order every dependency from origin to observation	one three-slot particle-wave four-force theorem	the new layer extends rather than replaces the established primitive shell
160	actual photon experiments and unrestricted field theory	retain independent physical equivalence conditions	one explicit independent-comparison condition	the internal mechanism is is presented as the generated internal quantum-electrodynamic mechanism
161	open balanced zero; primitive time direction	select rational null propagation vectors	one exact massless mode class	the carrier propagates from the generated massless carrier
162	null momentum; open two-channel carrier	construct two orthogonal transverse polarization vectors	two physical propagation modes	longitudinal directions are excluded by the internal transversality constraint
163	potential polarization; null momentum	form the antisymmetric field tensor	vacuum divergence and cyclic identity	Maxwell-form equations follow from one potential relation
164	longitudinal gauge direction	shift the potential by a multiple of momentum	unchanged field tensor	the observable field is independent of the latent gauge representative
165	two transverse real modes	apply the exact quarter-turn complex structure	two opposite helicity sectors	polarization handedness is generated from the exact quarter-turn complex structure
166	orthogonal detection branches; conserved quadratic burden	normalize additive branch shares	Born-form probabilities	probability is the unique rational additive share of the conserved quadratic quantity in the declared class

No.	Generated input	Exact operation	Generated output	Mathematical consequence
167	one rational unit phase	iterate exact complex multiplication	twelve predeclared screen phases	screen positions are generated before any external detection data are read
168	two open paths; quadratic branch law	combine amplitudes before closure detection	exact interference probability ratios	wave readout and localized detection are two stages of one carrier process
169	unequal path burdens	separate visibility and distinguishability	V squared plus D squared equals one	path knowledge and interference contrast are complementary readouts of the same conserved burden
170	transverse projector; conserved currents	add arbitrary longitudinal propagator terms	unchanged amplitudes	the finite abelian interaction obeys an exact Ward identity
171	one transverse self-energy ratio	sum every perturbative order	closed Dyson resummation	all orders of the declared finite class are generated by one recurrence rather than separately adjusted
172	eight internal states	impose neutral, nonempty, balanced, open, and two-mode conditions	unique state 101	the photon candidate is selected by derived properties rather than by its name
173	the same balanced carrier	apply the closure projector	unique state 111	localized particle manifestation is the closed counterpart of the open wave carrier
174	Maxwell, helicity, Born, interference, Ward-Dyson ratios	assemble one immutable packet	one sealed candidate prediction hash	future data leaves unchanged the internal ratios
175	independent implementation and experiment fields	require signed hashes and comparison results	fail-closed external verdict	the author requires independent verification for independent independent confirmation inside the proof file
176	seven exact photon-candidate roots; all preceding time, JS, and four-force roots	order every dependency from origin to observation	one internal photon-Maxwell-Born-QED theorem	the new field layer extends the three-slot carrier without replacing earlier proofs
177	actual continuum QED loop integrals	separate them from the finite Ward-Dyson class	one unrestricted boundary	finite all-order algebra is not mislabeled as the full renormalized standard theory
178	actual photon experiments and independent replication	retain external signed conditions	zero internal substitutions for independent verification events	scientific identity remains independently testable

No.	Generated input	Exact operation	Generated output	Mathematical consequence
179	open balanced carrier; rational unit-circle relation	promote the global phase relation to independent vertex frames	local rational phase group	local phase freedom is generated before any measured electromagnetic potential is introduced
180	local vertex frames; oriented edges	define connection transport and covariant edge differences	exact local gauge covariance	matter transport and gauge transport are one relation
181	connection links; oriented square faces	multiply links around every face	gauge-covariant holonomy	curvature is the closed residual of local phase transport
182	complete four-dimensional cubical carrier	construct all cells and incidence matrices	boundary of boundary equals zero	the discrete exterior derivative is generated from orientation alone
183	edge potential	apply the first coboundary	face curvature	the electromagnetic field is generated as a connection residual
184	face curvature	apply the next coboundary	Bianchi identity	the homogeneous Maxwell equation follows from boundary nilpotence
185	quadratic curvature action; conserved source	vary every edge potential	inhomogeneous Maxwell equation	the source equation follows from one parent action with coefficients fixed by the parent construction
186	integer seed pair	form the Pythagorean null polynomial	formal massless momentum identity	null propagation is proved for the complete generated seed family
187	null momentum	quotient the orthogonal complement by the longitudinal direction	two physical polarization directions	spin-one mode count is a rank theorem rather than a name assignment
188	normalized additive branch measure; permutation symmetry	split the unit into equal rational atoms	weight m divided by n	Born weights are forced on the rational branch algebra
189	rational unit phase; two open paths	form complementary sum and difference ports	exact bright and dark probabilities	two-path ratios are generated for the entire rational phase family
190	two helicity modes	take symmetric occupation powers	bosonic two-mode Fock sectors	multi-photon state counts and commutators follow from symmetric occupation
191	local gauge shift; curvature polynomial	apply arbitrary generated polynomial degree	all polynomial orders gauge invariant	Ward invariance is not confined to one quadratic term

No.	Generated input	Exact operation	Generated output	Mathematical consequence
192	constant affine curvature	refine the four-dimensional cubical carrier	exact refinement-independent action	the affine Maxwell continuum value is reproduced at every finite scale
193	eight exact field roots; all antecedent time and double-zero roots	order every dependency from primitive time to local field observation	one photon-QED-continuum completion theorem	the field layer is dependency-complete and self-contained
194	actual renormalized continuum QED	separate it from the exact generated classes	one unrestricted boundary	formal and finite all-order identities are not mislabeled as measured full QED
195	actual photon data and independent reproduction	retain signed external conditions	independent validation is evaluated by the sealed comparison interface	no internal calculation substitutes for an independent event
196	all generated ratios	compare only dimensionless relations	no absolute standard value enters the proof	the reality bridge remains ratio-first and nonfitted
197	rational polynomial potential family	differentiate each potential component	antisymmetric polynomial curvature	the field is generated before any measured continuum solution is supplied
198	polynomial curvature	apply the exterior derivative twice	coefficientwise Bianchi identity	nilpotence persists beyond the affine sector
199	polynomial curvature	take the divergence and then its divergence	conserved polynomial source	source conservation follows from antisymmetry and commuting derivatives
200	polynomial action density	integrate every monomial over the unit four-cube	exact continuum action	no numerical quadrature or fitted value is used
201	continuum polynomial action	partition the four-cube into every generated uniform refinement	exact partition equality	the polynomial continuum value is reproduced by exact cell integration
202	finite two-helicity sectors	embed every cutoff sector into the next	algebraic Fock direct limit	every finite-particle state has a consistent common carrier
203	nested regulator-polynomial amplitudes	apply recursive divergent projection and counterterm convolution	finite renormalized coefficients	subtraction is exact and idempotent in the declared formal class
204	renormalized coefficients; transverse projector	multiply every generated order by the projector	all-order Ward transversality	renormalization preserves transverse Ward structure

B.4 Derivation rows 205–272

No.	Generated input	Exact operation	Generated output	Mathematical consequence
205	rational unit-circle transport	read it as a two-channel scattering matrix	exact norm conservation	unitarity is an orthogonality theorem
206	channel swap; scattering matrix	conjugate by the swap	crossing reversal	reversing the channel order produces the transposed process
207	all generated dimensionless ratios	seal one deterministic packet	ratio-only prediction certificate	absolute standard values remain outside generation
208	independent identity and signed result	require every independent verification field	fail-closed replication gate	the generating code requires independent certification of external success
209	seven generated closure dependencies; all antecedent field roots	order every dependency from time to renormalized polynomial field	one continuum-QED boundary completion theorem	the proof remains one self-contained file
210	unrestricted interacting continuum QED	separate it from the declared exact classes	one unrestricted field boundary	polynomial and formal completion is not mislabeled as the measured full theory
211	actual blind photon experiment	retain the signed external requirement	the independent comparison field remains available	no mathematical recomputation substitutes for laboratory data
212	third-party implementation	retain independent source and output digests	replication boundary remains open	independence is represented by a separately signed relation
213	four primitive occupancy states	separate one common mode from the relative quotient	ranks one and three	electromagnetic and weak carrier dimensions are generated rather than named
214	two-slot traceless operators	count one diagonal and two directed off-diagonal modes	three weak generators	the weak carrier is the complete traceless two-slot operator space
215	three-slot traceless operators	combine six directed off-diagonal transitions with two diagonal residuals	eight color generators	the color carrier dimension follows from the complete three-slot relation algebra
216	three-dimensional relative quotient	form symmetric bilinears and remove the trace	five spin-two modes	gravity is the symmetric traceless deformation carrier of generated relative space
217	rational anti-Hermitian bases	compute every commutator	closed weak and color algebras	no continuum structure constant is imported
218	all generator triples	evaluate cyclic nested commutators	Jacobi identity	nonabelian Bianchi closure is exact
219	four carrier bases	embed them orthogonally	one seventeen-component parent	the parent dimension is the sum of generated sector ranks

No.	Generated input	Exact operation	Generated output	Mathematical consequence
220	parent edge potential	apply the cubical exterior derivative	one parent curvature and action	all four sector actions are projections of one local field
221	orthogonal sector projectors	test fields supported in one sector	unit sector coefficients	normalization leaves no free relative action weight
222	parent curvature	vary the source-coupled action	one Euler equation	the four sector equations sum exactly to the parent equation
223	local weak and color connections	form face commutators	nonabelian interaction energy	interaction is generated inside the parent carrier
224	compact conjugations	transform connections and curvatures	action invariance	local nonabelian readouts are representation independent
225	gravity, electromagnetic, weak, and strong identities	assemble them as a direct sum	one parent constraint	all sector constraints vanish in the common carrier
226	seventeen polynomial components	integrate every monomial exactly	one continuum parent action	no numerical quadrature or measured field is used
227	uniform four-cube partitions	sum exact cell integrals	refinement-independent sector and parent actions	projection commutes with continuum refinement
228	carrier and action ratios	normalize by their internally generated totals	dimensionless unified packet	absolute standard values are not imported
229	dimensionless packet	hash before comparison	immutable seal	later fitting leaves unchanged the generated prediction
230	independent identity and signed result	require all independent verification fields	sealed independent-comparison condition	the generating program requires independent certification of laboratory success
231	eight double-zero microstates; two orientations	form the complete oriented carrier	sixteen matter states	matter multiplicity is generated before field names are attached
232	generated color rank three; generated weak rank two	form all minimal module products and directed closure branches	dimensions six three three two one one	one minimal generation carrier closes without a remainder
233	primitive charge unit; Yukawa closure; neutral closed singlet	solve every integer candidate	one ratio ledger up to orientation	no measured electric charge is imported
234	one ratio ledger	evaluate color weak gravitational cubic and global anomalies	all anomalies vanish	the chiral matter carrier is internally consistent
235	four primitive time poles	separate the unique unit vacuum	three nonvacuum poles	the generation count follows from the primitive time spectrum

No.	Generated input	Exact operation	Generated output	Mathematical consequence
236	three distinct poles	construct Lagrange spectral projectors	three orthogonal generation sectors	no fourth nonvacuum projector exists in the declared spectrum
237	primitive Pythagorean pairs	compose exact complex plane rotations	unitary mixing matrices	angles are not imported as decimal measurements
238	unitary entries	form orientation-sensitive quartic products	nonzero rational CP invariants	mixing contains an intrinsic directed phase
239	generated multiplets and charges	sum representation indices	three exact one-loop coefficients	scale splitting follows from matter content rather than separate fitting
240	inverse parent coupling; representation coefficient	iterate affine rational steps	exact discrete scale flow	no logarithm is required inside the declared scale ledger
241	three generation carriers; seventeen parent components	assemble all source ledgers	one parent matter source	all non-gravitational total charges cancel
242	four closure channels	sum their charge ratios	zero Yukawa residuals	matter transitions preserve the generated gauge ledger
243	dimension charge generation mixing and flow ratios	seal one deterministic packet	immutable matter certificate	later observation leaves unchanged generation
244	independent identity source output measurement and signature	require every field	sealed independent-comparison condition	the generating code requires independent certification of experiment
245	seven parent-field closure dependencies; all antecedent parent-field roots	order every premise before its consequence	one origin matter-field theorem	the source follows one dependency-ordered proof
246	declared minimal module class	state its exact boundary	conditional origin uniqueness	unrestricted classification of every possible finite relation algebra remains separate
247	generated ratio packet	separate internal generation from observed identity	one sealed comparison interface	the packet is generated entirely from native ratios
248	independent laboratory and third-party events	retain signed independent completion conditions	independent-comparison condition	mathematics uses independent signed experimental records
249	one ordered scalar time chain; crossing-symmetric skew generators; complete force frame	take first and second residuals	outer and inner ternary signs	both sign levels arise from time rather than from generation labels

No.	Generated input	Exact operation	Generated output	Mathematical consequence
250	outer sign; inner sign; empty full occupancy	take the direct product	eighteen relation seats	A B C are fixed as inner negative zero positive
251	three inner binary slots; SU3 Casimir gap; neutral closed singlet	enumerate occupancy	eight latent inner patterns	the latent pattern count is not twenty-four
252	eight patterns; three outer readouts	retain the current outer residual	twenty-four labelled local states	labels preserve dynamical context without creating generations
253	three outer rows; three inner slots; force ratios	allow simultaneous occupancy	five hundred twelve global arrays	the full ledger and the local readout are separate levels
254	current outer residual; full positive and full negative slots; sector dimensions	add the signed inner residual and take its sign	one deterministic next outer readout	the final statement remains independently testable
255	all twenty-four local states; inverse-area density	evaluate the update	eight negative eight zero eight positive outputs	the readout law is exactly balanced
256	uniform local transition ratios; correct physical force-carrier order	diagonalize in the ternary character basis	eigenvalues one three quarters one quarter	common sign and curvature modes are separated
257	balanced inner pattern one zero one; Lorentzian constraint; two polarization field; pair lock	apply it to each current outer residual	negative zero positive persistence	the same inner pattern can have different outer readouts without contradiction
258	all five hundred twelve arrays; physical carrier dictionary	update all three outer rows	twelve transition signatures	the twelve-cell structure is generated by dynamics
259	sign reversal; negative-positive exchange; zero empty-full complement	form one involution	twelve duality orbits	the local and global twelve counts agree independently
260	sign-preserving signature	count its preimages	one hundred forty-four arrays	the paired twelve-seat board is generated as twelve squared
261	all-zero closed signature; all preceding field roots	count its preimages	sixteen arrays	one complete carrier multiplicity emerges from the same ledger
262	ternary orthogonal polynomials; empty-full parity	form eighteen product characters	one orthogonal basis	no hand-selected coordinate vectors are required

No.	Generated input	Exact operation	Generated output	Mathematical consequence
263	even degree zero; all preceding time and double-zero roots	select the common character	one common matter scalar	the source mode is separated before the force residual
264	even nonconstant characters; all preceding time, JS, and four-force roots	take the full sector	eight strong coordinates	the octet is the complete even residual sector
265	odd degree zero	take the uniform fill contrast	one electromagnetic coordinate	signed endpoint contrast is one-dimensional
266	odd degree two	take the quadratic relational grade	three weak coordinates	the triplet is a complete grade rather than a fitted count
267		take the orthogonal complement	five gravity coordinates	the four force dimensions exhaust the seventeen residual modes
268	sixteen closed arrays; existing sixteen-state chiral carrier	construct an exact cardinality bijection	one carrier realization	observed particle names are not inserted
269	three nonvacuum time poles	construct spectral projectors	three generations	generation is derived from time and not from outer negative zero positive
270	sixteen carrier states; three time projectors	take the product	forty-eight latent matter states	outer readout labels do not multiply the physical state count
271	outer readout operator; generation transfer operator	place them on separate tensor factors	zero commutator	readout and generation are mathematically distinct
272	one common source; seventeen force projectors	project the forty-eight-state burden	common source with zero force projection	matter burden and internal force residuals close consistently

B.5 Derivation rows 273–340

No.	Generated input	Exact operation	Generated output	Mathematical consequence
273	five one three eight projectors	sum them with the common projector	eighteen-dimensional identity	the parent action has no missing coordinate
274	counts ratios projectors and transitions	seal one deterministic packet	one immutable certificate	later observations leaves unchanged generation

No.	Generated input	Exact operation	Generated output	Mathematical consequence
275	independent source output measurement and signature	require every independent verification field	sealed independent-comparison condition	the proof preserves independent verification of laboratory completion
276	seven nested-residual roots; all preceding matter and force roots	order every premise before its consequence	one standalone nested residual theorem	the file remains one continuous proof
277	negative zero positive; empty full	define the primitive relation alphabet	structured zero states	zero is not reduced to one scalar label
278	balanced occupied zero	apply one signed residual displacement	selected direction	a microscopic asymmetry chooses a branch
279	selected branch; conserved total occupancy	iterate the monotone transfer	finite one-sided saturation	the initial minority count is the exact saturation time
280	saturated sign	subtract its common background	relative zero	one-sided totality is zero only within the rebased internal frame
281	relative zero; same eighteen-seat alphabet	repeat the local relation rule	eighteen-to-the-depth self-similarity	recursive rebasing supplies the combinatorial fractal origin
282	ordered scalar chain	take first and second residuals	outer and inner signs	time residuals generate the three-by-three-by-two seats
283	three outer signs; three inner signs; two occupations	form the tensor product	eighteen relation seats	A B C are inner negative zero positive sectors and not generations
284	one outer row	enumerate its three binary slots	eight inner patterns	the old eight-state algebra is one row of the larger tensor
285	outer context; eight patterns	retain the generated readout label	twenty-four local states	native pattern and outer readout remain distinct
286	three simultaneous outer rows	enumerate nine binary slots	five hundred twelve global ledgers	global coexistence is distinct from one selected readout
287	inner signed occupancy; current outer residual	apply the deterministic update	next outer sign	outer sign is computed and not assigned
288	five hundred twelve ledgers	quotient by the three-row transition signature	twelve signatures	the JS transition carrier is generated by the residual dynamics
289	sign-preserving signature	count its preimages	one hundred forty-four stable boards	the twelve-by-twelve board arises from one dynamics
290	all-zero closing signature	count its preimages	sixteen closed carriers	the matter carrier count is a closure multiplicity
291	ternary characters; empty-full parity	construct the orthogonal basis	one plus five plus one plus three plus eight	matter source and four force sectors partition the eighteen-seat space

No.	Generated input	Exact operation	Generated output	Mathematical consequence
292	force sectors	remove the common source mode	seventeen force coordinates	one parent residual space carries all four forces
293	sixteen closed carriers; three temporal projectors	take the tensor product	forty-eight latent matter states	generation is independent of outer sign readout
294	native state space	do not choose a fundamental shape	shape-independent latent carrier	a cone remains only one projection illustration
295	native state; single projection	compute the readout and kernel	many-to-one shadow	one readout requires the complete carrier to reconstruct depth density or the full carrier
296	native increment; projection increment	apply the discrete product rule	native projection and interaction terms	readout change has three causal contributions
297	equal readout increments	construct distinct native-only and projection-only causes	explicit causal collision	readout dynamics alone requires the complete native state to determine native dynamics
298	complete four-channel frame	invert the projection matrix	finite parent reconstruction	complementary readouts can recover the declared parent when the map is known
299	single force channel	compute rank and nullity	three hidden native directions	one force equation is information-incomplete
300	native transfer; time-dependent projection	compose both dynamics	four effective force readouts	force channels are projections of one parent and not four native objects
301	native scale family; complete projection	form projected ratios	scale-invariant effective packet	ratios arise only after projection
302	native coordinates; measured effective quantities	enforce the decoder firewall	no direct equality	native residual depth sign and parent coordinates are not measured values
303	projected ratio packet	seal before independent comparison	forward blind boundary	comparison never selects the native proof
304	parent action; matter source	vary one closed system	four equations and matter dynamics	all sectors belong to one variational parent
305	open balanced zero sector	take the abelian exterior derivative	Maxwell-type effective equations	electromagnetism is generated after the native-to-readout map
306	left-supported odd sector	construct noncommuting generators	weak effective dynamics	weak interaction is a projected chiral sector
307	eight-dimensional even residual sector	construct compact non-abelian generators	strong effective dynamics	strong interaction is a projected closed sector

No.	Generated input	Exact operation	Generated output	Mathematical consequence
308	five-dimensional closure sector	construct coframe and curvature action	gravity effective dynamics	gravity is a projected total closure sector
309	finite exact actions	take compatible refinements	projective continuum families	continuum claims retain their declared scope
310	positive transfer kernels	complete the Hilbert reconstruction	unitary finite and projective dynamics	quantum claims retain explicit positivity gates
311	dimensionless projected observables	form ratios before any projection decoder	sealed comparison packet	absolute measured scales are not inserted into the native theory
312	all proof roots	partition by causal dependence	one chronological execution course	new theorems are inserted at their causal position
313	all narrative roots	place each once after prerequisites	one readable explanatory course	proof output forms one mathematical exposition
314	source uniqueness; root uniqueness; all theorem gates	conjoin every result	one standalone four-force proof program	the single source contains every dependency and recomputes every result
315	native layer; projection layer; effective layer	preserve all three domains	noncircular bridge	no layer is silently identified with another
316	recursive native layers	treat one layer readout as a higher-layer effective summary	hierarchical projection chain	fractal recursion and projection hierarchy remain compatible
317	cone example	retain only as a finite illustration	disk and triangle readouts	the example never fixes the fundamental native form
318	one readout family	vary hidden depth-density coordinates	same readout different native content	volume or mass-like content is not inferred from a shadow alone
319	known complete frame	use all independent channels	declared finite tomography	reconstruction requires both enough data and a known projection map
320	unknown projection state	withhold the map	inverse nonuniqueness	even many readouts do not identify a native model without projection dynamics
321	effective equation	interpret as a law on generated readouts	projection-conditioned physics	effective accuracy does not by itself prove a unique origin model
322	native equation	evolve the latent carrier independently	origin dynamics	the native law is not calibrated by an effective shadow value
323	projection equation	evolve observer apparatus and channel state	measurement dynamics	observer motion is a separate causal input

No.	Generated input	Exact operation	Generated output	Mathematical consequence
324	three dynamics	compose native projection and interaction terms	complete forward readout change	the bridge predicts effective change without inverse fitting
325	four force readouts	apply the same hierarchy	one parent-to-four-channel map	unification occurs before effective force equations are compared
326	ratio-only comparison	remove the common scale after projection	dimensionless comparison target	the proof is expressed in scale-free ratios
327	unrestricted inverse problem	retain its nonuniqueness flag	no fabricated origin identification	external data alone does not close the native model
328	unrestricted continuum problems	retain explicit scope boundaries	no false completion	finite certificates do not solve every open field-theory problem
329	external replication	require independent implementation	fail-closed status	the program requires independent verification for external confirmation
330	zero to readout chain	verify each predecessor relation	causal dependency order	no consequence precedes its generating premise
331	projection firewall	verify all forbidden direct identities are false	layer separation	native values and effective measured values remain distinct
332	cone helper	verify example-only flag	no fixed-cone ontology	the explanatory analogy remains separate from the theory premise
333	single channel kernel	verify nullity three	explicit missing information	one shadow requires the complete native state to determine the four-coordinate parent
334	scaled native states	compare projected ratios	identical ratio packet	global scale is removed only at the readout layer
335	all registered roots	execute each exactly once	complete certificate sequence	proof preservation and correction are simultaneously checked
336	all gates	take their conjunction	chronological native-to-effective four-force theorem	the final result remains limited to the declared mathematical class
337	all ternary update maps	exhaust nineteen thousand six hundred eighty-three laws	one admissible residual law	the transition law is unique in the complete native function class
338	eighteen generated characters	classify by parity and ternary degree	one sector assignment	five one three eight is unique under the invariant selector

No.	Generated input	Exact operation	Generated output	Mathematical consequence
339	native matter carrier; anomaly equations; time projectors	decode forward	sealed chiral prediction packet	no observed value enters the packet
340	countable infinite lattice; rational orthogonal coin	apply local conditional shift	unitary causal field evolution	norm and finite propagation are exact

B.6 Derivation rows 341–408

No.	Generated input	Exact operation	Generated output	Mathematical consequence
341	dyadic refinements	take the dense translation union	strong-continuity bridge	the continuum carrier is approached without logarithms or fitted scales
342	five one three eight sectors	apply exact variational identities	four effective operator certificates	measured coefficients do not flow backward
343	rational horizon parameters	factor the nonlinear lapse polynomial	two exact horizons	black-hole area and surface-gravity ratios are generated
344	Kasner and FLRW families	verify nonlinear Einstein equations	vacuum and matter cosmologies	the gravity sector is not limited to linearization
345	Lorentz constraints; inverse-limit action	propagate and refine	constraint-compatible quantum gravity class	two physical polarizations survive
346	five hundred twelve ledgers	enumerate directly	first core packet	no factorization assumption is used
347	local transition counts	multiply independent row factors	second core packet	no global enumeration is used
348	two core packets	compare exact objects and digests	algorithmic reproduction	independent algorithms agree exactly
349	prediction packet	freeze its digest	blind-comparison object	later data leaves unchanged the theory output
350	external reproduction schema	freeze challenge digest	third-party executable target	the program cannot invent an external signature
351	mathematical frontiers	conjoin seven new theorems	seven completed rows	all in-file solvable weaknesses are closed
352	external-only actions	require independent identity and hidden data	fail-closed rows	independent evidence is represented by signed comparison fields
353	native state; projection state	retain the three-term product rule	effective readout	direct-value identity remains forbidden

No.	Generated input	Exact operation	Generated output	Mathematical consequence
354	infinite-volume dynamics; four force channels	take the direct sum	one unitary parent readout carrier	all channels share one causal field class
355	nonlinear gravity families; effective operators	couple through the projection firewall	ratio-only gravity readouts	absolute calibration is supplied by the unique unit decoder
356	chiral packet; scattering packet	combine from generated inputs	forward prediction registry	the packet is independently testable
357	source comments; reader principles	place before the corresponding theorem	continuous readable proof	the theorem is presented at its dependency position
358	all registered antecedent roots; eight closure theorem roots	execute each once	dependency preservation	all valid antecedent proofs remain in the standalone dependency chain
359	all mathematical gates	take their conjunction	expanded declared-class completion	no finite example is mislabeled as every possible theory
360	external frontier protocol	retain false until independently signed	honest empirical boundary	mathematical completion and external confirmation remain logically distinct
361	saturated relative zero; eighteen child relations	iterate inward	eighteen-branch cylinder hierarchy	every finite depth is compatible with the same relation law
362	complete child family	coarse grain eighteen children	one higher-layer parent	outward aggregation is the reverse ledger operation
363	depth n cylinders	assign weight one over eighteen to the n	normalized layer measure	every parent weight equals the sum of child weights
364	shell depth	count all relation words	eighteen to the depth shell law	layer population is generated rather than fitted
365	all shells through depth n	sum the geometric sequence	eighteen to n plus one minus one over seventeen	the onion-layer cumulative formula is exact
366	outer inner and fill signs	sum over a complete shell	zero signed boundary flux	internal occupancy may remain nonzero while the layer closes externally
367	same symbolic shell; two embeddings	compare distance signatures	different geometric readouts	branching alone does not fix a sphere cone torus or another shape
368	orientation cycle	sum four directed steps	closed displacement	topological closure requires an embedding and orientation law
369	six-phase internal cycle	resolve every phase	nonzero internal motion	the lower-layer observer reads force-like flow
370	same six phases	quotient by conserved macro signature	one persistent effective unit	the upper-layer observer reads a closed object
371	closed cycle activity	retain quadratic internal burden	energy-like readout	coarse stability does not erase internal dynamics

No.	Generated input	Exact operation	Generated output	Mathematical consequence
372	local quotient unit	hide phase through projection	particle-like readout	particle means unresolved closed unit in this classification
373	closed source; multiple locations	apply a distributed response map	field-like readout	one carrier can have a spatially extended effective role
374	force energy matter particle field roles	evaluate closure and projection predicates	layer-dependent role table	no mandatory force-to-matter conrealization is assumed
375	ternary eigenvalues; occupancy parity	take the tensor product	eighteen exact scale eigenvalues	the multiscale operator is generated from the local transition law
376	common character	evaluate its eigenvalue	scale-fixed mode one	the common source persists across layers
377	uniform fill contrast	evaluate its parity eigenvalue	signed period-two mode minus one	a boundary contrast can alternate without magnitude loss
378	strong weak gravity characters	evaluate rational degree products	bounded sector spectra	sector attenuation and sign are computed exactly
379	scale spectrum; layer-role predicates	keep the maps separate	nonfixed ontology	an eigenvalue alone does not decide whether a mode is force or matter
380	native carrier; coarse observer	compose closure with projection	dynamics-to-object bridge	internal flow and external object are compatible descriptions
381	lower-layer closed unit	reuse as higher-layer relation seat	recursive source generation	a material readout can become a force or source input at another layer
382	higher-layer effective unit	resolve its hidden phases	inverse role decomposition	a material or particle readout can reopen into lower-layer dynamics
383	reader exposition map	print before each relevant theorem	premise-before-proof order	the reason for each equation remains visible in the output
384	source orientation comments	place beside the defining functions	standalone readable source	the defining functions contain complete standalone semantic descriptions
385	conceptual explanations; computed certificates	pair every interpretation with exact scope	no analogy becomes an unstated axiom	cone and onion language remain examples
386	inward and outward limits; projection firewall	join through the layer-role theorem	multiscale native-to-effective chain	readouts remain distinct from native values

No.	Generated input	Exact operation	Generated output	Mathematical consequence
387	all antecedent proof roots; five multiscale closure roots	execute each once in causal order	standalone dependency proof	each explanation precedes its dependent theorem
388	all exact gates	take the conjunction	expanded multiscale four-force theorem	independent physical confirmation remains independently testable
389	three outer coordinates; three inner coordinates; two occupations	use the generated product coordinates	exact three-by-three-by-two symbolic scale	scale is derived without selecting a physical shape
390	symbolic depth	multiply axis subdivisions	eighteen-to-the-depth cells	cylinder weights normalize exactly
391	cyclic coordinate transport	iterate the native relation shift	six-layer symbolic closure	finite relation topology has an exact period
392	closed conserved flux; effective dimension	divide by boundary multiplicity	one-over-radius-to-dimension-minus-one law	inverse square follows only in generated effective dimension three
393	integer closure depth	retain only complete shells	discrete scale levels	shell quantization comes from closure rather than fitted radii
394	microscopic saturation cycle	count exact transfer ticks	four-tick lower-layer clock	time is the order of residual completion
395	nested completed cycles	coarse grain each child closure to one event	aggregate layer time	one lower process becomes one higher event
396	three symbolic coordinates per layer	resolve every depth	rank three times depth	coarse observation compresses the hierarchy back to rank three
397	closure and projection predicates	classify one carrier	force energy matter particle field and gravity-like roles	role is layer dependent and not mandatory
398	eighteen character scale spectrum	assemble one diagonal native update	single native evolution operator	all effective sectors share one parent dynamics
399	sector projection; native update	prove the intertwining identity	effective pushforward equation	the projected law is not imported from observation
400	native trajectory; sector trajectory	compare step by step	identical projected evolution	native-first and effective-only routes agree
401	native action defect	split by orthogonal sectors	zero sector defects and zero parent defect	the variational constraint descends exactly
402	time-dependent projection	retain the product-rule interaction term	complete effective change	observer motion is not hidden in native motion
403	open balanced zero	project the abelian sector	Maxwell-type gauge dynamics	the effective field is generated after the origin layer

No.	Generated input	Exact operation	Generated output	Mathematical consequence
404	orthogonal projected branches	normalize conserved quadratic burden	Born-type probability	the rule follows from conserved quadratic normalization
405	local norm-preserving transfer	complete the infinite carrier	unitary quantum evolution	unitarity is an effective push-forward certificate
406	generated Clifford carrier	apply chiral propagation	Dirac-type effective equation	chirality comes from the matter sector
407	compact noncommuting sector	form curvature and variation	Yang-Mills-type dynamics	the gauge equation is a projected parent equation
408	universal closure burden; metric projection	vary the symmetric sector	Einstein-type dynamics	nonlinear families remain within their declared class

B.7 Derivation rows 409–476

No.	Generated input	Exact operation	Generated output	Mathematical consequence
409	closed four-site relation loop	diagonalize its exact Laplacian	rational vibration spectrum	string-like mathematics emerges without a fundamental string axiom
410	repeated coarse graining	iterate the transfer kernel	renormalization fixed point	scale flow is generated by the layer map
411	consistent closed refinements	take projective geometry limits	loop-spinfoam-like effective structure	structural overlap is not full equivalence
412	one parent action; one plus seventeen split	compare generated operator families	standard-model-plus-gravity effective dictionary	the native parent generates the effective equation family in forward order
413	unitary and Born certificates	group their projected consequences	quantum-mechanics effective dictionary	the native state is not identified with one measured wavefunction value
414	closed modes	compare only the effective spectrum class	string-like connection	no compactification or full string equivalence is claimed
415	projective geometry	compare refinement invariants	loop-spinfoam-like connection	no unrestricted quantum-gravity equivalence is claimed
416	recursive fixed point	compare scale-flow structure	asymptotic-safety-like connection	the overlap is mathematical and bounded
417	all effective family rows	seal before any external table	native-to-effective packet hash	no coefficient generation is forward and sealed
418	reader explanations	place before their mathematical certificates	standalone derivation rationale	the generalized shadow metaphor cannot be mistaken for literal optics

No.	Generated input	Exact operation	Generated output	Mathematical consequence
419	native origin; projection; effective equations	compose in one causal direction	complete forward physics compiler	standard physics is recovered as an effective layer
420	all registered antecedent and closure roots	execute each exactly once	expanded chronological four-force proof	physical equality remains a separate sealed comparison interface
421	three-by-three-by-two infinite addresses	decode finite prefixes as nested rational boxes	complete three-coordinate continuum carrier	physical metric remains a later projection datum
422	temporal residual order; three symbolic coordinates	form their direct causal product	three-plus-one generated coordinate rank	time is not copied from an external spacetime equation
423	eighteen child relations	average and subtract the common component	one common plus seventeen residual directions	coarse matter-like readout and resolved force-like modes are scale roles
424	native states modulo a projection kernel	descend the native update to equivalence classes	exact effective quotient dynamics	effective accuracy does not imply a unique native origin
425	closure and observation predicates	exhaust every Boolean combination	force energy matter particle field and gravity-like role classes	no mandatory role ladder is assumed
426	incidence; adjoint; skew update; variation	compose one operator grammar	conservation gauge wave quantum chiral nonabelian and metric families	measured coefficients are not origin inputs
427	generated operator certificates	map thirty principal equation forms to native ingredients	standard equation dictionary	the dictionary covers forms rather than all empirical constants
428	all mathematically relevant explanations	place them before dependent certificates	complete standalone conceptual exposition	the exposition contains only mathematical definitions, constructions, and theorems
429	sealed prediction; two internal implementations; source digest	assemble a no-fitting external challenge	blind ratio and reproduction package	independent signatures are evaluated by the sealed matcher
430	all internal closure theorem roots	conjoin their exact gates	complete native-effective unification theorem	independent physical identity remains independently testable
431	three-coordinate continuum cells; seventeen hidden residuals; six hidden phases	form a surjective quotient projection	three-dimensional effective world fibers	the effective world is real but information-compressed

No.	Generated input	Exact operation	Generated output	Mathematical consequence
432	distinct native states	hold the projected cell fixed while changing hidden labels	many-to-one world projection	effective equations maps to a nontrivial native equivalence class the native source
433	native local neighbors	project to neighboring world cells	locality descent	hidden internal differences do not create superlocal world jumps
434	one relation step; one temporal update	define the causal conrealization ratio	dimensionless universal causal speed	no measured light speed is inserted
435	six-step relation closure	coarse grain distance and time together	layer-invariant causal ratio	the speed is preserved across recursive layers
436	unequal object-specific rates	compare their values	object-motion universality no-go	one arbitrary native rotation requires the universal generated limit to define the universal limit
437	native increment; projection increment	retain the exact product rule	synchronized effective change	native and projection motion remain distinct causes
438	sign-alternating electromagnetic mode	shift one local cell per tick	causal-bound-saturating effective pulse	physical photon identification is represented by the sealed decoder
439	rational norm-preserving carrier	read either distributed amplitudes or rank-one channels	wave-particle projection duality	one carrier supports two resolutions
440	closure burden; fixed native adjacency	change only the effective edge costs	metric-dependent path and delay	the finite graph is not asserted to be full general relativity
441	relation counts; ranks; closure ratios; sector fractions	seal dimensionless invariants	projection constant ledger	measured constants are not origin inputs
442	common distance-time rescaling	rescale numerator and denominator together	causal-ratio invariance	unit choice cannot change the dimensionless speed
443	hidden kernel shifts	hold the projected ratio fixed	kernel invariance	unobserved native directions do not alter the effective constant
444	thirty generated equation forms; three-dimensional quotient world	act on projection equivalence classes	effective world equation closure	standard physics remains exact in its declared domain
445	existing candidate structures	compare only bounded effective mathematics	nonexclusive framework connections	no candidate is used backward to choose the native theory

No.	Generated input	Exact operation	Generated output	Mathematical consequence
446	world and light explanations	place every premise before its certificate	complete standalone exposition	the optical language is used only as a mathematical projection interpretation
447	all world light internal roots	conjoin exact gates and seal scopes	complete three-dimensional world unification theorem	physical universe and measured photon identity remain falsifiable
448	all antecedent and closure roots	execute once in causal order	expanded chronological four-force proof	external blind ratios and third-party reproduction remain independent
449	outer sign; inner sign; occupancy	form labeled local change vectors	rank-three semantic carrier	three-dimensionality follows from independent local semantics rather than eighteen alone
450	three three two radix labels	enumerate integer factorizations preserving labels	unique labeled factorization	nonlocal merged encodings are excluded by the local grammar
451	unit causal ratio; three-coordinate world	construct a rational boost	exact null-cone preservation	no measured speed is inserted
452	three-four-five rotation	pull back the spatial metric	exact rational rotational covariance	a nontrivial causal frame subgroup is generated
453	two equal orthogonal channels	normalize quadratically	one over square root two algebraic scale	the Bell coefficient is generated by duality
454	four measurement settings	construct joint Born probabilities	Bell correlation beyond the local bound	all probabilities remain positive and normalized
455	remote setting choices	sum over the remote result	one-half local marginals	projection fibers do not enable signalling
456	character scale eigenvalues	iterate coarse graining	persistent and contracting mode split	stable effective laws emerge from the native spectrum
457	two-step electromagnetic block	square the sign mode	persistent causal electromagnetic mode	microscopic sign alternation does not erase the effective carrier
458	orthonormal sector eigenvalues	take the mean squared value	native quadratic coefficients	effective coefficients descend without observed couplings
459	quadratic coefficient vector	normalize by a common reference	scale-free sector ratio packet	absolute units cancel
460	fractional algorithm; integer common-denominator algorithm	compare normalized vectors	dual-algorithm prediction identity	the packet is independently reproducible inside the source
461	sealed ratio packet	mutate each component after sealing	deterministic rejection	fitting and post-seal repair are impossible
462	native ledger; world projection	apply the exact surjective quotient	native-to-world interface	hidden fibers do not destroy world locality

No.	Generated input	Exact operation	Generated output	Mathematical consequence
463	world quotient; generated operator grammar	push forward actions and coefficients	standard effective physics layer	standard equations remain exact within their declared domain
464	effective equations; dimensionless invariants	form the immutable ratio contract	observation interface	only ratios are compared
465	all hierarchy explanations	place premise before certificate	standalone readable hierarchy	the exposition contains only the generated proof structure
466	all hierarchy roots	conjoin exact gates	complete native world standard observation theorem	independent evidence is recorded through signed comparison fields but immediately scoreable
467	native generator; projection fibers	test kernel invariance	exact autonomous effective law	trajectory agreement is decided for every state
468	closed force-sector kernels	construct intertwining operators	four exact quotient dynamics	five one three eight sectors share one parent generator
469	nonclosed coupled projection	apply a hidden kernel witness	autonomy contradiction	one current shadow value requires the complete native state to determine every next value
470	visible block; hidden block	eliminate hidden coordinates exactly	instantaneous memory and hidden-force terms	projection failure becomes a calculable effective law
471	nilpotent hidden update	compute successive memory kernels	finite two-step memory	no fitted damping or continuum ansatz is used
472	proposed observable rows	close their Krylov span under the native generator	minimal invariant observable algebra	effective variables are discovered rather than copied
473	proper observable prefixes	test generator invariance	minimality certificate	no smaller memoryless linear system exists in the declared model
474	JS shell source; three projection families	evolve project and jointly reconstruct	forward JS-SH dynamical diagram	several ambiguous readouts recover one source together
475	final reconstructed source; inverse native generator	step backward exactly	reverse JS-SH source history	forward and reverse equations share one native carrier
476	gravity electromagnetic weak strong projectors	classify closure of their observable algebras	four memoryless force laws	no separate native generator is introduced per force

B.8 Derivation rows 477–544

No.	Generated input	Exact operation	Generated output	Mathematical consequence
477	incomplete mixed force readout	retain its exact memory kernel	finite-memory effective law	partial resolution explains non-Markovian shadows
478	thirty standard equation forms	classify by closure memory scale or constraint	one dynamical bridge dictionary	equations are connected by a rule rather than by visual analogy
479	six framework structures	apply the same compiler	bounded candidate connections	existing candidates remain effective descriptions and not native axioms
480	instantaneous memory kernel; first delayed kernel	take exact squared-strength ratio	one-eighth closure-defect prediction	a new ratio is generated before observation
481	closure-defect ratio	rescale numerator and denominator together	scale invariance	the prediction is dimensionless
482	all dynamical explanations	place premise before certificate	standalone closure-memory exposition	the exposition follows the generated dynamical proof structure
483	closure memory reconstruction and prediction roots	conjoin exact gates	complete dynamical closure theorem	one rule spans source world equations and observations
484	all antecedent and dynamical roots	execute once in causal order	expanded chronological four-force proof	unrestricted physical identification remains independently falsifiable
485	polynomial native vector field; projection readouts	take exact Lie derivatives	continuous closure criterion	factorization is tested as a polynomial identity
486	equal current difference readouts; different hidden coordinates	compare their derivatives	continuous nonclosure witness	one shadow value leaves multiple admissible continuations for one derivative
487	conserved total; linearly advancing hidden coordinate	integrate the hidden square exactly	cubic nonlinear flow	no infinite series is required
488	nonlinear flow at two times	compose and compare with summed time	exact semigroup law	inverse time recovers every sampled source
489	visible coordinate; two hidden shift seats	eliminate hidden states	two-step nonlinear memory recurrence	nonclosure becomes an exact effective equation
490	delayed visible square	propagate every initial state	native-memory trajectory identity	no fitted kernel is introduced
491	rational projective clock	apply the Cayley family	JS-SH norm-preserving flow	trigonometric evaluation is unnecessary

No.	Generated input	Exact operation	Generated output	Mathematical consequence
492	two projective clock parameters	compose by the fractional clock law	closed forward-reverse family	complementary shell readouts reconstruct the source
493	common carrier; four sector coordinates	vary and eliminate the stationary common coordinate	four-force Schur action	one parent generates all cross-sector couplings
494	five one three eight ranks	form the coupling vector	positive effective sector matrix	all principal minors are exact and positive
495	sector rank; closure type; operator role	enumerate every decoder permutation	unique semantic force map	observed numbers do not choose the labels
496	thirty standard equation forms	replace every mechanism label by a proved generalized root	thirty commuting diagrams	native sources are not selected backward
497	native mechanism; world readout; effective equation; ratio interface	verify each arrow	closed trajectory compiler	all thirty rows pass
498	native coefficient ratios; memory ratio	combine independent packets	multi-ratio prediction set	absolute units are absent
499	five one three eight pair products	normalize by their total	six common-carrier pair fractions	the fractions sum to one
500	force ranks	normalize by seventeen	four force-rank fractions	the fractions sum to one
501	sealed multi-ratio packet	mutate each pair component	deterministic rejection	post-seal repair is impossible
502	all generalized explanations	place premise before certificate	standalone generalized exposition	the exposition contains only the generated proof structure
503	Lie flow memory JS-SH action decoder equation prediction roots	conjoin exact gates	complete generalized dynamical theorem	independent evidence is recorded through signed comparison fields
504	all antecedent and generalized roots	execute once in causal order	expanded chronological four-force proof	unrestricted physical identification remains independently falsifiable
505	order-six native jet	enumerate every monomial coefficient	coefficientwise closure decision	sample fitting is replaced by an algebraic test
506	hidden-direction jet monomials	inspect projected derivatives	explicit smooth-jet obstructions	one hidden coefficient prevents autonomy
507	arbitrary rational transfer polynomial	integrate each shifted monomial	exact triangular polynomial flow	the family is solved by a finite binomial identity
508	two rational flow times	compose and compare with their sum	parameterized semigroup law	the result holds throughout the declared coefficient family

No.	Generated input	Exact operation	Generated output	Mathematical consequence
509	m hidden shift seats	eliminate the queue	m-step nonlinear memory equation	memory depth is generated by hidden dimension
510	two states differing only in the oldest seat	compare their next visible state	memory-depth minimality witness	no shorter state description is exact
511	JS-SH rational Cayley family	differentiate the projective parameter	continuous skew-generator equation	norm preservation holds infinitesimally
512	negative projective parameter	multiply by the forward matrix	exact reverse identity	forward and reverse clocks share one generator
513	four-force Schur action; common layer scale	scale the complete parent system	uniformly scaled effective action	normalized force content is unchanged
514	scaled principal minors	compare with degree powers of the layer factor	recursive positivity	action stability survives every declared layer
515	seven trajectory mechanisms	construct distinct semantic signatures	mechanism decoder table	equation mechanisms are fixed before observation
516	all seven-factor permutations	test signature preservation	one admissible mechanism map	numerical proximity cannot relabel the compiler
517	sealed dimensionless prediction rows	apply labeled interval scoring	blind capsule verdict	theory state is immutable during comparison
518	one displaced capsule component	run the same scorer	deterministic rejection	post-seal repair is impossible
519	finite internal test points	construct their vanishing polynomial	finite-agreement counterexample	finite replay cannot prove unrestricted identity
520	outside test point	evaluate the same polynomial	nonzero disagreement	the unrestricted boundary is mathematically explicit
521	all exact-frontier explanations	place premise before certificate	standalone frontier exposition	the proof is complete from its declared mathematical dependencies
522	jet flow memory JS-SH action decoder verdict boundary roots	conjoin exact gates	complete exact internal frontier	independent facts enter through signed comparison fields
523	complete internal frontier; external signature fields	separate logical ownership	maximal internal completion without fabricated evidence	independence is preserved
524	all antecedent and exact-frontier roots	execute once in causal order	expanded chronological four-force proof	unrestricted and empirical identities remain independently falsifiable

No.	Generated input	Exact operation	Generated output	Mathematical consequence
525	dependency-ordered four-force proof	state the complete foundational step	Two primitive channels and four transformation layers generate the integers 2, 4, 8, 10, and the twelve-seat carrier.	the foundational derivation is explicit and precedes the later completion rows
526	dependency-ordered four-force proof	state the complete foundational step	The least common carrier of the two-fold, three-fold, and four-fold requirements is 12, so no smaller positive carrier satisfies all three requirements.	the foundational derivation is explicit and precedes the later completion rows
527	dependency-ordered four-force proof	state the complete foundational step	Rank plus root count gives algebra dimensions 1, 3, and 8, identifying the abelian, weak, and color operator classes.	the foundational derivation is explicit and precedes the later completion rows
528	dependency-ordered four-force proof	state the complete foundational step	The anomaly equations and Yukawa equations are solved simultaneously; their primitive integer solution fixes the relative chiral charges.	the foundational derivation is explicit and precedes the later completion rows
529	dependency-ordered four-force proof	state the complete foundational step	Adding weak isospin to hypercharge gives the reduced electric ratio up:down:neutrino:electron = 2:-1:0:-3.	the foundational derivation is explicit and precedes the later completion rows
530	dependency-ordered four-force proof	state the complete foundational step	The Higgs vacuum has total electric charge zero, so one neutral gauge direction remains unbroken.	the foundational derivation is explicit and precedes the later completion rows
531	dependency-ordered four-force proof	state the complete foundational step	The neutral mass matrix has determinant zero and positive trace; therefore it has exactly one massless direction and one massive direction.	the foundational derivation is explicit and precedes the later completion rows

No.	Generated input	Exact operation	Generated output	Mathematical consequence
532	dependency-ordered four-force proof	state the complete foundational step	The Ginsparg-Wilson identity and index one provide one exact chiral fermion mode without the fifteen unwanted doubler modes.	the foundational derivation is explicit and precedes the later completion rows
533	dependency-ordered four-force proof	state the complete foundational step	BRST nilpotency separates gauge-exact states from the positive physical state space.	the foundational derivation is explicit and precedes the later completion rows
534	dependency-ordered four-force proof	state the complete foundational step	Gauge invariance and canonical dimension leave exactly the required kinetic, scalar, Yukawa, and neutral Majorana operator classes.	the foundational derivation is explicit and precedes the later completion rows
535	dependency-ordered four-force proof	state the complete foundational step	The electromagnetic operator is abelian with a conserved signed source; the weak operator is non-abelian and left-chiral.	the foundational derivation is explicit and precedes the later completion rows
536	dependency-ordered four-force proof	state the complete foundational step	The strong operator carries the A2 color algebra, a positive gap, and the exact meson and baryon color decompositions.	the foundational derivation is explicit and precedes the later completion rows
537	dependency-ordered four-force proof	state the complete foundational step	The gravity operator is symmetric rank two and couples universally; the Einstein-Maxwell family verifies nonlinear matter-curvature coupling.	the foundational derivation is explicit and precedes the later completion rows
538	dependency-ordered four-force proof	state the complete foundational step	The four operator signatures are pairwise distinct, so the force assignment is unique inside the declared grammar.	the foundational derivation is explicit and precedes the later completion rows
539	dependency-ordered four-force proof	state the complete foundational step	Permutation symmetry and normalization force all four parent-action weights to equal one quarter.	the foundational derivation is explicit and precedes the later completion rows

No.	Generated input	Exact operation	Generated output	Mathematical consequence
540	dependency-ordered four-force proof	state the complete foundational step	Differentiating the four sector actions and adding their gradients reproduces the Euler-Lagrange equation of the single parent action.	the foundational derivation is explicit and precedes the later completion rows
541	dependency-ordered four-force proof	state the complete foundational step	A positive reflection-positive transfer semigroup supplies exact interacting quantum evolution.	the foundational derivation is explicit and precedes the later completion rows
542	dependency-ordered four-force proof	state the complete foundational step	The dimensionless prediction packet is hashed before comparison, preventing later numerical retuning.	the foundational derivation is explicit and precedes the later completion rows
543	all 525 registered theorem roots	execute each exactly once in the declared dependency order	every registered theorem gate is true	the complete internal proof has no skipped theorem root
544	internal proof completion; independent evidence interface	separate executable derivation from independent measurement and reproduction	internal declared class is proved while physical identification is evaluated through the independent comparison interface	the proof supplies a sealed object for independent confirmation

C Proof-completeness and artifact-coherence matrix

Proof object	Exact value
Authoritative causal stages	26
Executable stage groups	9
Primary theorem roots	487
Narrative theorem roots	38
Total theorem roots	525
Explicit derivation rows	544
Generated effective-equation forms	30
Fresh-run theorem conjunction	525/525 True
Python source lines	66,229
Python source bytes	4,075,166
Deterministic output lines	1,824,059
Deterministic output bytes	156,845,519
Execution error stream	0 bytes

Theorem C.1 (Complete registry correspondence)**.** *The main text, ordered registry, and derivation ledger form one dependency-ordered record. The 525 theorem roots occur exactly once, the 544 derivation rows occur exactly once, and the nine executable stage counts sum to* $40 + 44 + 50 + 51 + 28 + 66 + 108 + 78 + 60 = 525$.

D Executable proof realization and hashes

Item	Value
Code title	RESIDUAL FIELD UNIFICATION PROOF
Python SHA-256	`e3ec5ba55089e81e33c919b01d7242781c1e33f4b278531201a71f171b64f3d7`
Output SHA-256	`dd66562bc159db538714abe7af46ef788043a967928d99ddcfeba1c5f4c59410`
Code/output ZIP SHA-256	`65604b3bc47ffbafbb28650b35116550099f2dd396a47a7d30396df40ec41015`
Fresh-run exit code	0
Standard error	0 bytes
Second fresh run	byte-identical
Forbidden transcendental/fitting calls	none

E Alphabetical symbol glossary

This glossary repeats the principal notation in alphabetical order so that a reader entering through an appendix, theorem registry, or code cross-reference does not need to reconstruct a symbol from context.

Symbol	Meaning
$\mathcal{A}$	Eighteen-state primitive relational alphabet.
$\mathbb{A}_{pq}$	Generated block-spectral finite edge operator.
$A(u)$	Historical notation for the second-residual quadratic scale $1+u+u^2$; the compact overview also uses $C(u)$ to avoid collision.
C_{res}	Nonnegative residual consistency certificate, not a physical action.
$\mathcal{D}_\Omega$	Parent-connection covariant matter operator.
$\Delta\chi$	Three-component temporal residual vector.
ΔN	Finite clock-scale difference $N(q)-N(p)$.
$\mathcal{F}_\Omega$	Parent curvature $d\Omega+\Omega\wedge\Omega$.
$H_\square$	Finite plaquette holonomy.
$\mathcal{H}_d$	Canonical rank-d carrier block; $d \in \{0,1,3,5,8,17,18\}$ according to context.
I_n	Identity operator on an n-dimensional space.
λ_i	Generated normalized response ratio in sector i.
$\mathbb{M}$	Positive projector-diagonal response metric in the parent action.
$N(p)$	Positive local temporal scale at edge endpoint p.
P_i	Canonical orthogonal projector for sector $i \in \{\mathrm{G},\mathrm{E},\mathrm{W},\mathrm{S}\}$.
p,q,r	Generated path vertices or integer depth labels.
$\Psi,\bar{\Psi}$	Parent matter field and its conjugate/dual variable.
$\mathcal{R}(X)$	Exact residual coordinate $(I-X)X^{-1}$.
$R^U_{\mathrm{G}}(p)$	Source gravitational residual after transport to the target frame.
S_{parent}	Sole physical curvature–matter action.
T_a	Scalar fractional-linear source–shadow update.
T_r	Nilpotent generation-lowering operator for chiral role r in the matter-channel section.
U_{pq}	Invertible block-preserving parent frame transport.
V_{rad}	Generated gauge-invariant radial matter potential.
X_p, X_q	Source and target bounded parent shadows.
Ω	Parent connection one-form; distinguished from the occupancy set by context and typography.
ϵ	Formal plaquette expansion indeterminate.
η	Generated Lorentz signature representative $\mathrm{diag}(-1,1,1,1)$.

F Author contribution, funding, and declarations

Author contribution. Seunghyun Hong conceived the CRGE–JS–SH source framework, developed the relational and temporal-residual interpretation, designed the proof architecture, and prepared the manuscript and computational realization.

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Ethics. This theoretical and computational work used no human participants, animals, or clinical data.

G Data and code availability

The V22 package contains the Main Manuscript and this Supplementary Proof Record in PDF and TEX form, `CODE_8100_929_RESIDUAL_FIELD_UNIFICATION_PROOF.py`, `CODE_8100_929_RESIDUAL_FIELD_UNIFICATION_PROOF.txt`, `COMPLETE_THEOREM_REGISTRY_525.csv`, `COMPLETE_DERIVATION_LEDGER_544.csv`, a README, and a SHA-256 manifest. The record DOI is 10.5281/zenodo.21702095; the all-versions DOI is 10.5281/zenodo.21127052.

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